

EE104 Fundamentals of Electric Circuits (Lesson 12)

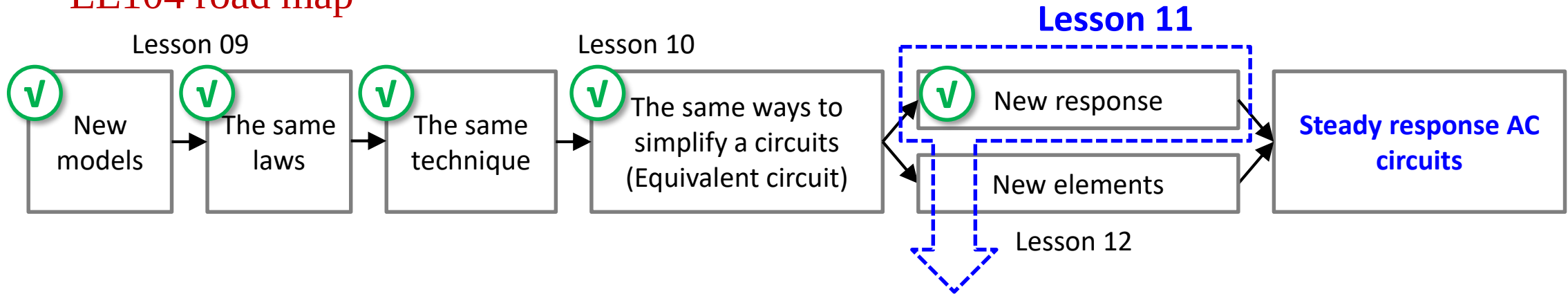
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Review of Lessons 11(a)

- EE104 road map



Frequency response

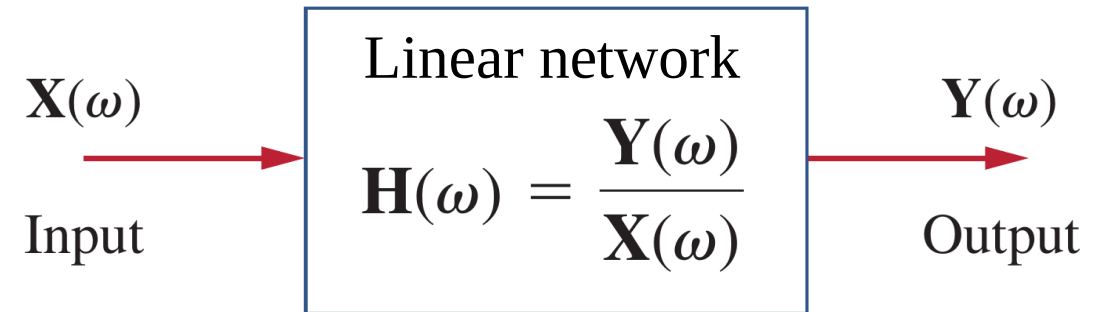
- Transfer function
- Passive filters: lowpass and highpass filters
- Resonance + bandpass and bandstop filters
- Active filters
- High-order filter

Review of Lessons 11(b)

- Frequency response:

a) Frequency response is the quantitative measure of the output spectrum of a circuit or device in response to a stimulus, and is used to characterize the dynamics of the system. It is a measure of magnitude and phase of the output as a function of frequency, in response to the input.

b) 频率响应是指电路或系统对不同频率的激励信号所作出响应。



电压增益 $H(\omega) = \text{Voltage gain} = \frac{V_o(\omega)}{V_i(\omega)}$

电流增益 $H(\omega) = \text{Current gain} = \frac{I_o(\omega)}{I_i(\omega)}$

转移阻抗 $H(\omega) = \text{Transfer Impedance} = \frac{V_o(\omega)}{I_i(\omega)}$

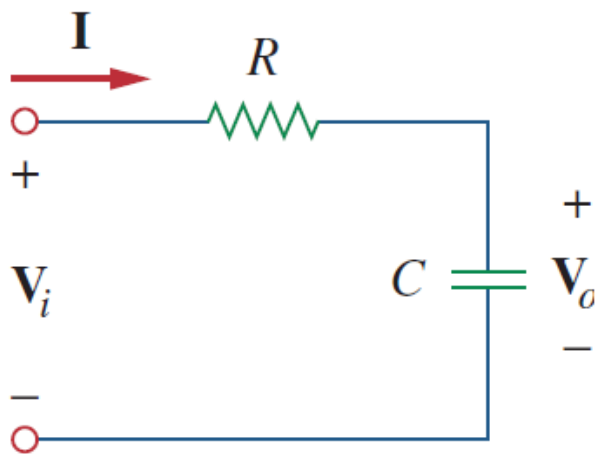
转移导纳 $H(\omega) = \text{Transfer Admittance} = \frac{I_o(\omega)}{V_i(\omega)}$

Review of Lessons 11(c)

- Frequency response: magnitude response + phase response

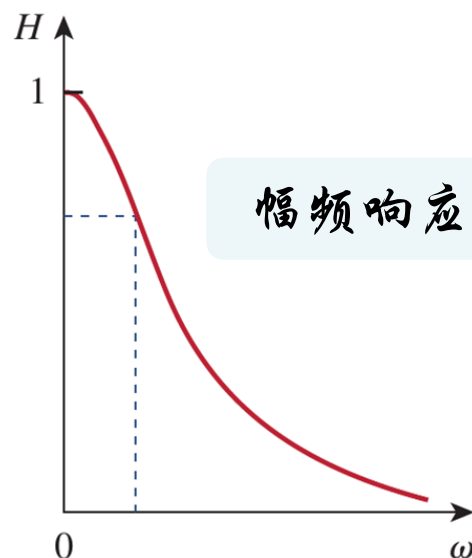
Frequency response

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_s} = \frac{1}{1 + j\omega RC}$$



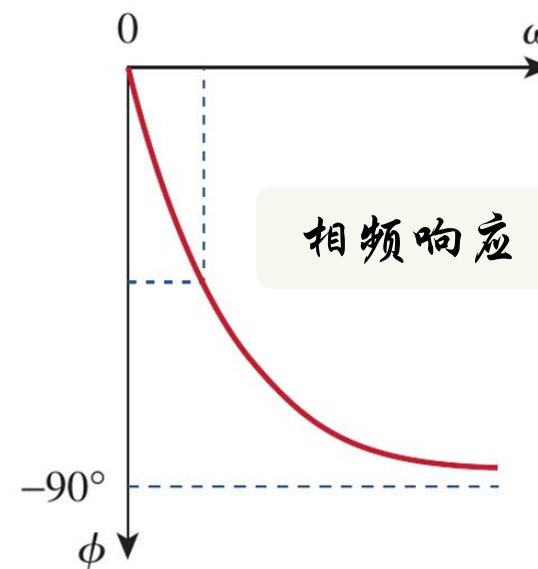
Magnitude response

$$H = \frac{|\mathbf{V}_o|}{|\mathbf{V}_i|} = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$



Phase response

$$\theta = -\tan^{-1}(\omega RC)$$



Review of Lessons 11(d)

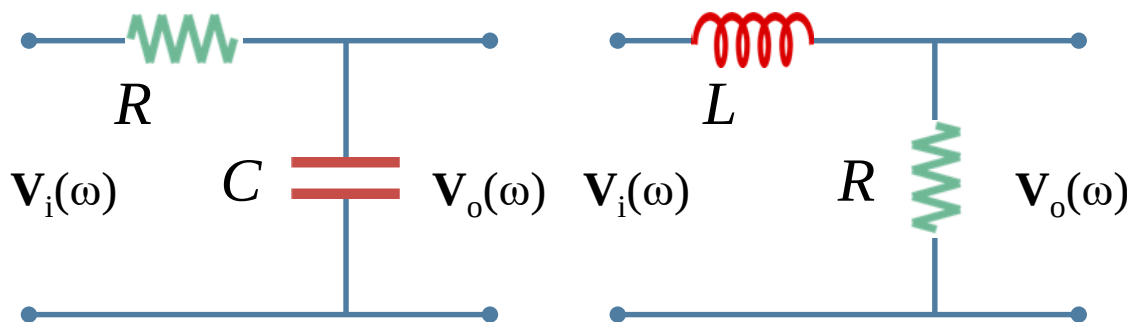
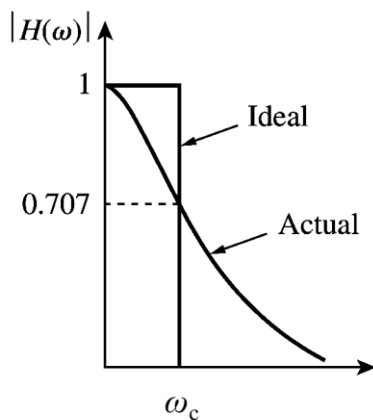
- Passive lowpass and highpass filters

Magnitude response

Topologies

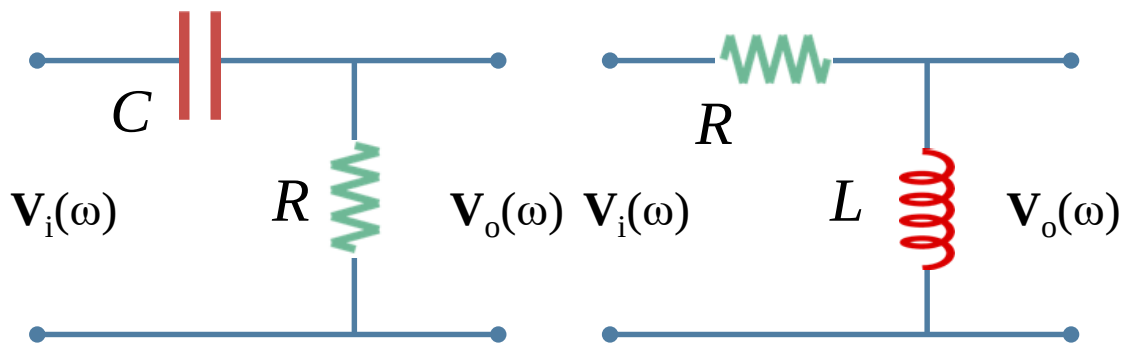
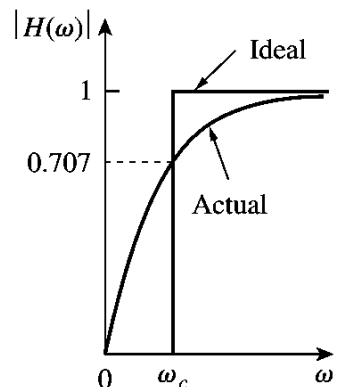
Cutoff frequency

Lowpass filters



$$\omega_c = \frac{1}{RC} \Rightarrow |X_c| = R$$

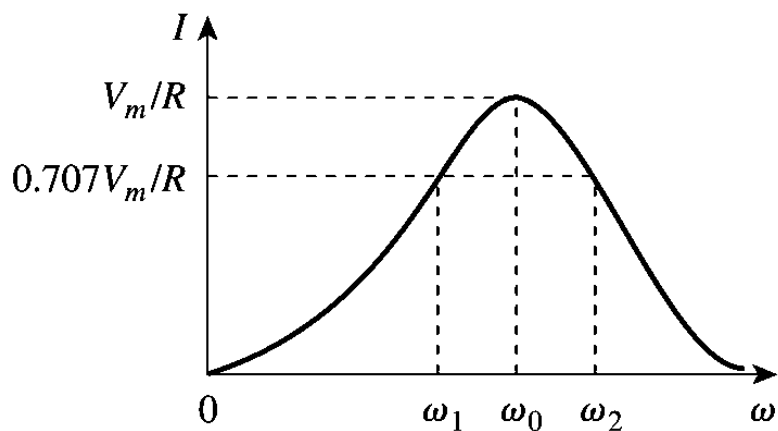
Highpass filters



$$\omega_c = \frac{R}{L} \Rightarrow |X_L| = R$$

Review of Lessons 11(e)

• Resonance (谐振)



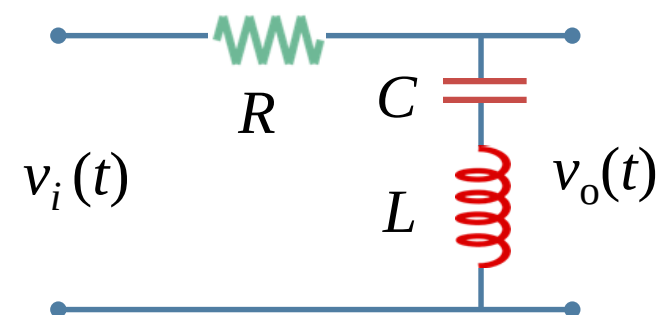
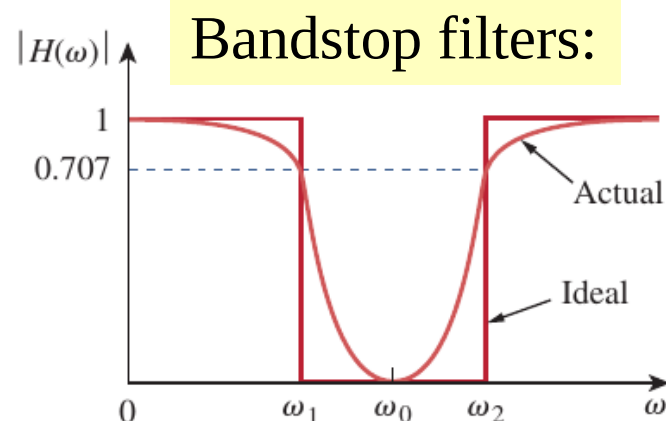
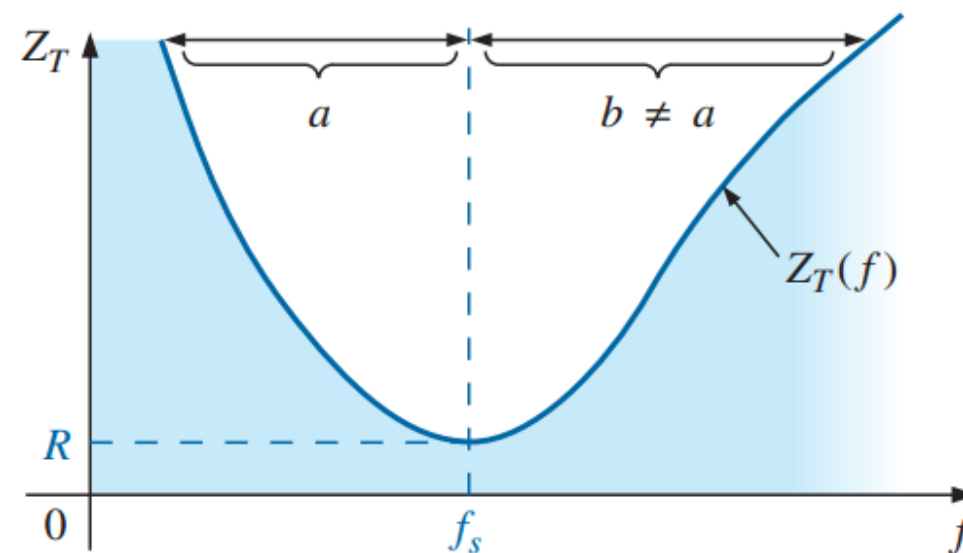
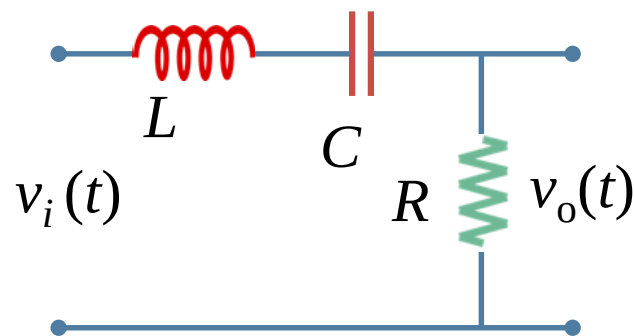
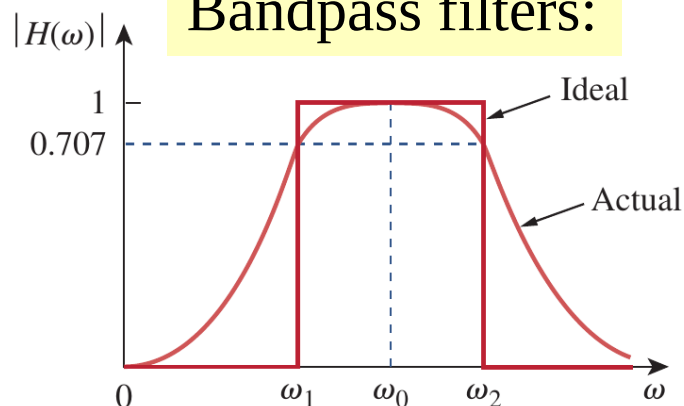
Resonance is a condition in an RLC circuit in which **the capacitive and inductive reactances are equal in magnitude**, thereby resulting in a purely resistive impedance.

Characteristic	Series circuit	Parallel circuit
<u>Resonant frequency, ω_0</u>	$\frac{1}{\sqrt{LC}}$	$\frac{1}{\sqrt{LC}}$
<u>Quality factor, Q</u>	$\frac{\omega_0 L}{R}$ or $\frac{1}{\omega_0 RC}$	$\frac{R}{\omega_0 L}$ or $\omega_0 RC$
<u>Bandwidth, B</u>	$\frac{\omega_0}{Q}$	$\frac{\omega_0}{Q}$
<u>Half-power frequencies, ω_1, ω_2</u>	$\omega_0 \sqrt{1 + \left(\frac{1}{2Q}\right)^2} \pm \frac{\omega_0}{2Q}$	$\omega_0 \sqrt{1 + \left(\frac{1}{2Q}\right)^2} \pm \frac{\omega_0}{2Q}$
For $Q \geq 10$, ω_1, ω_2	$\omega_0 \pm \frac{B}{2}$	$\omega_0 \pm \frac{B}{2}$

Review of Lessons 11(f)

- Passive bandpass and bandstop filters:

Bandpass filters:

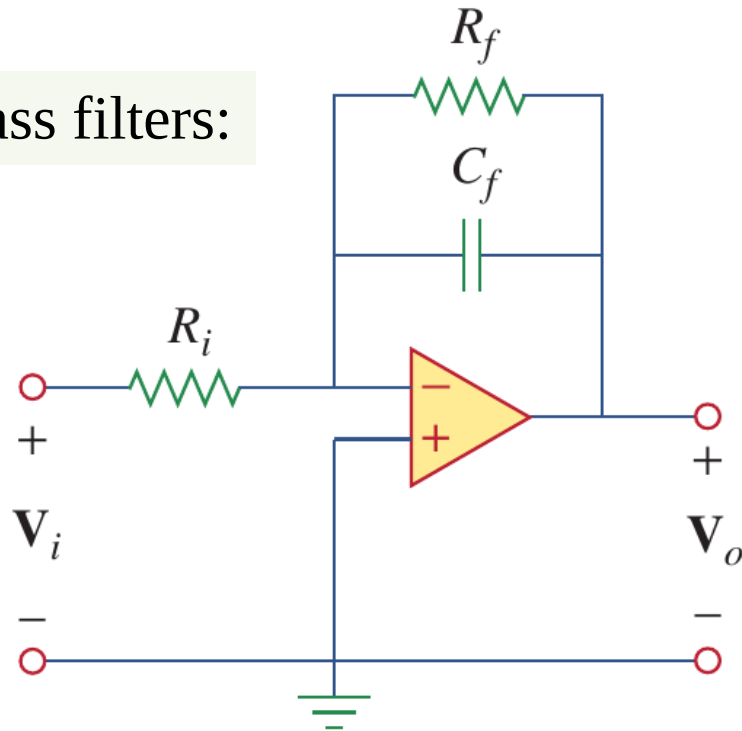


Again, this is all about the impedance.

Review of Lessons 11(g)

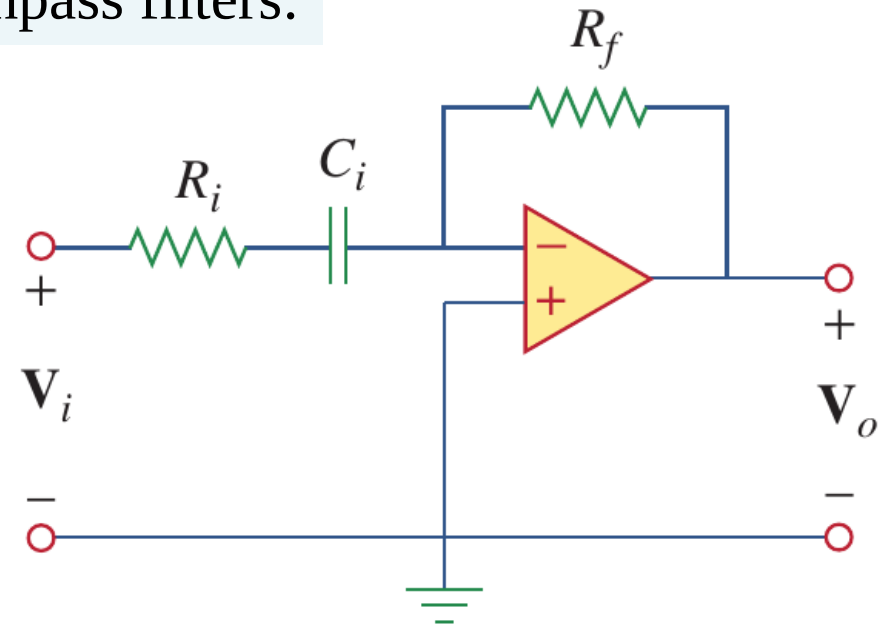
- Active lowpass and highpass filters:

Lowpass filters:



$$\mathbf{H}(\omega) = -\frac{\mathbf{Z}_f}{\mathbf{Z}_i} = -\frac{R_f}{R_i} \left(\frac{1}{1 + j\omega C_f R_f} \right), \omega_c = \frac{1}{C_f R_f}$$

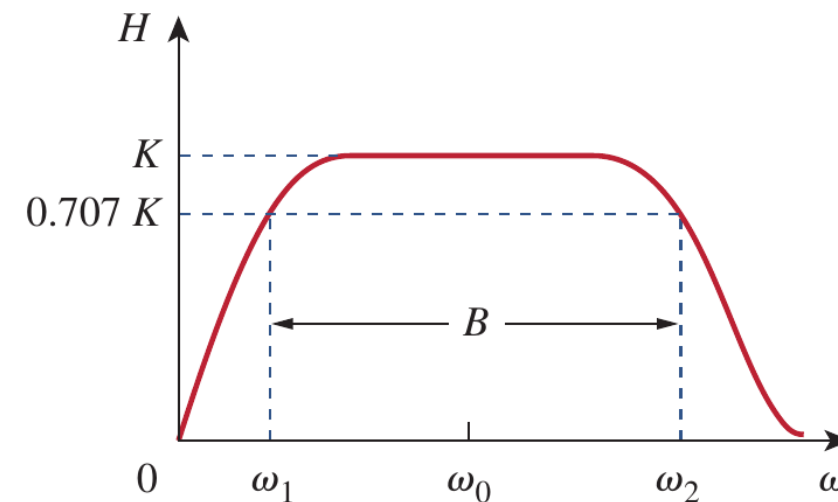
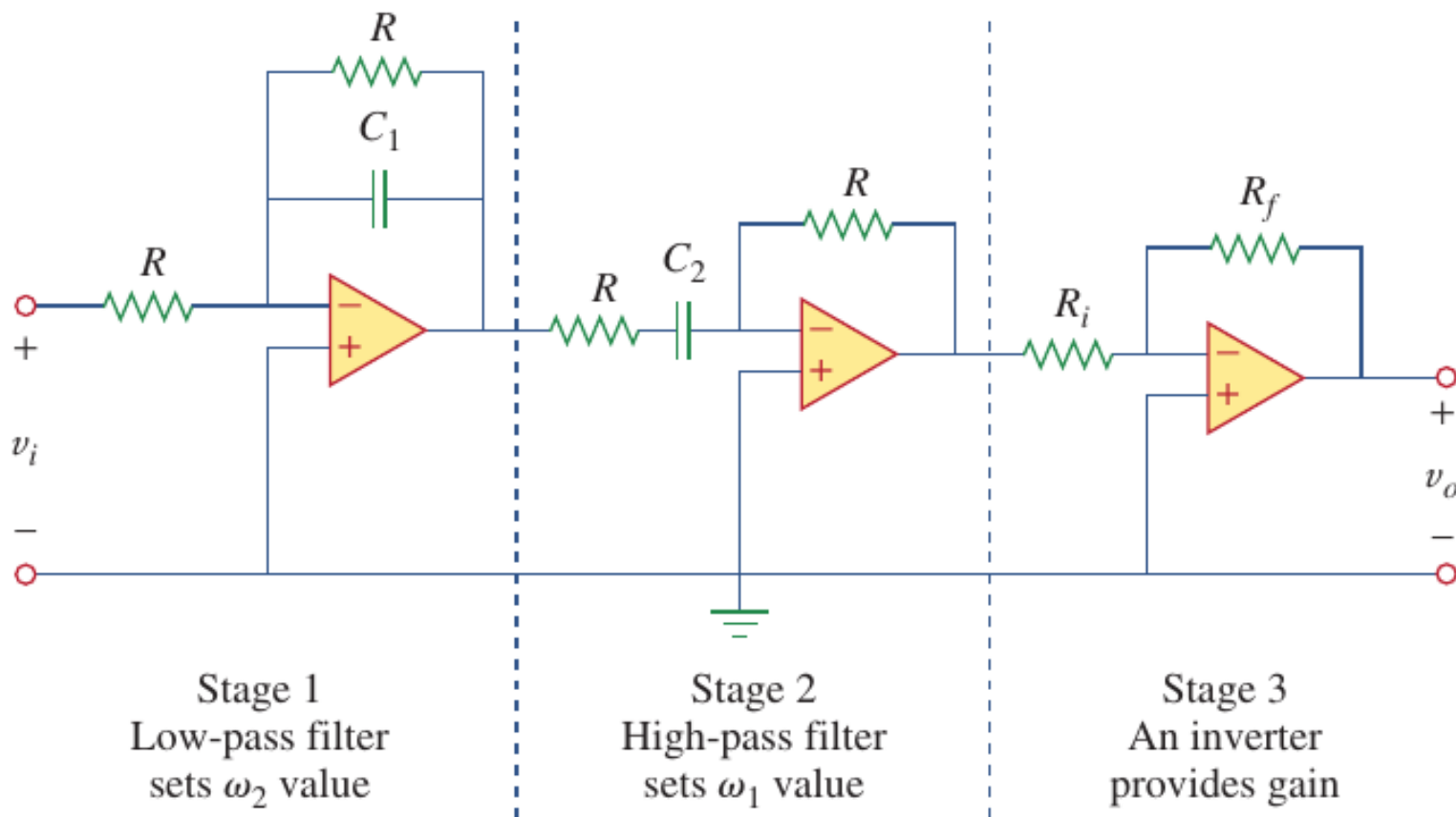
Highpass filters:



$$\mathbf{H}(\omega) = -\frac{\mathbf{Z}_f}{\mathbf{Z}_i} = -\frac{R_f}{R_i} \left(\frac{1}{1 + \frac{1}{j\omega C_i R_i}} \right), \omega_c = \frac{1}{C_i R_i}$$

Review of Lessons 11(h)

- Active bandpass filters:



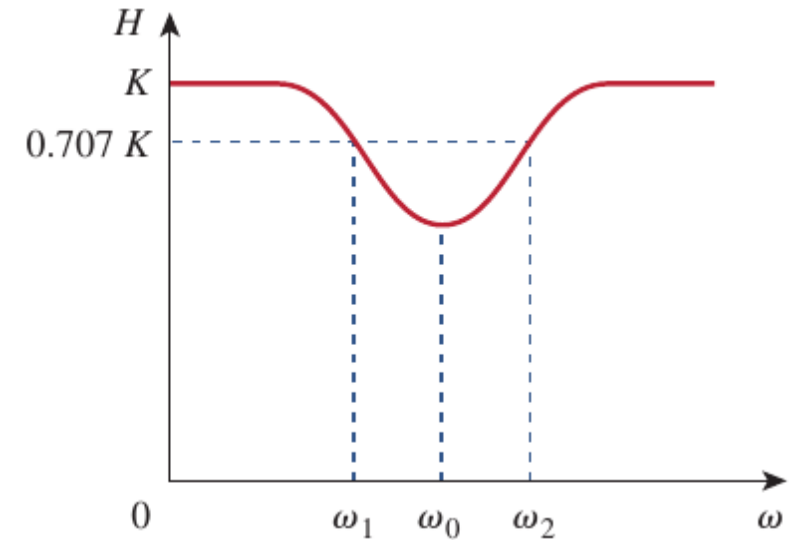
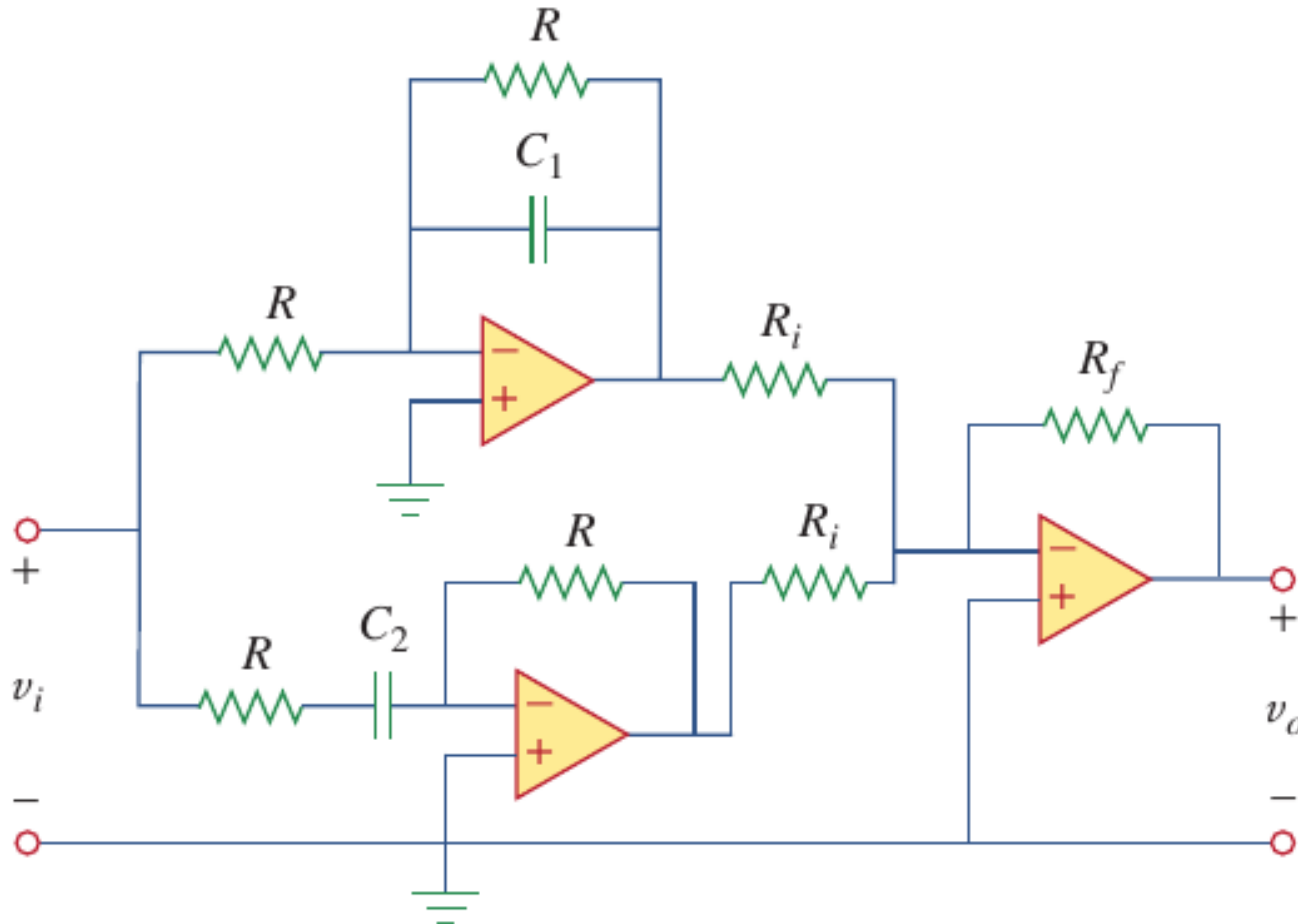
$$\omega_1 = \frac{1}{RC_2} = \frac{1}{R_{high}C_{high}}$$

$$\omega_2 = \frac{1}{RC_1} = \frac{1}{R_{low}C_{low}}$$

$$|\mathbf{H}(\omega)|_{max} = K = \frac{R_f}{R_i} \frac{\omega_2}{\omega_2 + \omega_2}$$

Review of Lessons 11(i)

- Active bandstop filters:



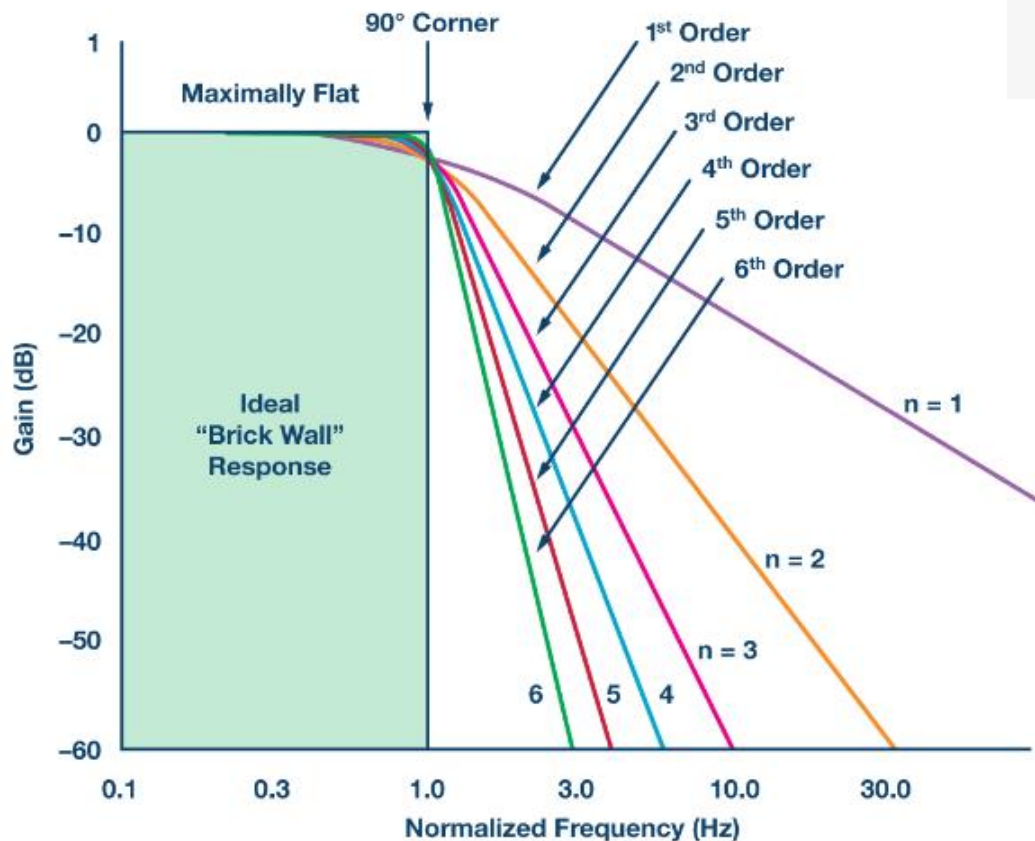
$$\omega_1 = \frac{1}{RC_1} = \frac{1}{R_{low}C_{low}}$$

$$\omega_2 = \frac{1}{RC_2} = \frac{1}{R_{high}C_{high}}$$

$$K = \frac{R_f}{R_i}, \quad |H(\omega)|_{min} = \frac{R_f}{R_i} \frac{\omega_1}{\omega_1 + \omega_2}$$

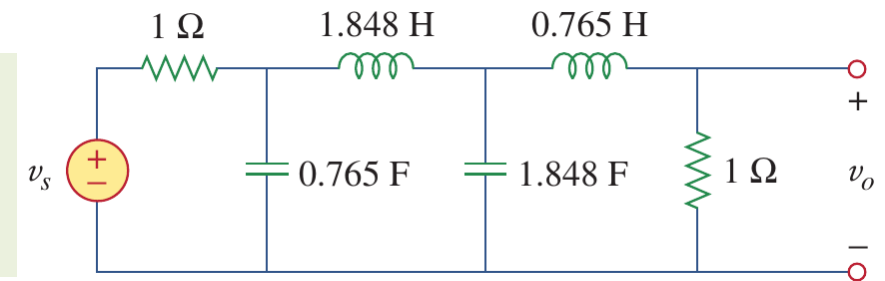
Review of Lessons 11(j)

- High-order filters:

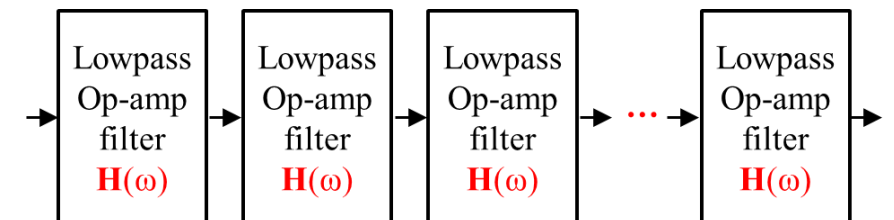


a) Higher-order filters provided greater roll off rates between pass band and stop band, achieve required levels of attenuation or sharpness of cutoff.

Cascaded passive filters (the next slide)



Cascaded active filters



$$\omega_{c,eff} = \omega_c \sqrt[n]{2} - 1$$

Review of Lessons 11(k)

- Scaling (比例变换):

Magnitude scaling 幅值的比例变换

Increasing all Z in the network by a factor (K_m) while keeping the same frequency response.

$$R' = K_m R, \quad L' = K_m L$$
$$C' = \frac{C}{K_m}, \quad \omega' = \omega$$

Frequency scaling 频率的比例变换

Shifting the frequency response of a network while leaving the element impedance the same.

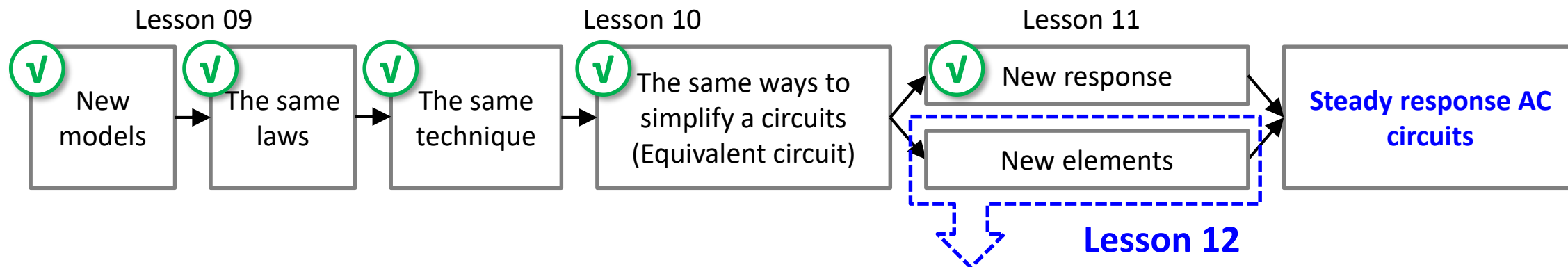
$$R' = R, \quad L' = \frac{L}{K_f}$$
$$C' = \frac{C}{K_f}, \quad \omega' = K_f \omega$$

Magnitude & frequency scaling 幅值与频率的比例变换

$$R' = K_m R, \quad L' = \frac{K_m}{K_f} L$$
$$C' = \frac{1}{K_m K_f} C, \quad \omega' = K_f \omega$$

Overview of Lessons 12(a)

- EE104 road map



Transformers

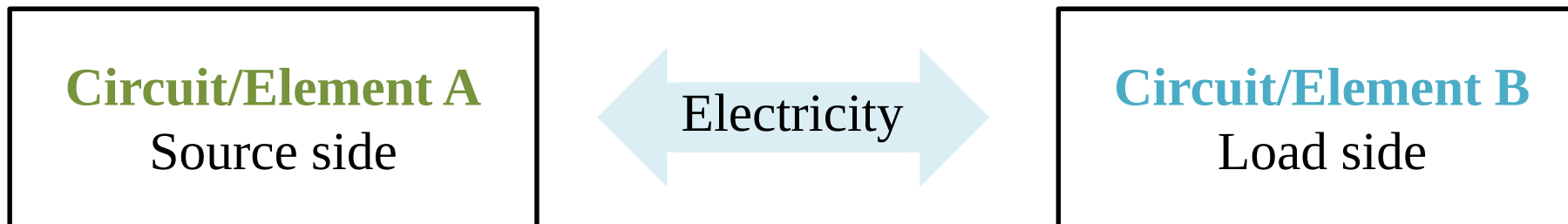
- Coupling
- Mutual inductance
- Linear transformers
- Ideal transformers

Lesson 12 Magnetically coupled circuits

12.1	Coupled circuits	Slides 15-21
12.2-12.7	Mutual inductance	Slides 23-60
12.8	Linear transformers	Slides 62-69
12.9	Ideal transformers	Slides 71-83
12.10	Summary and assignment	Slides 85-86

12.1 Coupled circuits (a)

- Coupled circuits



- a) How to pass power/signals from the power network to the load network?
- b) **Coupling** is the act or process of linking two or more circuits so that electricity can be transferred between them.
- c) 耦合是指两个或两个以上的电路元件或电路网络的输入与输出之间存在紧密配合与相互影响，并通过相互作用从一侧向另一侧传输能量的现象。耦合电路就是指参与耦合过程的电路。
- d) In some cases, 将前级电路的能量递至后级电路亦称为耦合。

12.1 Coupled circuits (b)

- An example:

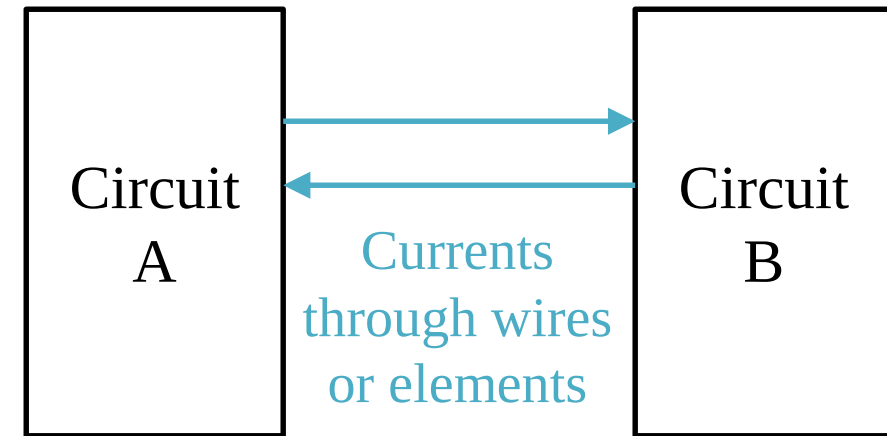


To charge your phone, the source side and the load side must be coupled, such that:

- a) the power can be transferred from the source side to the load side.
- b) the signal can be transferred from the load to the source side to modulate or terminate the coupling.

12.1 Coupled circuits (c)

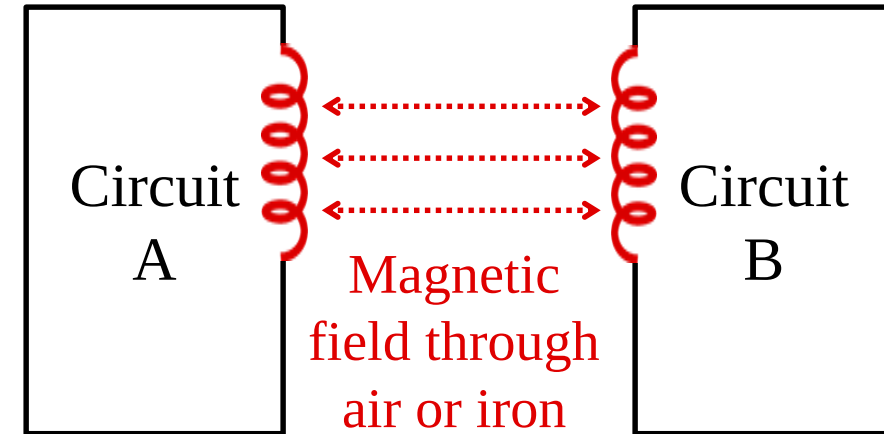
- electrical coupling (电耦合) or conductive coupling (传导耦合):



- All circuits we have learned so far can all be regarded as **conductively coupled**, i.e., one branch in the circuit affects other branches through current conduction.
- In such case, the electricity (power or signals) is transferred by the collective movement of charges, or in simple words we can say **the electricity is transferred directly**.

12.1 Coupled circuits (d)

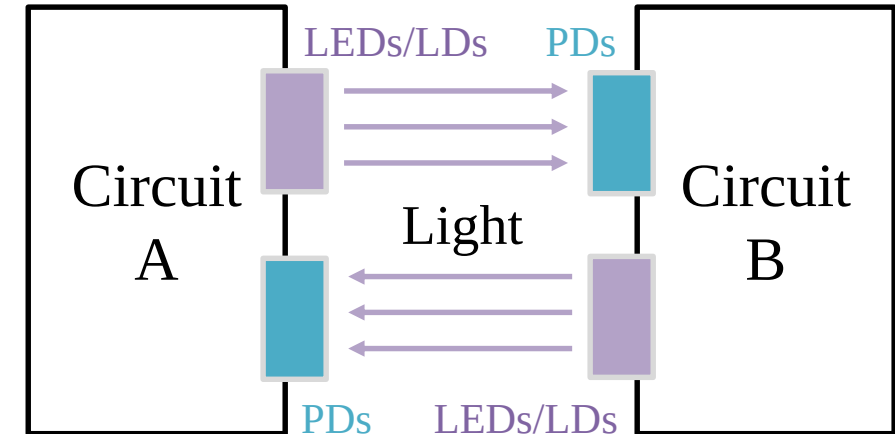
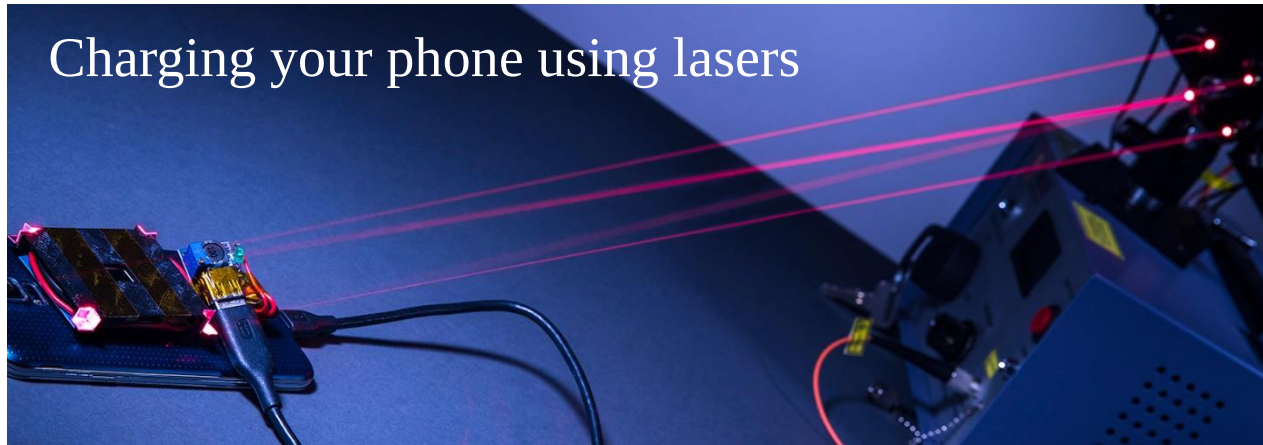
- Magnetic coupling (磁耦合):



- a) The electricity can be also transferred using many other ways. For example, one can pass power or signals using the magnetic field generated by one network to affect the other, based on inductors.
- b) **Magnetic coupling is an indirect approach.** The electricity in one network is first transferred into magnetic field and then back into electricity in the other network.
- c) An important advantage of such coupling technique is that the power or the signal can be passed without the networks touching each other, while the efficiency is generally lower.

12.1 Coupled circuits (e)

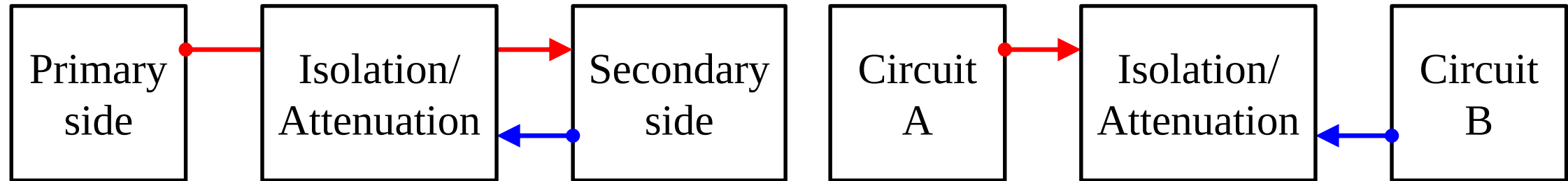
- Photonic coupling (光耦合):



- a) Similar to using magnetic fields to couple two circuits, light is also an approach to couple two circuits, i.e. sending power or signals.
- b) A good example is silicon photonics for data transfer.
- c) <https://www.intel.com/content/www/us/en/architecture-and-technology/silicon-photonics/silicon-photonics-overview.html>

12.1 Coupled circuits (f)

- Coupling isn't always good - interference/crosstalk:



- In electronics, [crosstalk](#) is a phenomenon by which a signal transmitted on one circuit creates an undesired effect in another circuit or channel. Crosstalk is usually caused by undesired capacitive, inductive, or conductive coupling from one circuit or channel to another.
- In many applications we need unidirectional or close-to-zero transfer of electricity between two or more circuits, and the undesirable portion of the coupling (often called as interference or crosstalk) is harmful to the functionality of the system.
- Isolation is needed to decrease the crosstalk ([Recall: voltage followers and relays](#)).

12.1 Coupled circuits (g)

- An interesting example: how to stop the annoying drones?



In July 2014, a drone narrowly missed colliding with an Airbus A320 in London's Heathrow airport, which is rated as a "serious risk of collision," the top rating CAA can give.

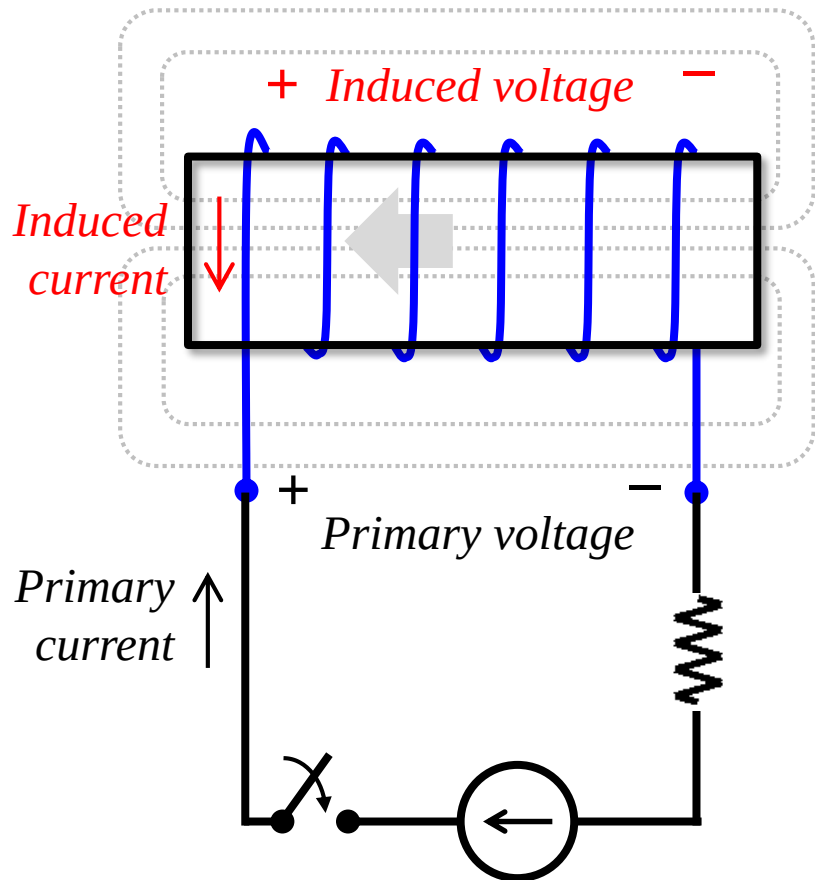
In Jan. 2015, a drone carrying more than six pounds of methamphetamine crashed in Mexico near the US border. According to the DEA, drones are becoming a common means to transport drugs over the border.

Lesson 12 Magnetically coupled circuits

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12.2-12.7	Mutual inductance	Slides 23-60
12.8	Linear transformers	Slides 62-69
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12.2 Mutual inductance (a)

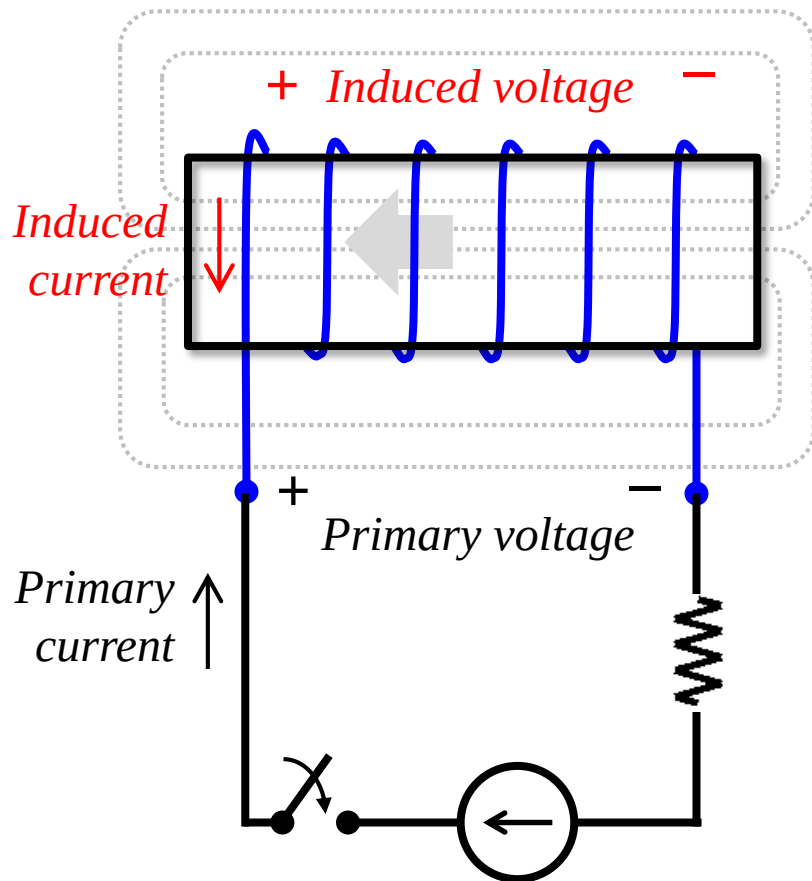
- Self inductance (自感)



- Magnetic fields are built as a current passes.
(Right hand grip rule)
- An increasing current results in a larger magnetic flux (ϕ), inducing an electromagnetic force (induced EMF).
(Faraday law of electromagnetic induction)
- The induced EMF creates a flux to oppose the increasing flux.
(Lenz's Law)
- The direction of the EMF will be opposite to the source EMF
- The inductor absorbs the electricity and turns it into magnetic fields.

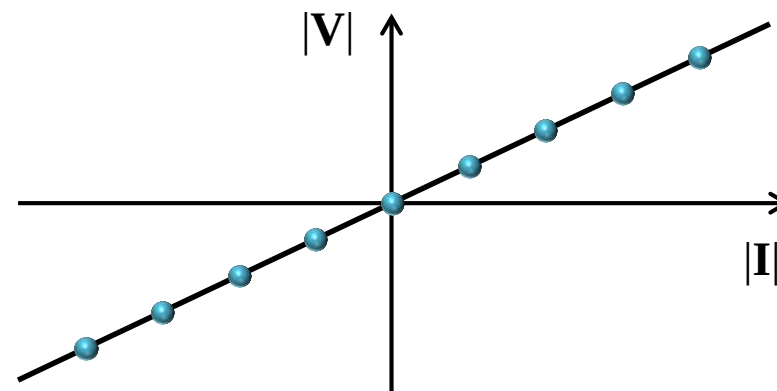
12.2 Mutual inductance (b)

- Self inductance (自感)



$$v = \frac{Nd\phi}{dt} = \frac{Nd\phi}{di} \frac{di}{dt} = L \frac{di}{dt}$$

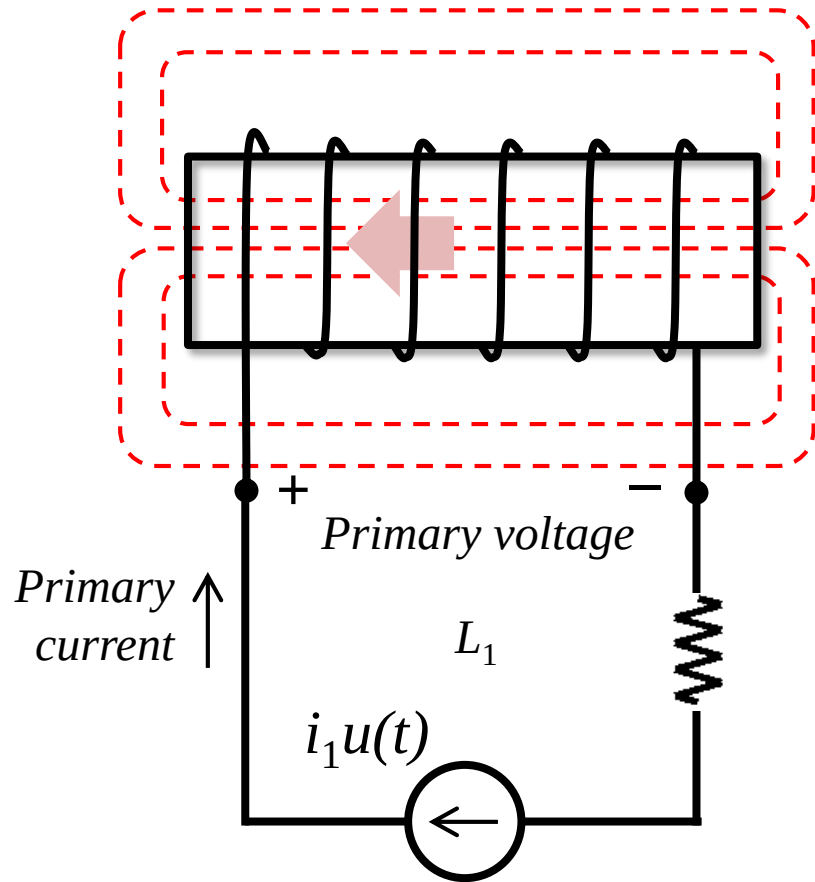
If you measure $|\mathbf{V}|$ and plot it as a function of the $|\mathbf{I}|$, at a fixed frequency ω , then you have:



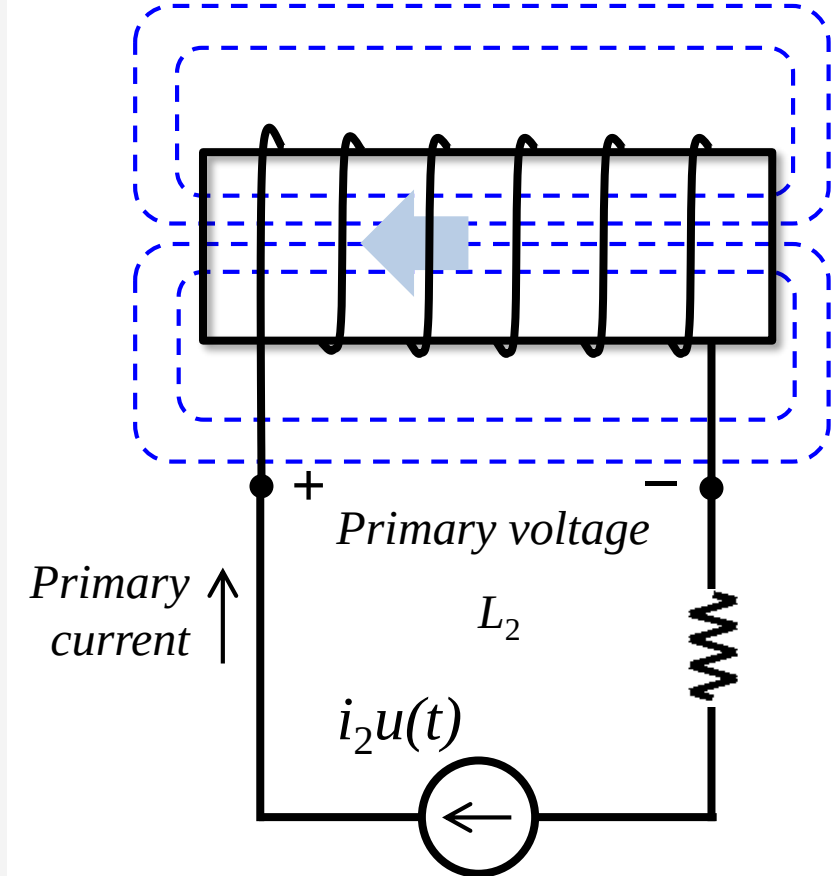
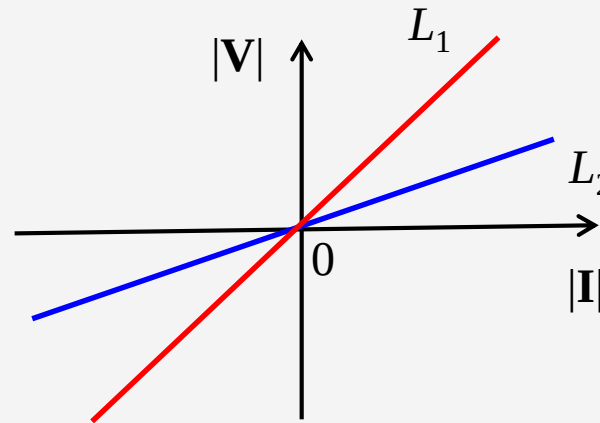
$$|\mathbf{V}| = \omega |\mathbf{I}| L$$

12.2 Mutual inductance (c)

- Mutual inductance (互感)-01

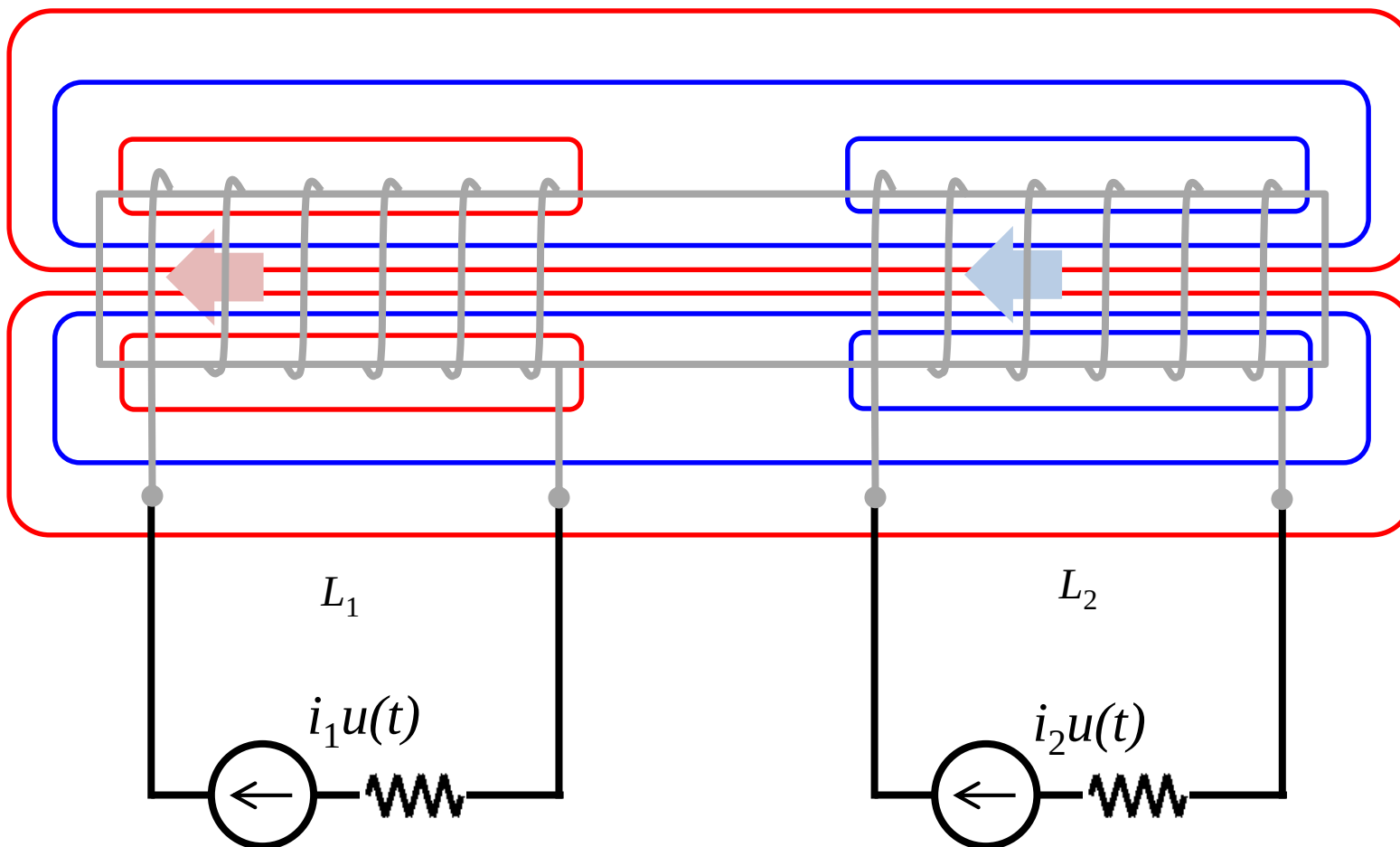


If there are two inductors, L_1 and L_2 , extremely apart away from each other. The same results of $|\mathbf{V}| = \omega L |\mathbf{I}|$ can be seen as in discrete device measurements.



12.2 Mutual inductance (d)

- Mutual inductance (互感)-02



Flux generated by coils 1 and 2:

$$\phi_1 = \phi_{11} + \phi_{12} \text{ \& } \phi_2 = \phi_{22} + \phi_{21}$$

Then v across L_1 is:

$$\begin{aligned} v_1 &= N_1 \frac{d(\phi_1 + \phi_{21})}{dt} \\ &= N_1 \frac{d\phi_1}{dt} + N_1 \frac{d\phi_{21}}{dt} \\ &= N_1 \frac{d\phi_1}{di_1} \frac{di_1}{dt} + N_1 \frac{d\phi_{21}}{dt} \\ &= N_1 \frac{d\phi_1}{di_1} \frac{di_1}{dt} + N_1 \frac{d\phi_{21}}{di_2} \frac{di_2}{dt} \\ &= L_1 \frac{di_1}{dt} + M_{21} \frac{di_2}{dt} \end{aligned}$$

12.2 Mutual inductance (e)

- Mutual inductance (互感)-03

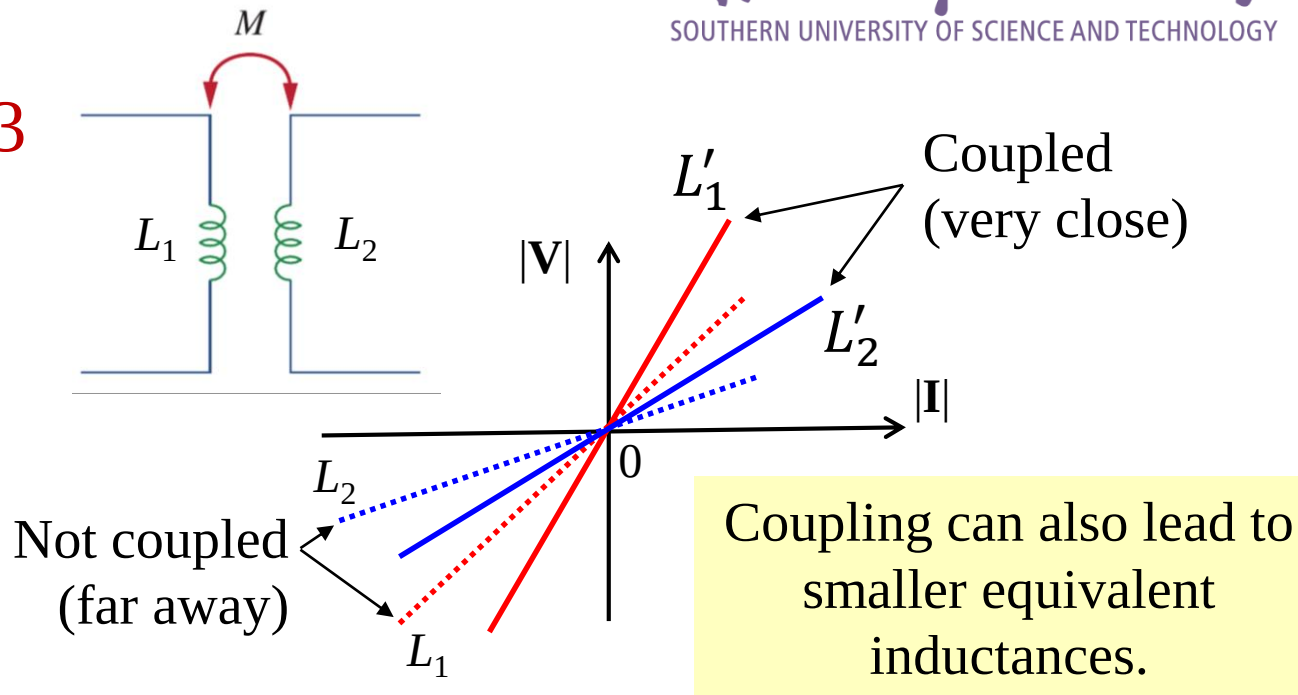
Similarly we can find v_2 :

$$\begin{cases} v_1 = v_{11} + v_{21} = L_1 \frac{di_1}{dt} + M_{21} \frac{di_2}{dt} \\ v_2 = v_{22} + v_{12} = L_2 \frac{di_2}{dt} + M_{12} \frac{di_1}{dt} \end{cases}$$

a) $M_{12} = M_{21} = M$

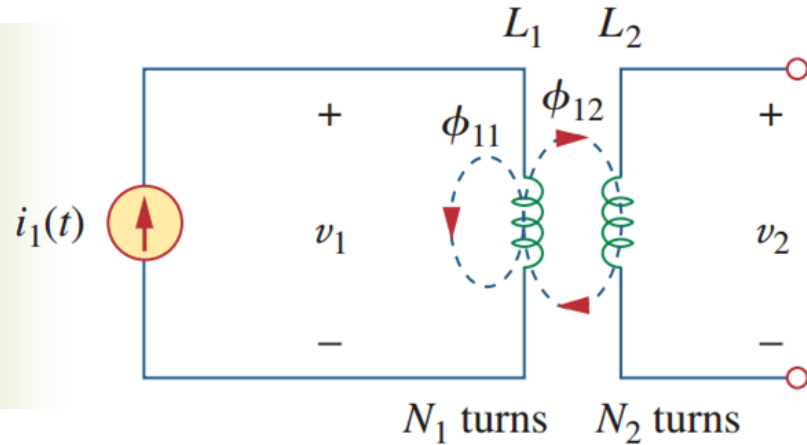
b) M is referred as the **mutual inductance** between the two coils, measured in henrys (H), showing the ability of one inductor to induce a voltage across a neighboring inductor.

c) Mutual magnetic coupling only exists when the inductors or coils are in close proximity, and the circuits are driven by time-varying sources. (Recall that inductors act like short circuits to DC).



12.2 Mutual inductance (f)

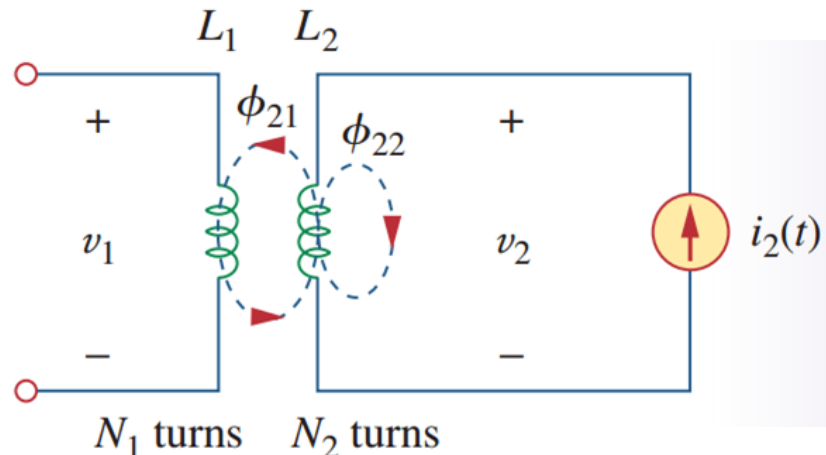
• Mutual inductance (M)-04



Coil 1 $\rightarrow$ Coil 2

$$v_{11} = L_1(di_1/dt)$$

$$v_{12} = M_{12}(di_1/dt)$$



Coil 2 $\rightarrow$ Coil 1

$$v_{22} = L_2(di_2/dt)$$

$$v_{21} = M_{21}(di_2/dt)$$

The same results can be obtained using the superposition theorem:

$$\begin{cases} v_1 = v_{11} + v_{21} = L_1 \frac{di_1}{dt} + M_{21} \frac{di_2}{dt} \\ v_2 = v_{22} + v_{12} = L_2 \frac{di_2}{dt} + M_{12} \frac{di_1}{dt} \end{cases}$$

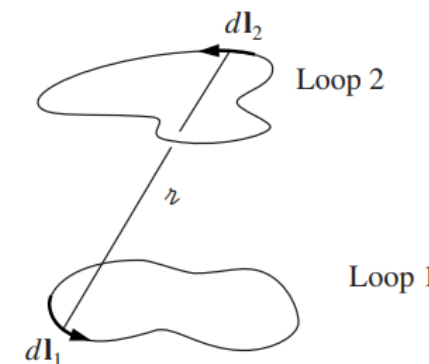
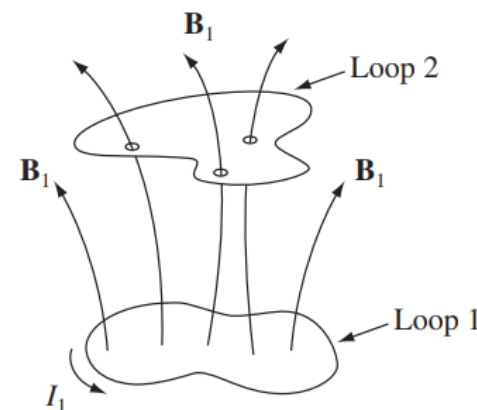
For AC:

$$\begin{cases} \mathbf{V}_1 = \mathbf{V}_{11} + \mathbf{V}_{21} = j\omega L_1 \mathbf{I}_1 + j\omega M_{21} \mathbf{I}_2 \\ \mathbf{V}_2 = \mathbf{V}_{22} + \mathbf{V}_{12} = j\omega L_2 \mathbf{I}_2 + j\omega M_{12} \mathbf{I}_1 \end{cases}$$

12.2 Mutual inductance (g)

- $M_{12} = M_{21} = M$

- a) Mutual inductance depends only on the geometrical factor of the two coils such as the number of turns, radius of two coils and on the properties of a material medium such as magnetic permeability of the medium surrounding the coils.
- b) Textbook p416-p417
- c) If interested: *Introduction to electrodynamics*, Pages 321 to 323, David J. Griffiths using Biot-Savart law, magnetic potential, Stokes' theorem.
- d) If interested: Nillson's book: p156-p158.
- e) If interested: 对易定理, 《电路第5版》, 邱关源, p99 - p102.



$$\mathbf{B}_1 = \frac{\mu_0}{4\pi} I_1 \oint \frac{d\mathbf{l}_1 \times \hat{\mathbf{r}}}{r^2}, \quad \Phi_2 = \int \mathbf{B}_1 \cdot d\mathbf{a}_2.$$

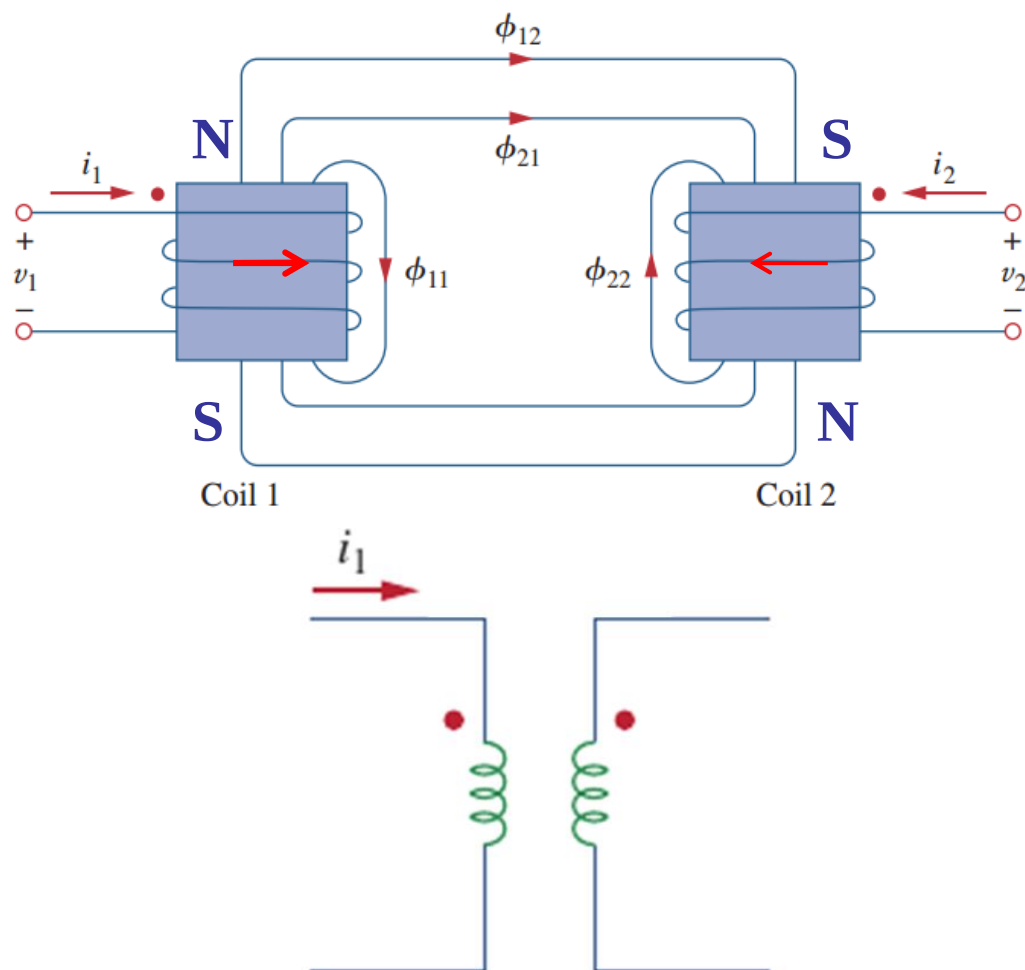
$$\Phi_2 = \int \mathbf{B}_1 \cdot d\mathbf{a}_2 = \int (\nabla \times \mathbf{A}_1) \cdot d\mathbf{a}_2 = \oint \mathbf{A}_1 \cdot d\mathbf{l}_2$$

$$\mathbf{A}_1 = \frac{\mu_0 I_1}{4\pi} \oint \frac{d\mathbf{l}_1}{r},$$

$$\Phi_2 = \frac{\mu_0 I_1}{4\pi} \oint \left(\oint \frac{d\mathbf{l}_1}{r} \right) \cdot d\mathbf{l}_2 \quad M_{21} = \frac{\mu_0}{4\pi} \oint \oint \frac{d\mathbf{l}_1 \cdot d\mathbf{l}_2}{r}$$

12.3 Dot convention (a)

- Dot convention (同名端规则):



- a) If a current enters the dotted terminal of one coil, the reference polarity of the mutual voltage in the second coil is positive at the dotted terminal of the second coil.
- b) If a current leaves the dotted terminal of one coil, the reference polarity of the mutual voltage in the second coil is negative at the dotted terminal of the second coil.
- c) 当两个电流分别从两个线圈的对应端子同时流入或流出，若所产生的磁通相互加强时，则这两个对应端子称为互感线圈的同名端。

12.3 Dot convention (b)

- Example 12.01

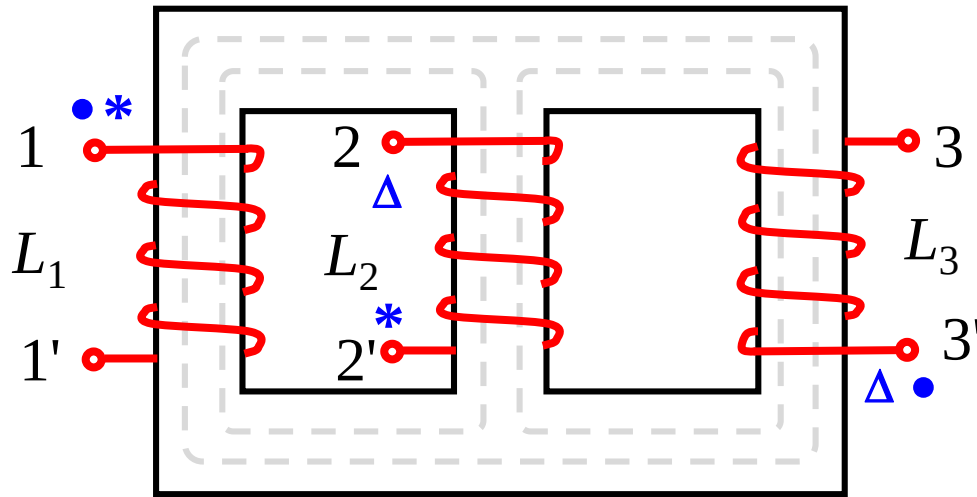
Example 12.01: determine the dot markings between L_1 and L_2 , between L_2 and L_3 , as well as between L_1 and L_3 .

Solution:

Terminals 1 and 3' for L_1 and L_3

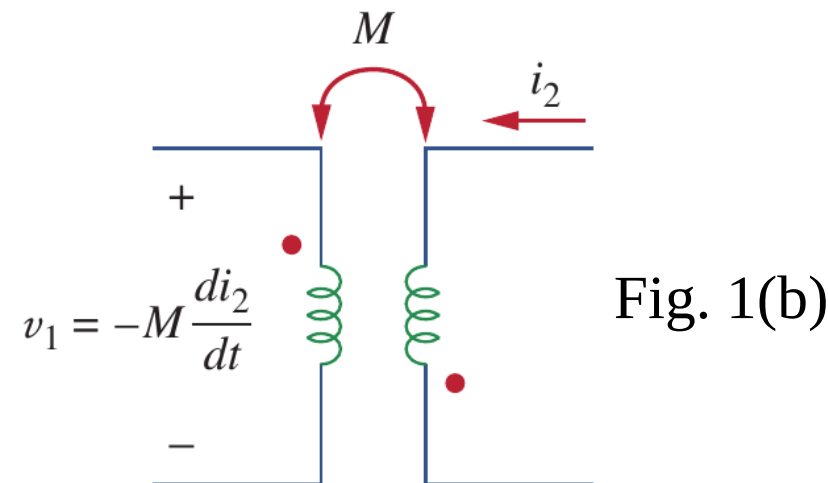
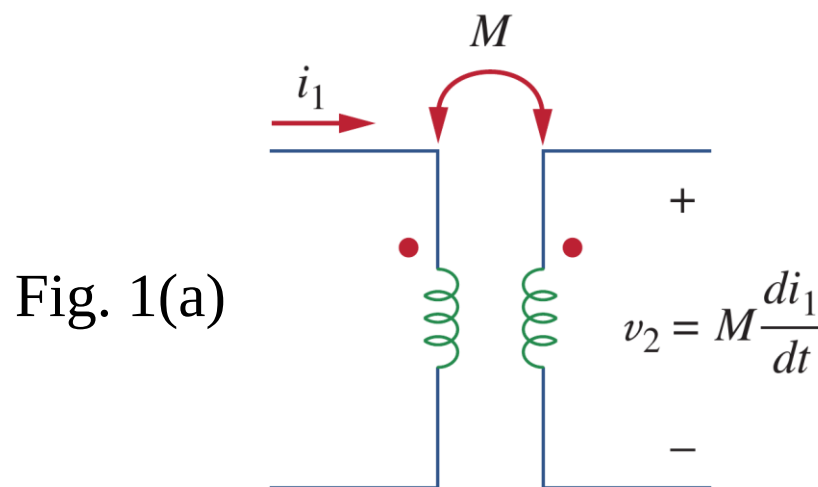
Terminals 1 and 2' for L_1 and L_2

Terminals 2 and 3' for L_2 and L_3



12.3 Dot convention (c)

- Don't forget the reference direction:

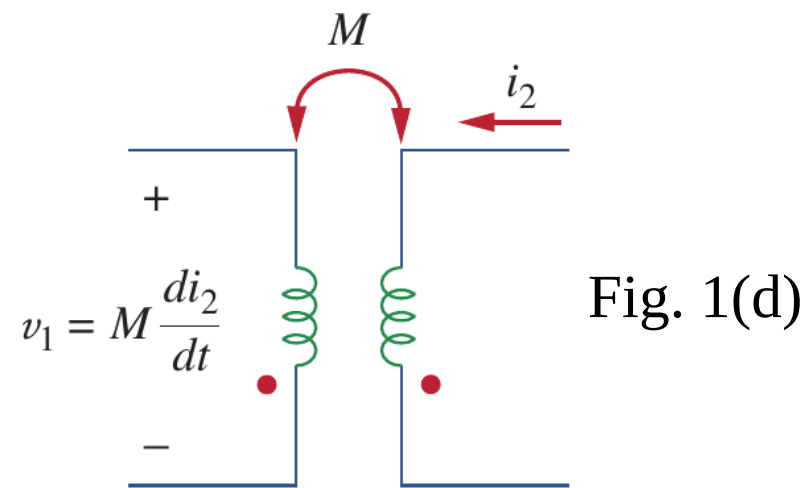
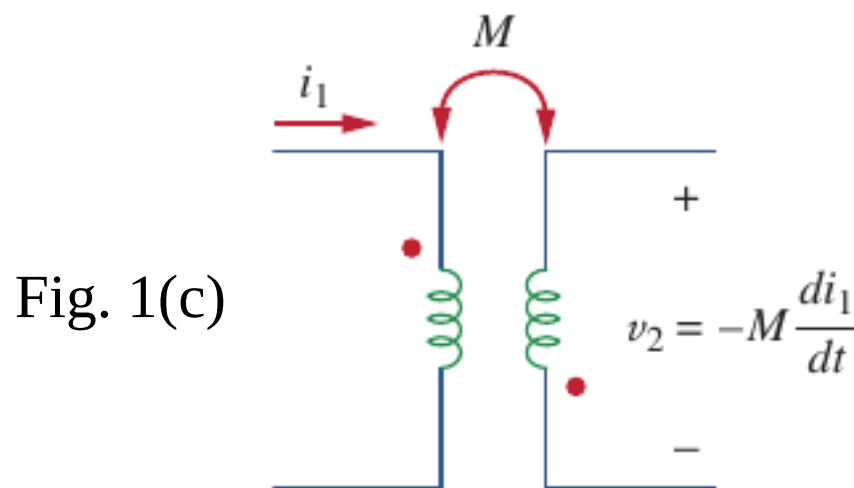


The reference polarity of the mutual voltage depends on the reference direction and the dots on the coupled coils.

- If the dot in the secondary side is at the positive-reference terminal, and the current enters the dotted side in the primary side, then the induced voltage is positive, as in Fig. 1(a).
- If the dot in the secondary side is at the positive-reference terminal, and the current leaves the dotted side in the primary side, then the induced voltage is negative, as in Fig. 1(b).

12.3 Dot convention (d)

- Don't forget the reference direction:

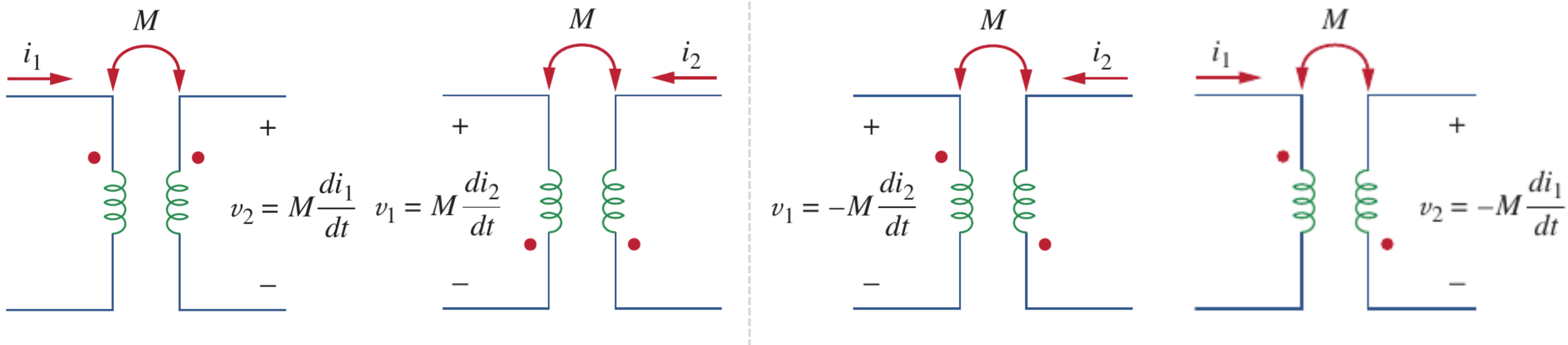


The reference polarity of the mutual voltage depends on the reference direction and the dots on the coupled coils.

- c) If the dot in the secondary side is at the negative-reference terminal, and the current enters the dotted side in the primary side, then the induced voltage is negative, as in Fig. 1(c).
- d) If the dot in the secondary side is at the negative-reference terminal, and the current leaves the dotted side in the primary side, then the induced voltage is positive, as in Fig. 1(d).

12.3 Dot convention (e)

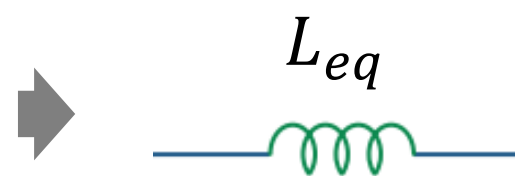
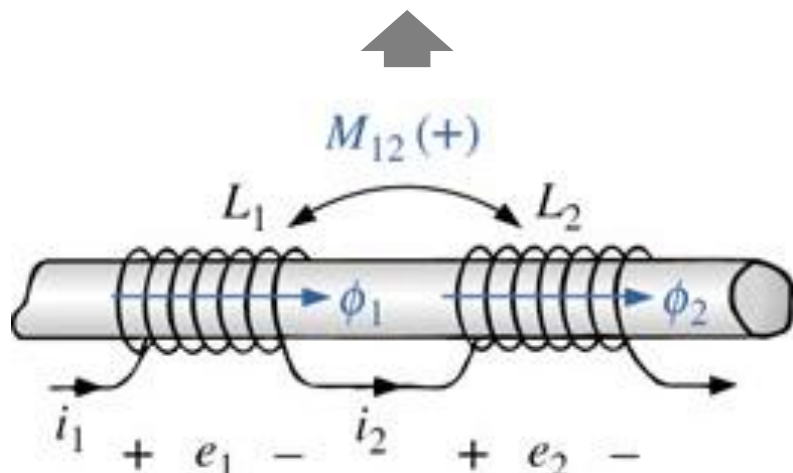
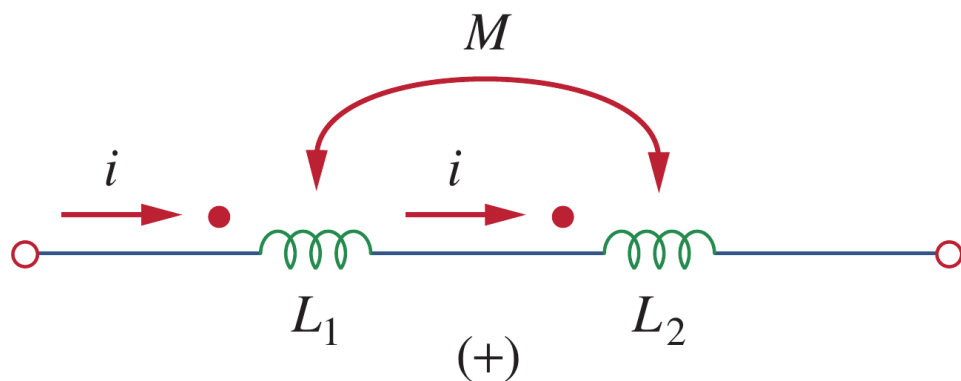
- Dot convention:



- KEEP IN MIND that both the dot and the reference direction are needed to determine the polarity of the induced voltage.
- You can use this to find out the dot marking. Think about how.

12.4 Equivalents of coupled inductors (a)

- Series-aiding connection (同向串联/顺接):



Equivalent circuits

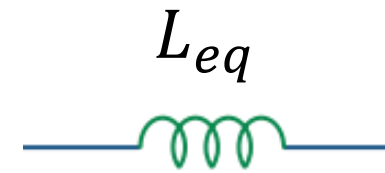
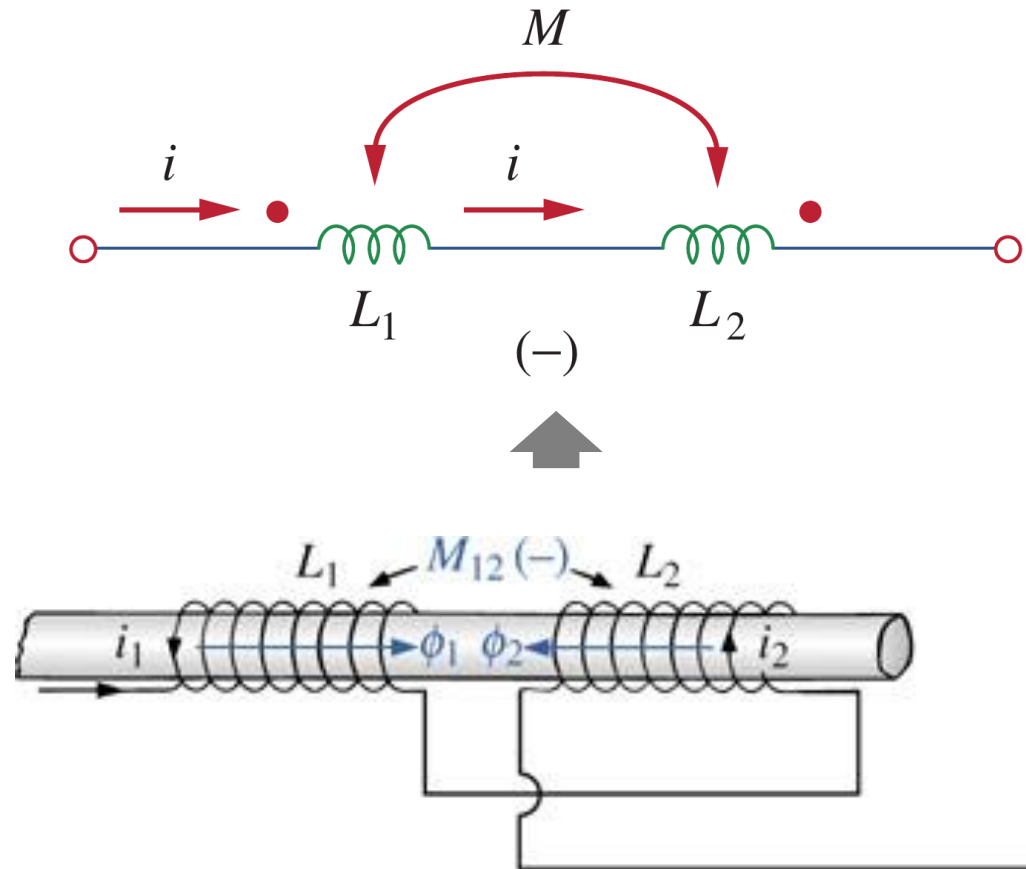
去耦等效电路

$$\begin{aligned} v &= v_1 + v_2 \\ &= (v_{11} + v_{21}) + (v_{22} + v_{12}) \\ &= L_1 \frac{di}{dt} + M_{21} \frac{di}{dt} + L_2 \frac{di}{dt} + M_{12} \frac{di}{dt} \\ &= (L_1 + L_2 + 2M) \frac{di}{dt} \end{aligned}$$

$$\Rightarrow L_{eq} = L_1 + L_2 + 2M$$

12.4 Equivalents of coupled inductors (b)

- Series-opposing connection (反向串联/反接):



Equivalent circuits

去耦等效电路

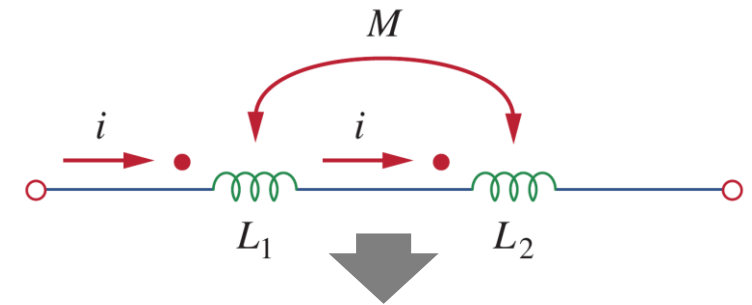
$$\begin{aligned} v &= v_1 + v_2 \\ &= (v_{11} - v_{21}) + (v_{22} - v_{12}) \\ &= L_1 \frac{di}{dt} - M_{21} \frac{di}{dt} + L_{22} \frac{di}{dt} - M_{12} \frac{di}{dt} \\ &= (L_1 + L_2 - 2M) \frac{di}{dt} \\ \Rightarrow L_{eq} &= L_1 + L_2 - 2M \end{aligned}$$

12.4 Equivalents of coupled inductors (c)

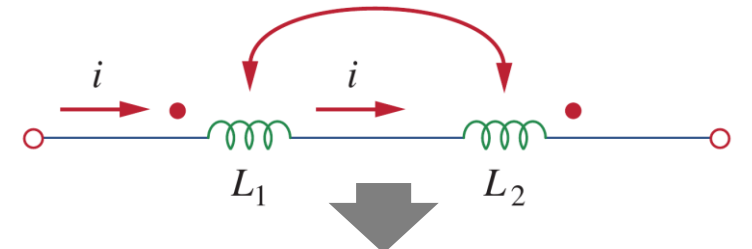
- How to measure the mutual inductance M :

- Based on series-aiding and series-opposing connections, we can measure the mutual inductance between two inductors, assuming the dotted terminals are known.
- Connect the two inductors in series-aiding connection, and measure the equivalent inductance, $L = L_1 + L_2 + 2M$
- Connect the two inductors in series-opposing connection, and measure the equivalent inductance, $L' = L_1 + L_2 - 2M$
- Then we have the mutual inductance $M = (L - L')/4$.

Step 1: $L = L_1 + L_2 + 2M$



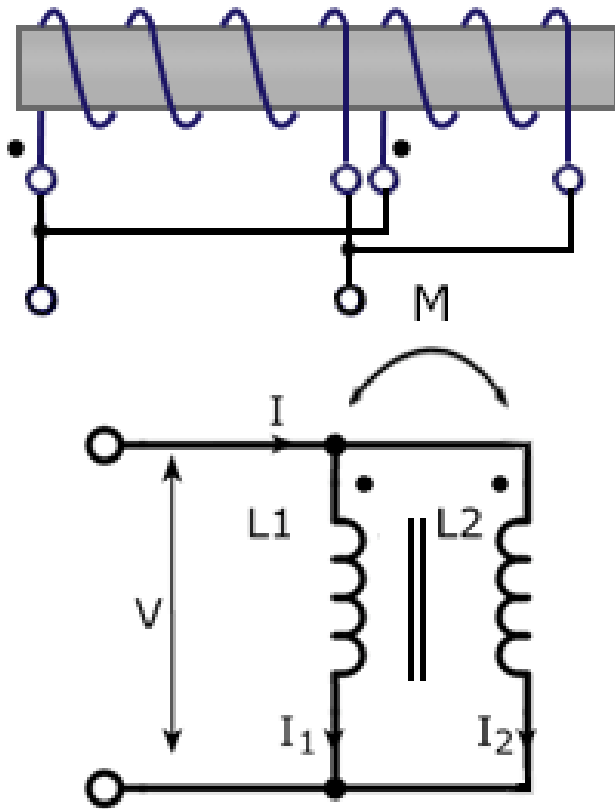
Step 2: $L' = L_1 + L_2 - 2M$



Step 3: $M = (L - L')/4$

12.4 Equivalents of coupled inductors (d)

- Parallel-opposing connection (同侧并联):



$$v = v_1 = v_2$$

$$\begin{cases} L_1 \frac{di_1}{dt} + M_{21} \frac{di_2}{dt} = L_2 \frac{di_2}{dt} + M_{12} \frac{di_1}{dt} \\ i_1 + i_2 = i \end{cases}$$

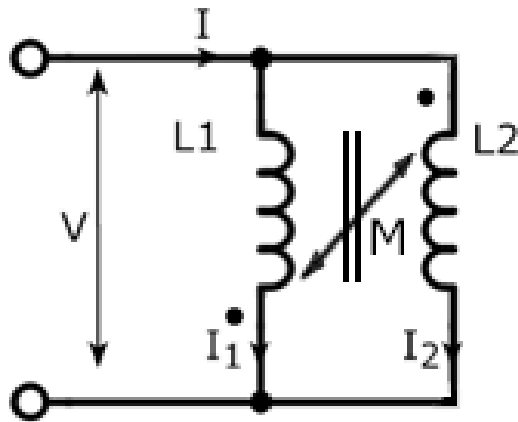
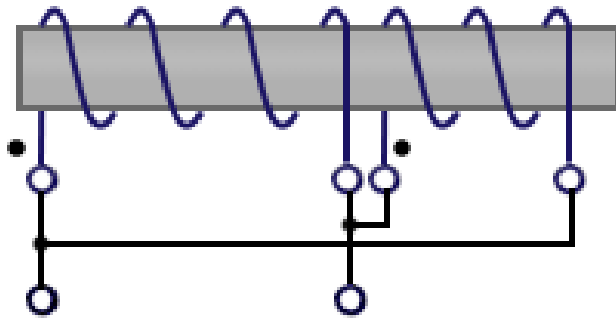
$$\Rightarrow i = \frac{(L_1 L_2 - M^2)}{L_1 + L_2 - 2M} \frac{di}{dt} \Rightarrow L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M} \geq 0$$

$$\text{If } k = 1, L_1 L_2 = M^2 \Rightarrow \begin{cases} L_{eq} = 0 \text{ when } L_1 \neq L_2 \\ L_{eq} = L \text{ when } L_1 = L_2 = L \end{cases}$$

*The k will be introduced later, $M = k\sqrt{L_1 L_2}$

12.4 Equivalents of coupled inductors (e)

- Parallel-opposing connection (异侧并联):



$$v = v_1 = v_2$$

$$\begin{cases} L_1 \frac{di_1}{dt} - M_{21} \frac{di_2}{dt} = L_2 \frac{di_2}{dt} - M_{12} \frac{di_1}{dt} \\ i_1 + i_2 = i \end{cases}$$

$$\Rightarrow i = \frac{(L_1 L_2 - M^2)}{L_1 + L_2 + 2M} \frac{di}{dt} \Rightarrow L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 + 2M} \geq 0$$

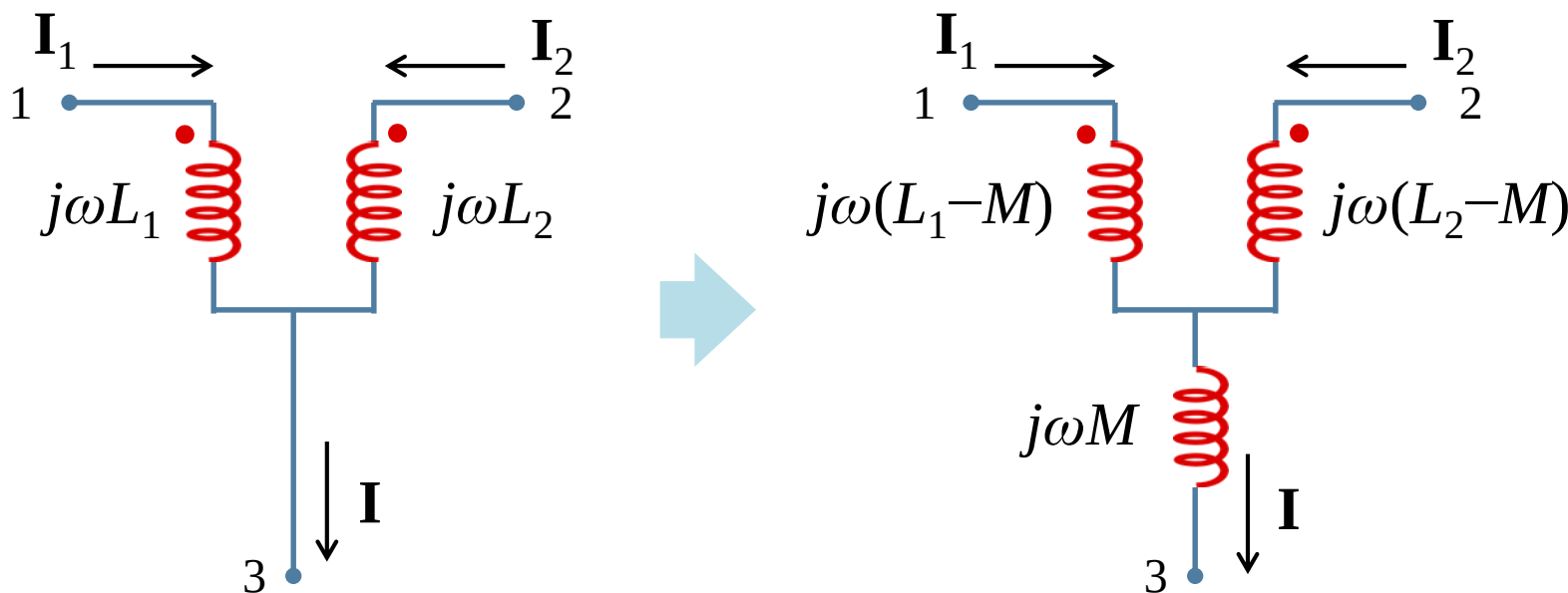
$$\text{If } k = 1 \Rightarrow L_1 L_2 = M^2 \Rightarrow L_{eq} = 0$$

$$\text{If } k = 0 \text{ \& } L_1 = L_2 = L \Rightarrow L_{eq} = L/2$$

*The k will be introduced later, $M = k\sqrt{L_1 L_2}$

12.4 Equivalents of coupled inductors (f)

- Parallel-aiding connection (同侧并联):



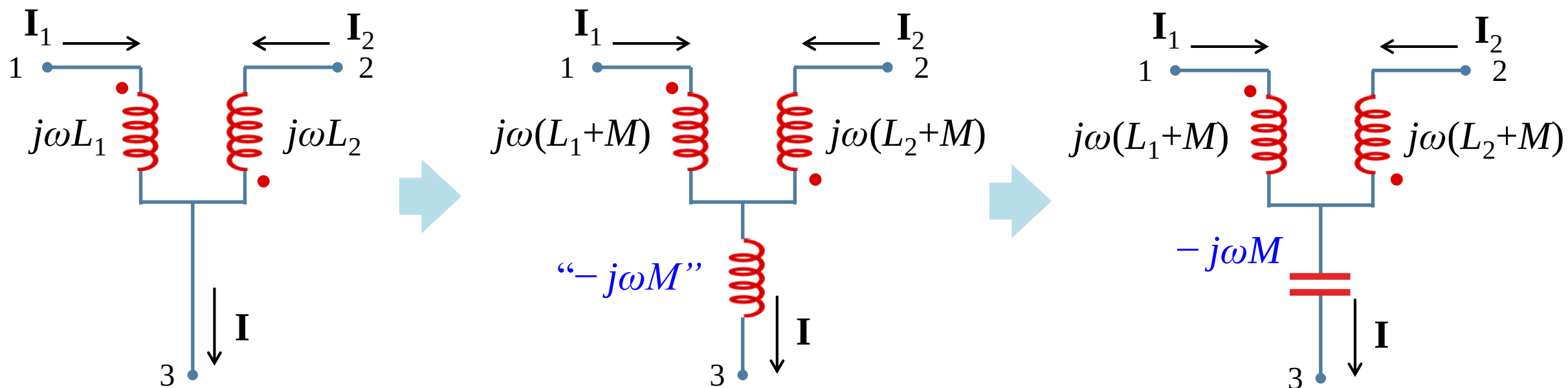
$$I = I_1 + I_2$$

$$j\omega L_1 I_1 + j\omega M I_2 = V_{13} = j\omega(L_1 - M)I_1 + j\omega M(I_1 + I_2)$$

$$j\omega L_2 I_2 + j\omega M I_1 = V_{23} = j\omega(L_2 - M)I_2 + j\omega M(I_1 + I_2)$$

12.4 Equivalents of coupled inductors (g)

- Parallel-opposing connection (异侧并联):



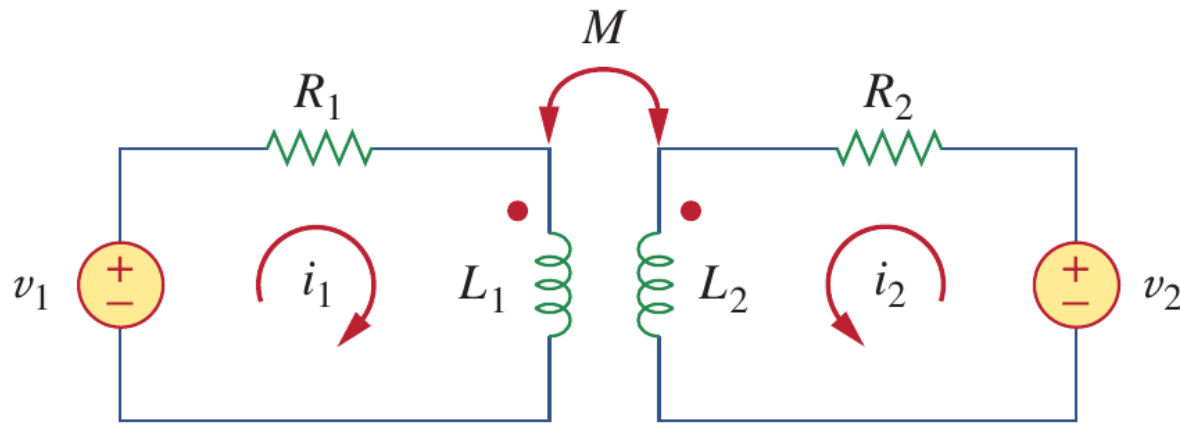
$$\mathbf{I} = \mathbf{I}_1 + \mathbf{I}_2$$

$$j\omega L_1 \mathbf{I}_1 - j\omega M \mathbf{I}_2 = \mathbf{V}_{13} = j\omega(L_1+M)\mathbf{I}_1 - j\omega M(\mathbf{I}_1 + \mathbf{I}_2)$$

$$j\omega L_2 \mathbf{I}_2 - j\omega M \mathbf{I}_1 = \mathbf{V}_{23} = j\omega(L_2+M)\mathbf{I}_2 - j\omega M(\mathbf{I}_1 + \mathbf{I}_2)$$

12.4 Equivalents of coupled inductors (h)

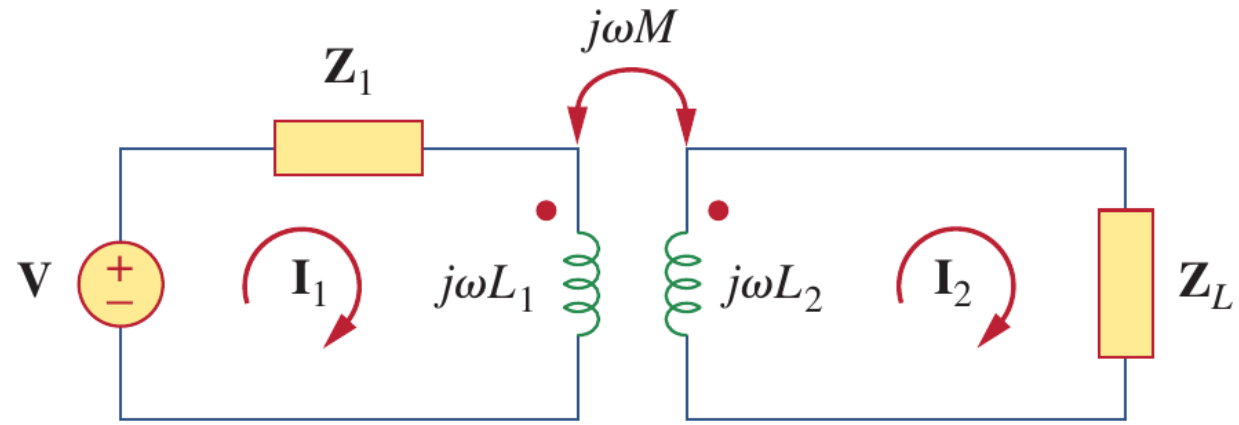
- Transformer connection (变压器连接):



$$\begin{cases} \mathbf{V}_1 = (R_1 + j\omega L_1)\mathbf{I}_1 + j\omega M\mathbf{I}_2 \\ \mathbf{V}_2 = (R_2 + j\omega L_2)\mathbf{I}_2 + j\omega M\mathbf{I}_1 \end{cases}$$

Self impedance Mutual impedance

Why is this positive?



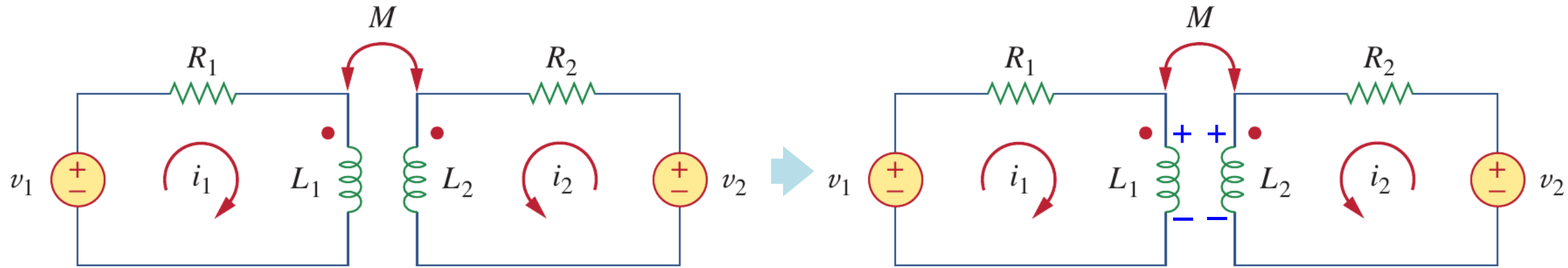
$$\begin{cases} \mathbf{V}_1 = (\mathbf{Z}_1 + j\omega L_1)\mathbf{I}_1 - j\omega M\mathbf{I}_2 \\ 0 = (\mathbf{Z}_L + j\omega L_2)\mathbf{I}_2 - j\omega M\mathbf{I}_1 \end{cases}$$

Self impedance Mutual impedance

Why is this negative?

12.4 Equivalents of coupled inductors (i)

- Transformer connection (变压器):



$$\begin{cases} \mathbf{V}_1 = (R_1 + j\omega L_1)\mathbf{I}_1 + j\omega M\mathbf{I}_2 \\ \mathbf{V}_2 = (R_2 + j\omega L_2)\mathbf{I}_2 + j\omega M\mathbf{I}_1 \end{cases}$$

Self impedance Mutual impedance

Why is this positive?

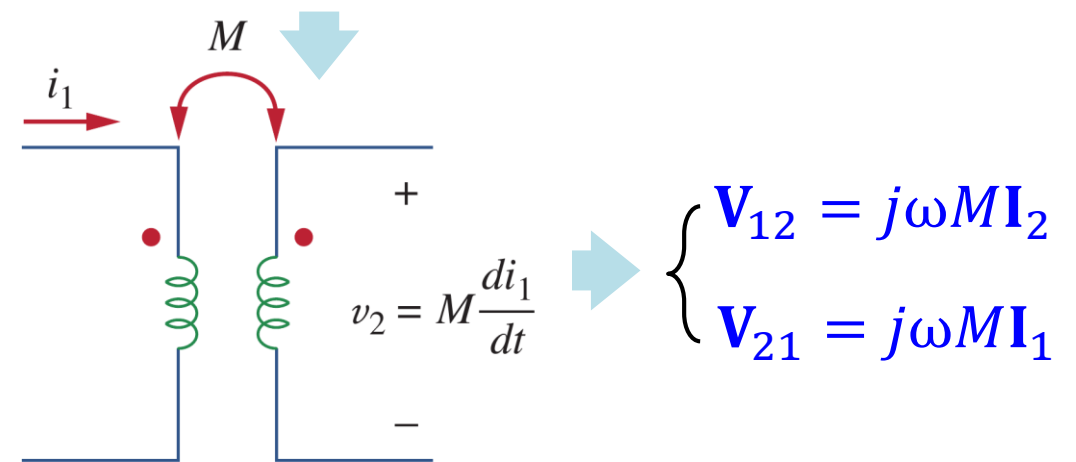
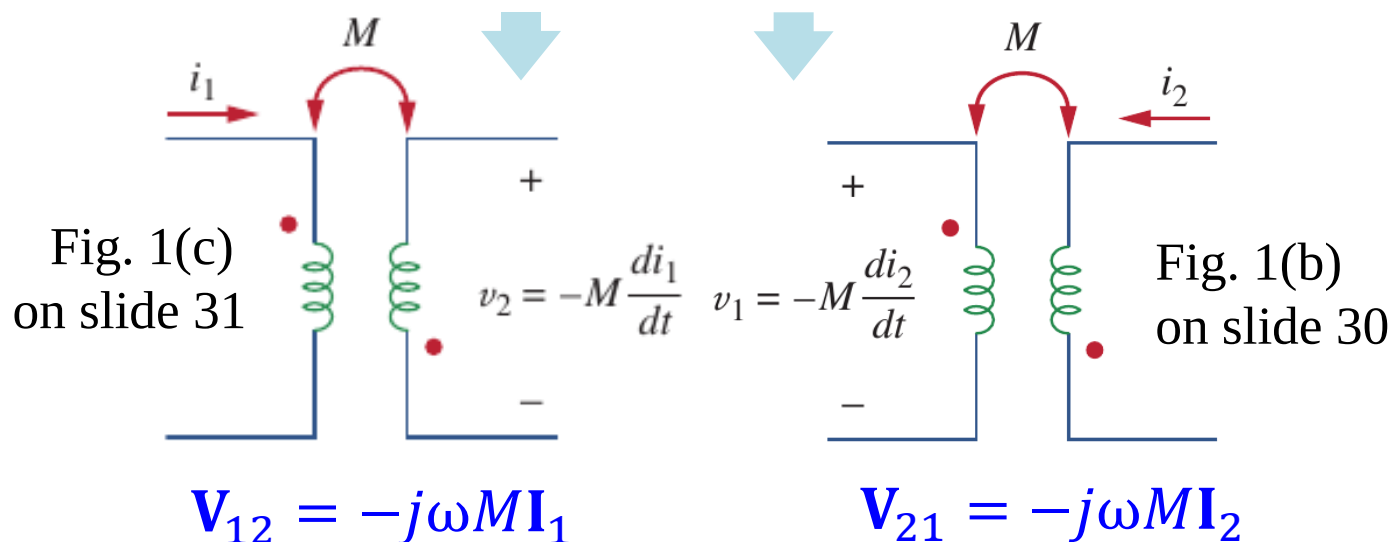
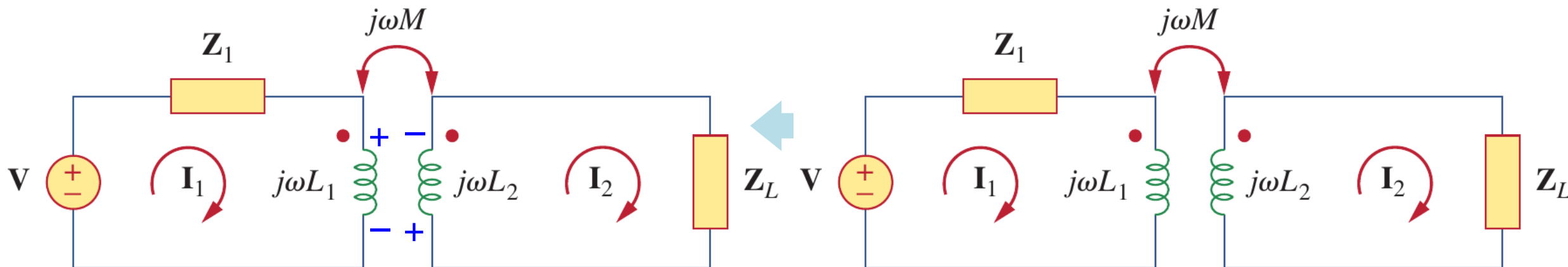


Fig. 1(a) on slide 30

12.4 Equivalents of coupled inductors (j)

- Transformer connection (变压器):



$$\begin{cases} V_1 = (Z_1 + j\omega L_1)I_1 - j\omega M I_2 \\ 0 = (Z_L + j\omega L_2)I_2 - j\omega M I_1 \end{cases}$$

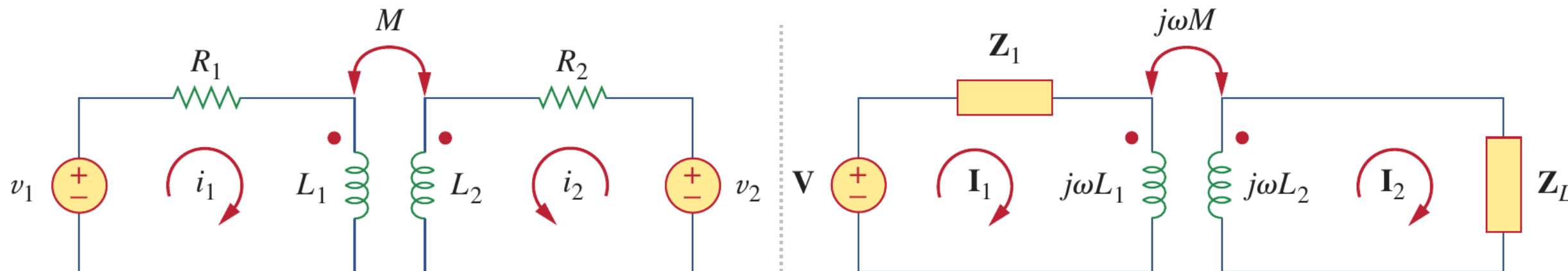
Self impedance

Mutual impedance

Why is this negative?

12.4 Equivalents of coupled inductors (k)

- Transformer connection (变压器):



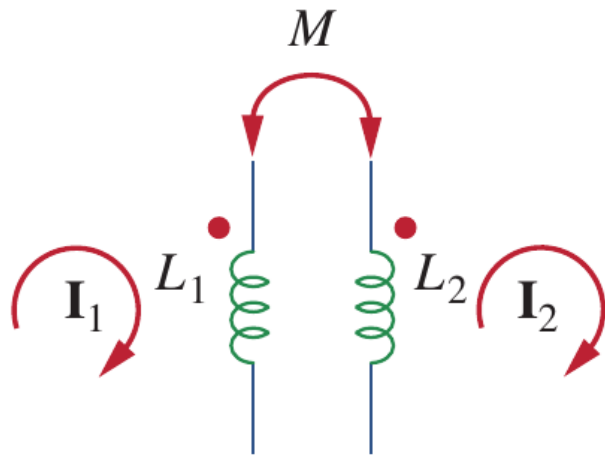
$$\begin{cases} \mathbf{V}_1 = (R_1 + j\omega L_1)\mathbf{I}_1 + j\omega M\mathbf{I}_2 \\ \mathbf{V}_2 = (R_2 + j\omega L_2)\mathbf{I}_2 + j\omega M\mathbf{I}_1 \end{cases}$$

$$\begin{cases} \mathbf{V}_1 = (\mathbf{Z}_1 + j\omega L_1)\mathbf{I}_1 - j\omega M\mathbf{I}_2 \\ 0 = (\mathbf{Z}_L + j\omega L_2)\mathbf{I}_2 - j\omega M\mathbf{I}_1 \end{cases}$$

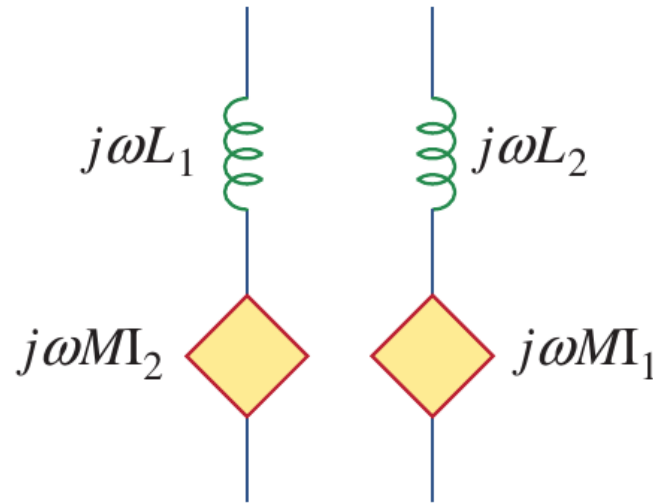
Note: always give a second thought on the sign and value of the induced voltages.

12.4 Equivalents of coupled inductors (I)

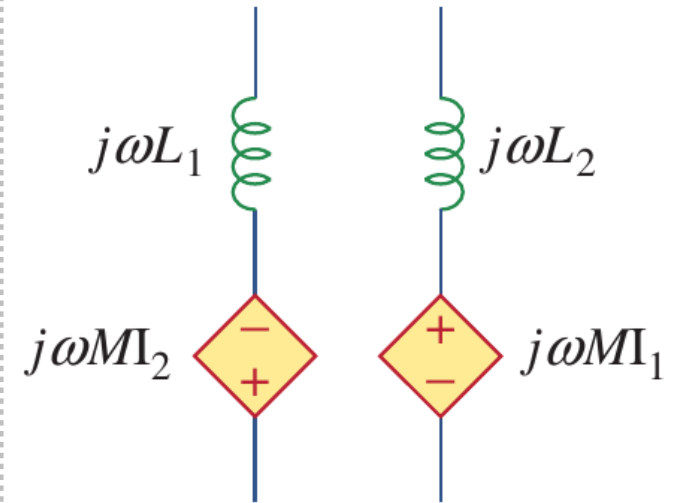
- Transformer connection (变压器):



Step 1: select your ref. current directions.



Step 2: draw the equivalent circuit and determine the values of induced voltages.



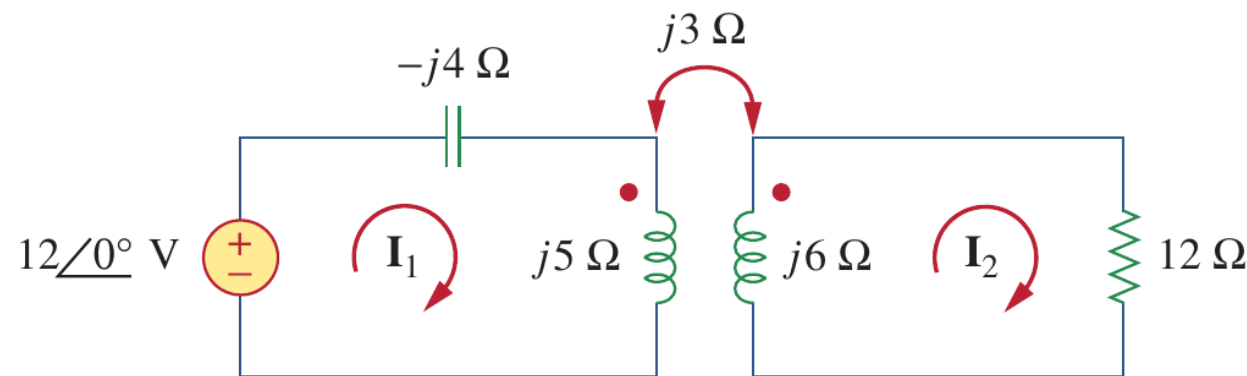
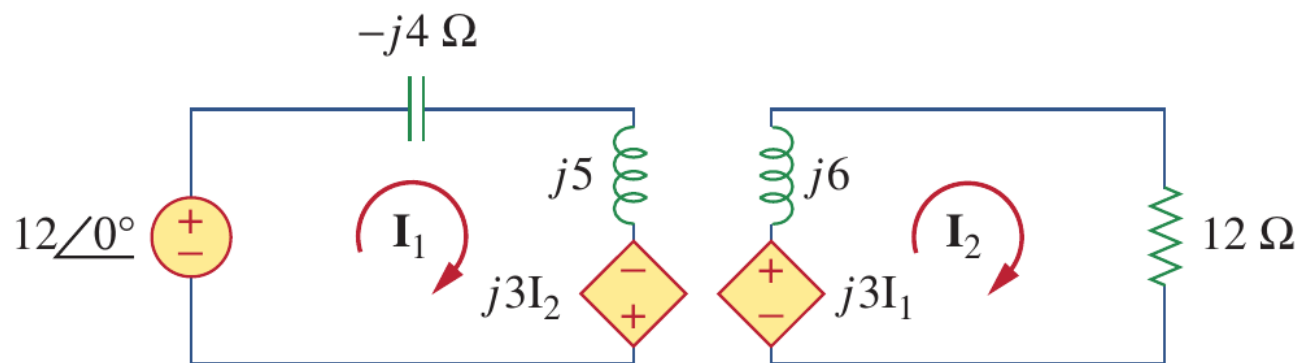
Step 3: determine the polarities of the induced voltages based on reference directions.

12.4 Equivalents of coupled inductors (m)

- Example 12.02:

Example 12.02: calculate the phasor currents $\mathbf{I}_1$ and $\mathbf{I}_2$ in the following circuit.

Solution: the equivalent circuit is as follows.



KVL for loop 1:

$$-12 + (-j4 + j5)\mathbf{I}_1 - j3\mathbf{I}_2 = 0$$

KVL for loop 2:

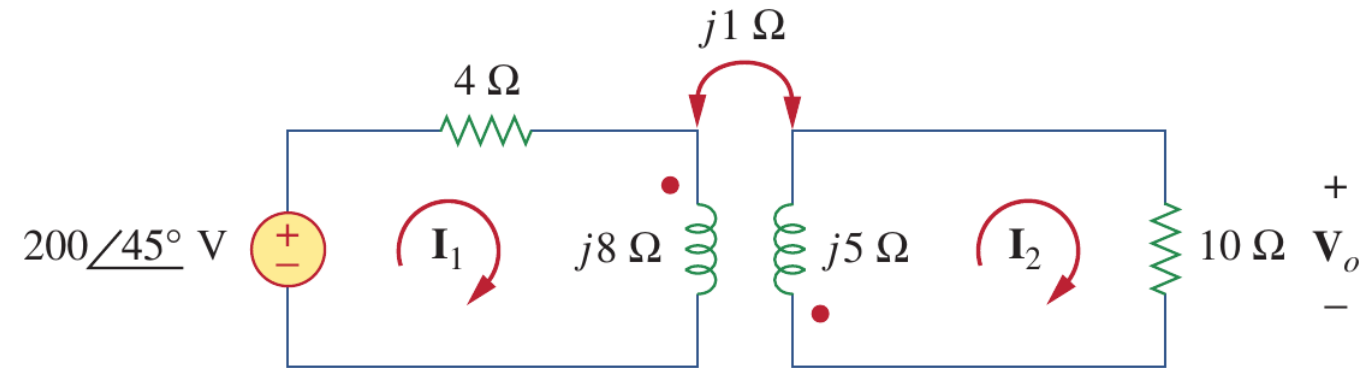
$$-j3\mathbf{I}_1 + (12 + j6)\mathbf{I}_2 = 0$$

$$\begin{cases} \mathbf{I}_2 = \frac{12}{4 - j} = 2.91 \angle 14.04^\circ \text{ A} \\ \mathbf{I}_1 = (2 - j4)\mathbf{I}_2 = 13.01 \angle -49.39^\circ \text{ A} \end{cases}$$

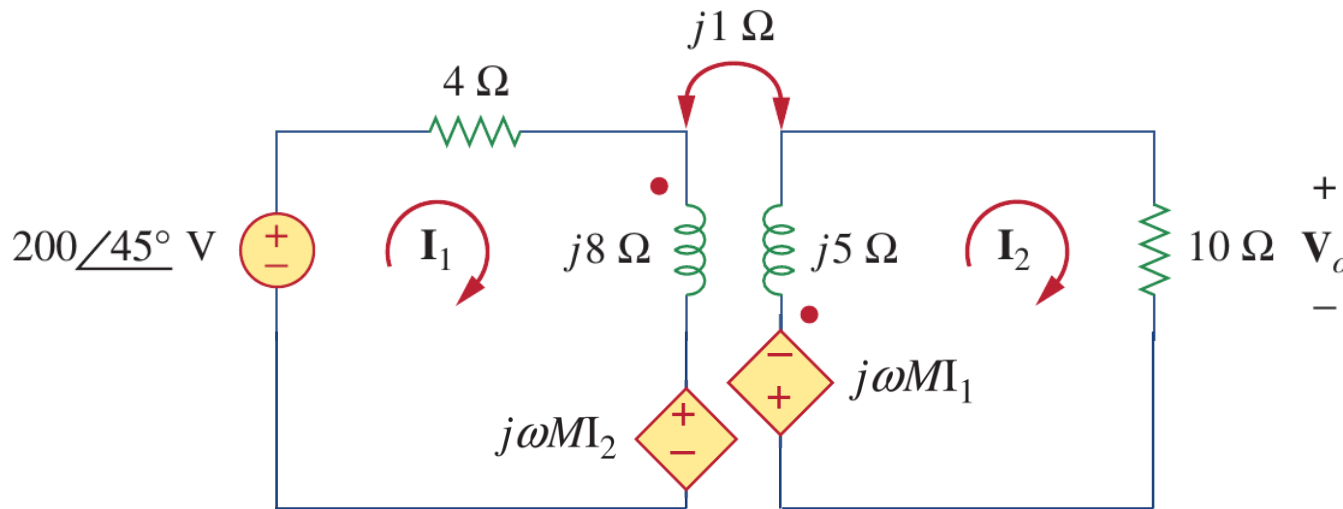
12.4 Equivalents of coupled inductors (n)

- Example 12.03:

Example 12.03: determine the voltage V_o in the following circuit.



Solution: (1) Draw the equivalent circuit but with the polarities of the dependent voltage sources undetermined, then determine the polarities of the induced voltages.



- In the source side: I_1 flows into the dotted terminal.
- In the load side: I_2 flows into the dotted terminal.

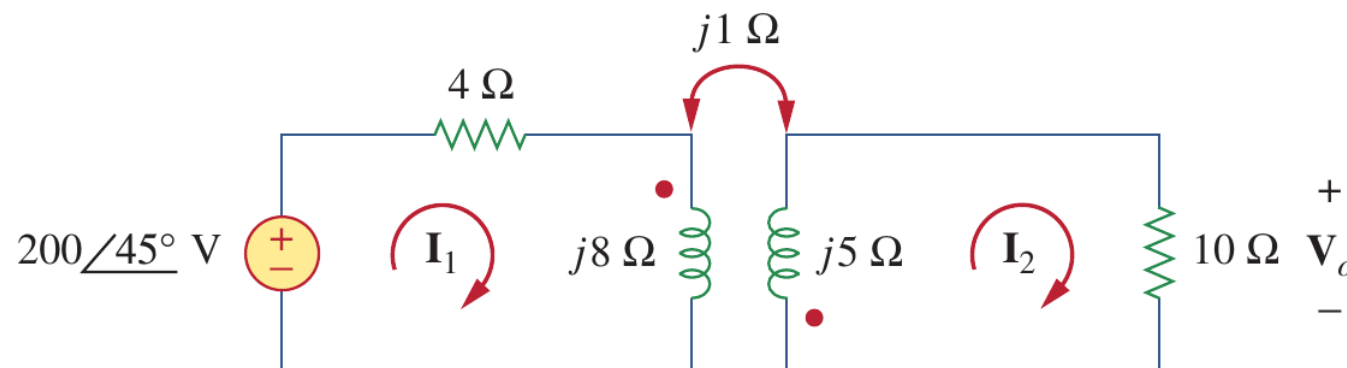
$$\text{Therefore } V_{12} = j\omega M I_1 = j I_1$$

$$V_{21} = j\omega M I_2 = j I_2$$

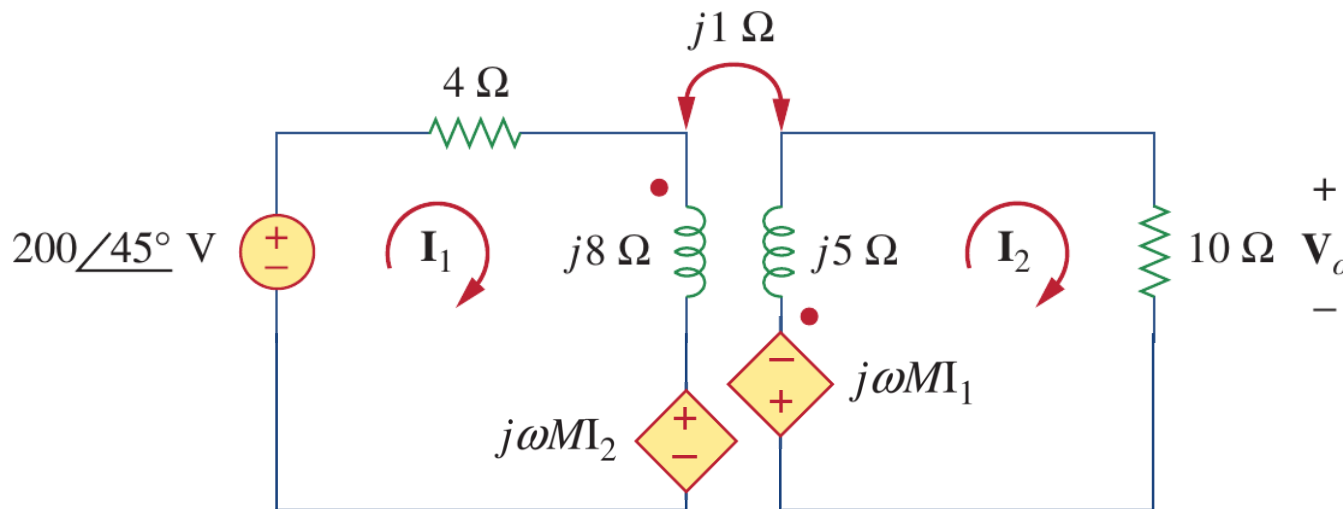
12.4 Equivalents of coupled inductors (o)

- Example 12.03:

Example 12.03: determine the voltage V_o in the following circuit.



Solution: (2) Now we know the polarities of induced voltages in the equivalent circuit.



$$V_{12} = j\omega M I_1 = j I_1 \quad \& \quad V_{21} = j\omega M I_2 = j I_2$$

$$\begin{cases} 200\angle 45^\circ = (4 + j8)I_1 + jI_2 \\ 0 = (10 + j5)I_1 + jI_2 \end{cases}$$

$$\Rightarrow I_1 = (10j - 5)I_2$$

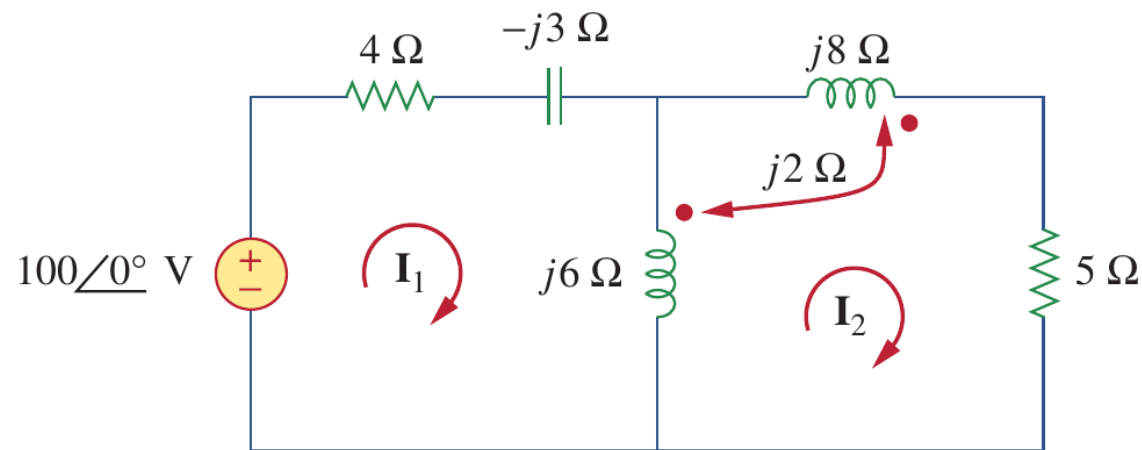
$$\Rightarrow I_2 \approx 2\angle -135^\circ \text{ A}$$

$$V_o \approx 20\angle -135^\circ \text{ V}$$

12.4 Equivalents of coupled inductors (p)

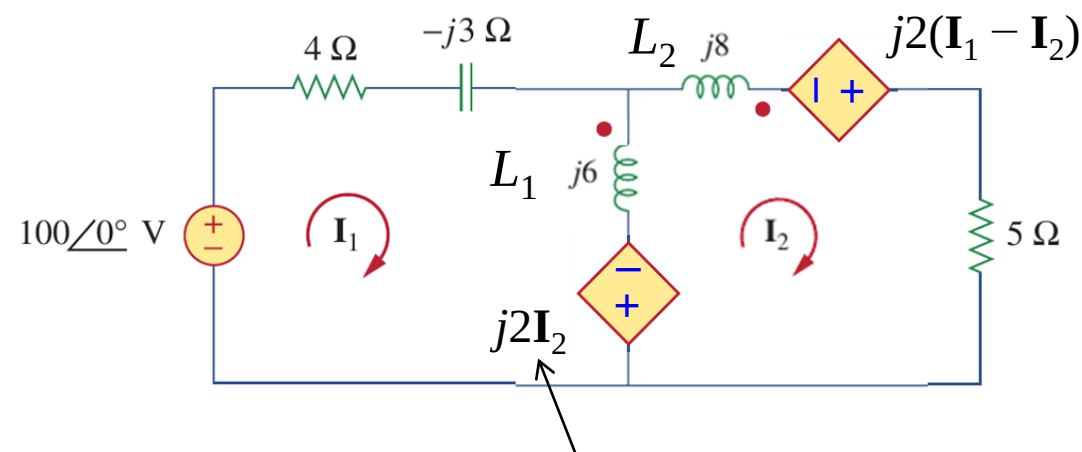
- Example 12.04:

Example 12.04: calculate the mesh current in the following circuit.



Solution: (1) The equivalent circuit is as follows

The current through L_1 is $\mathbf{I}_1 - \mathbf{I}_2$



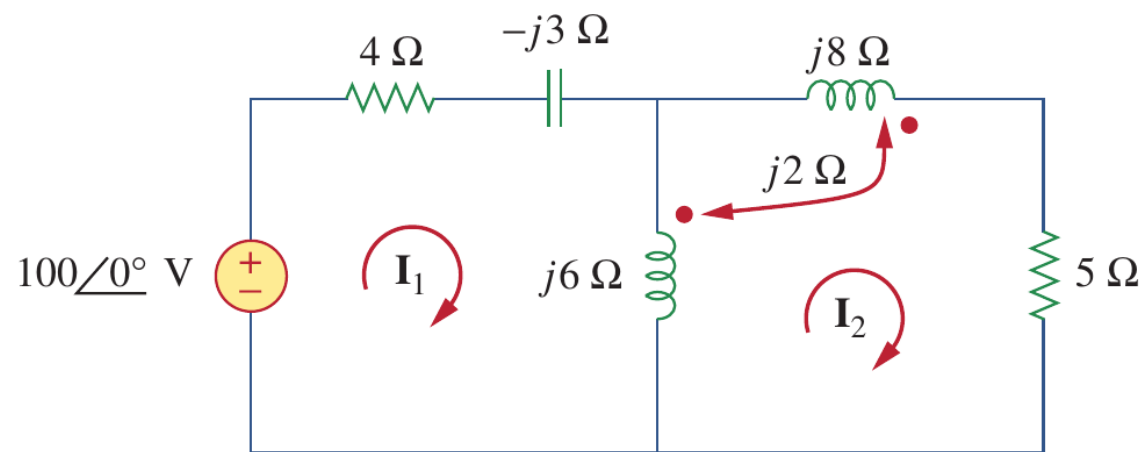
The current through L_2 is $\mathbf{I}_2$

$\mathbf{I}_1 - \mathbf{I}_2$ enters dotted side of L_1 and $\mathbf{I}_2$ leaves the dotted side of L_2 , so the induced voltages $\mathbf{V}_{12}$ and $\mathbf{V}_{21}$ are negative

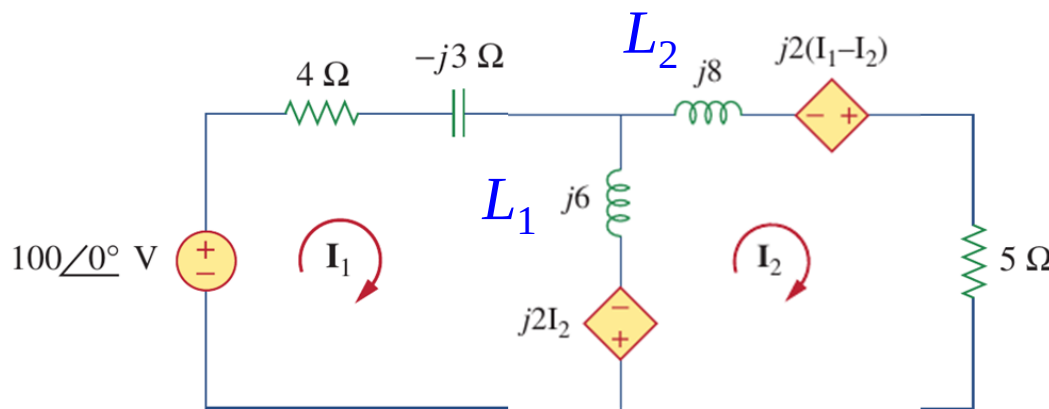
12.4 Equivalents of coupled inductors (q)

- Example 12.04:

Example 12.04: calculate the mesh current in the following circuit.



Solution: (2) Use mesh analysis to solve the circuit:



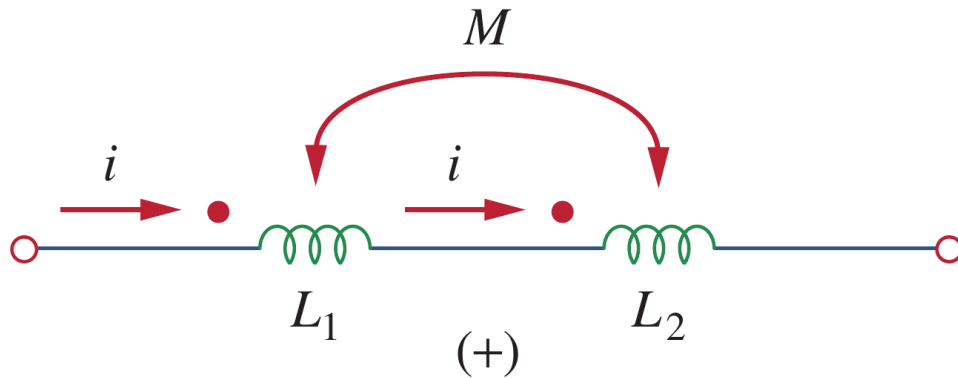
$$\begin{cases} 100 = (4 - j3 + j6)I_1 - j6I_2 - j2I_2 \\ 0 = (j6 + j8 + j2 \times 2 + 5)I_2 - j6I_1 - j2I_1 \end{cases}$$

$$\Rightarrow \begin{cases} 100 = (4 + j3)I_1 - j8I_2 \\ 0 = (5 + j18)I_2 - j8I_1 \end{cases}$$

$$\Rightarrow I_1 = 20.3 \angle 3.5^\circ \text{ A} \quad I_2 = 8.693 \angle 19^\circ \text{ A}$$

12.5 Energy in mutual inductance (a)

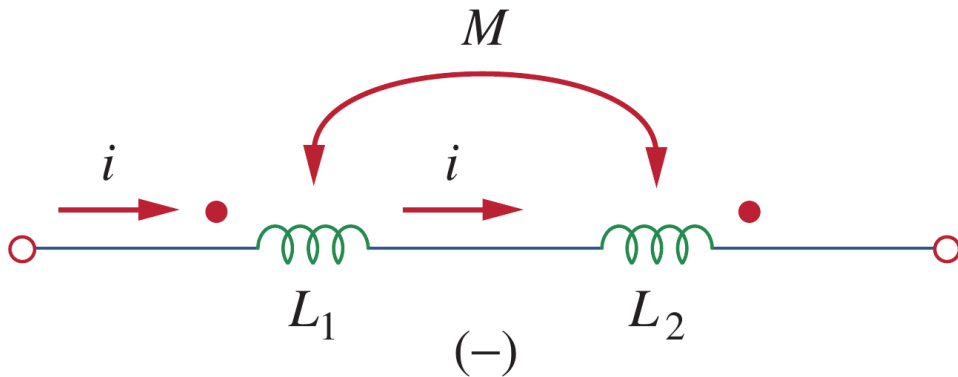
- Series connection:



Series-aiding:

$$L_{eq} = L_1 + L_2 + 2M$$

$$w = \frac{1}{2} L_{eq} i^2 = \frac{1}{2} L_1 i^2 + \frac{1}{2} L_2 i^2 + M i^2$$



Series-opposing:

$$L_{eq} = L_1 + L_2 - 2M$$

$$w = \frac{1}{2} L_{eq} i^2 = \frac{1}{2} L_1 i^2 + \frac{1}{2} L_2 i^2 - M i^2$$

12.5 Energy in mutual inductance (b)

- Parallel-aiding connection:

$$w = \int_{-\infty}^t p_{\text{total}} dt$$

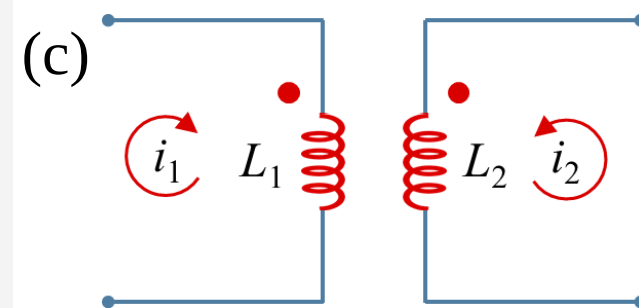
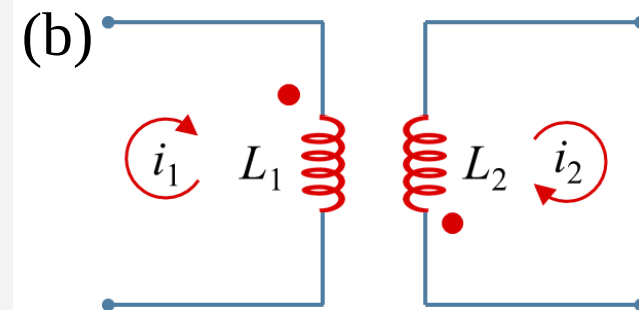
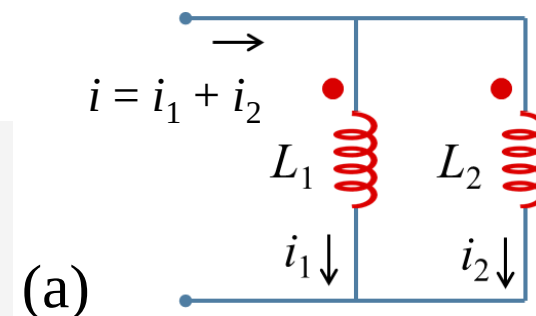
*Initial currents are zero

$$w = \int_{-\infty}^t i_1 v_1 dt + \int_{-\infty}^t i_1 v_{21} dt + \int_{-\infty}^t i_2 v_2 dt + \int_{-\infty}^t i_2 v_{12} dt$$

$$= \int_{-\infty}^t i_1 L_1 \frac{di_1}{dt} dt + \int_{-\infty}^t i_1 M \frac{di_2}{dt} dt + \int_{-\infty}^t i_2 L_2 \frac{di_2}{dt} dt + \int_{-\infty}^t i_2 M \frac{di_1}{dt} dt$$

$$= \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 + M \int_{-\infty}^t i_1 \frac{di_2}{dt} dt + M \int_{-\infty}^t i_2 \frac{di_1}{dt} dt$$

$$w = \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 + M i_1 i_2$$



12.5 Energy in mutual inductance (c)

- Parallel-opposing connection:

$$w = \int_{-\infty}^t p_{\text{total}} dt$$

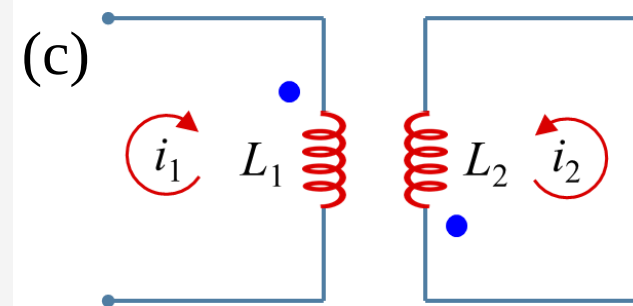
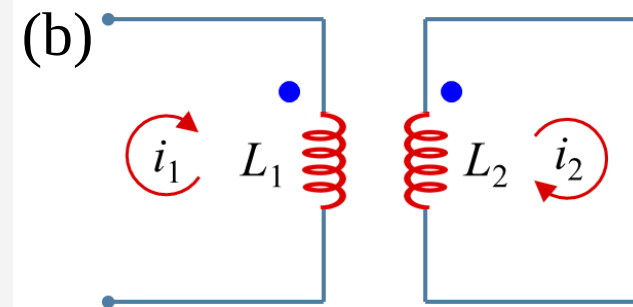
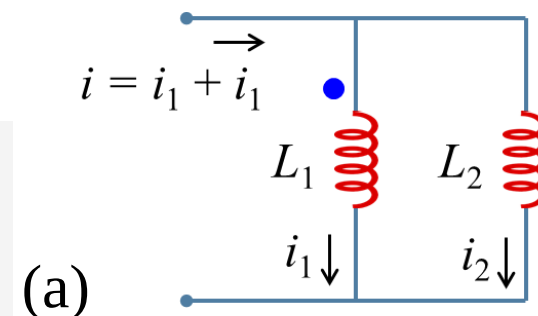
*Initial currents are zero

$$w = \int_{-\infty}^t i_1 v_1 dt - \int_{-\infty}^t i_1 v_{21} dt + \int_{-\infty}^t i_2 v_2 dt - \int_{-\infty}^t i_2 v_{12} dt$$

$$= \int_{-\infty}^t i_1 L_1 \frac{di_1}{dt} dt - \int_{-\infty}^t i_1 M \frac{di_2}{dt} dt + \int_{-\infty}^t i_2 L_2 \frac{di_2}{dt} dt - \int_{-\infty}^t i_2 M \frac{di_1}{dt} dt$$

$$= \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 - M \int_{-\infty}^t i_1 \frac{di_2}{dt} dt - M \int_{-\infty}^t i_2 \frac{di_1}{dt} dt$$

$$w = \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 - M i_1 i_2$$



12.5 Energy in mutual inductance (d)

- Series-opposing parallel connection:

(a) For aiding connection:

$$w = \frac{1}{2}L_1i_1^2 + \frac{1}{2}L_2i_2^2 + Mi_1i_2$$

(b) For opposing connection:

$$w = \frac{1}{2}L_1i_1^2 + \frac{1}{2}L_2i_2^2 - Mi_1i_2$$

(c) $i_1 = i_2$ for series connections

Read textbook p564-p565

- a) When the currents do NOT both enter or leave the dotted terminals of the coils, the term related to M will have a negative sign.
- b) The energy stored in the circuit cannot be negative because the circuit is passive.

$$\frac{1}{2}L_1i_1^2 + \frac{1}{2}L_2i_2^2 - Mi_1i_2 \geq 0$$

$$\frac{1}{2}(i_1\sqrt{L_1} - i_2\sqrt{L_2})^2 + i_1i_2(\sqrt{L_1L_2} - M) \geq 0$$

$$\sqrt{L_1L_2} - M \geq 0 \text{ or } M \leq \sqrt{L_1L_2}$$

12.6 Coupling coefficient (a)

- Coupling coefficient (耦合系数):

$$\left. \begin{aligned} M &\leq \sqrt{L_1 L_2} \Rightarrow \frac{M}{\sqrt{L_1 L_2}} \leq 1 \\ M &\geq 0 \Rightarrow \frac{M}{\sqrt{L_1 L_2}} \geq 0 \end{aligned} \right\} \text{ Let } k = \frac{M}{\sqrt{L_1 L_2}} \text{ or } M = k\sqrt{L_1 L_2} \Rightarrow 0 \leq k \leq 1$$

- a) The coupling coefficient k is a measure of the magnetic coupling between two coils, which is the fraction of the total flux emanating from one coil that links the other coil.
- b) The mutual inductance M approaches the upper limit is specified by the k .

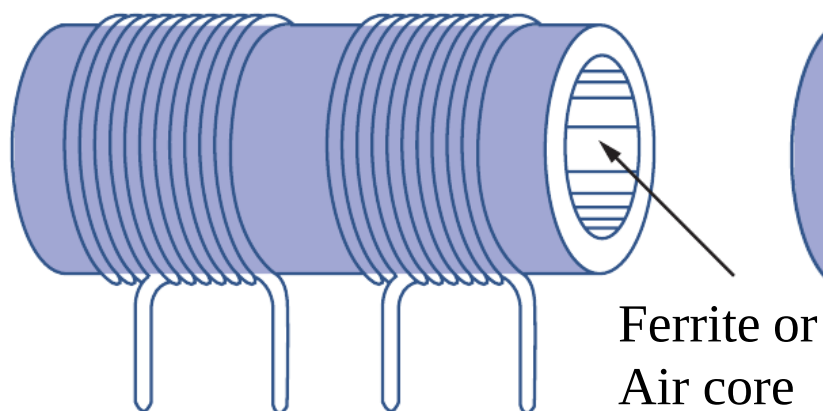
$$k = \frac{\phi_{12}}{\phi_1} = \frac{\phi_{12}}{\phi_{11} + \phi_{12}}$$
$$k = \frac{\phi_{21}}{\phi_2} = \frac{\phi_{21}}{\phi_{21} + \phi_{22}}$$

12.6 Coupling coefficient (b)

- Coupling coefficient (耦合系数):

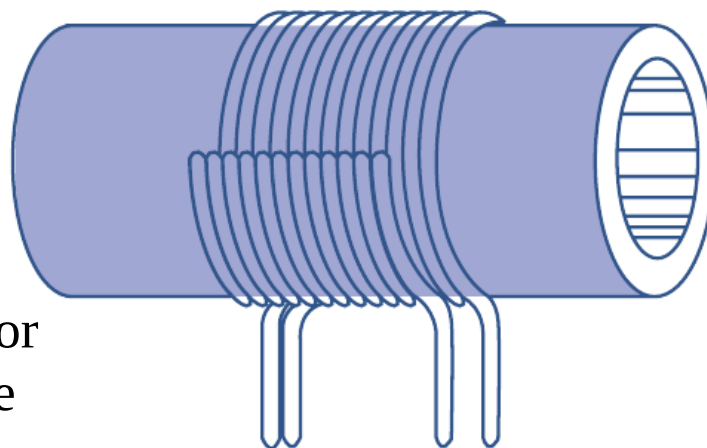
$$k < 0.5$$

Loosely coupled
松耦合/松散耦合



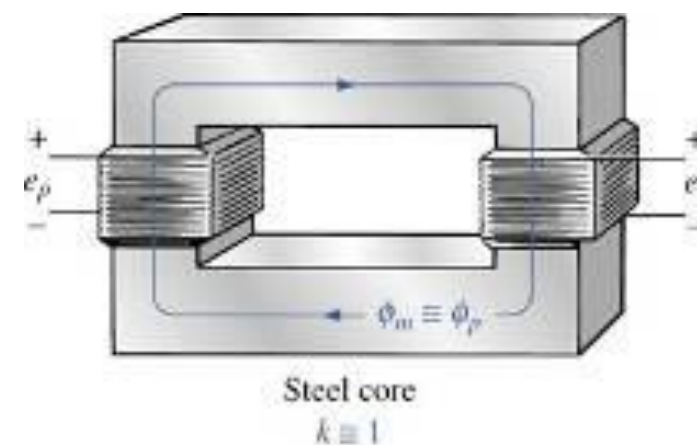
$$0.5 < k < 1$$

Tightly coupled
紧耦合/紧密耦合



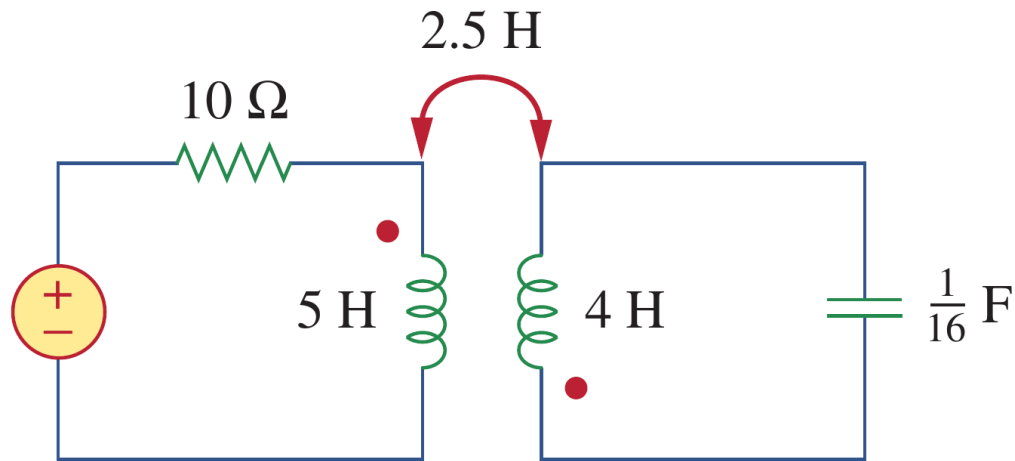
$$k = 1$$

Ideally coupled
全耦合/理想耦合



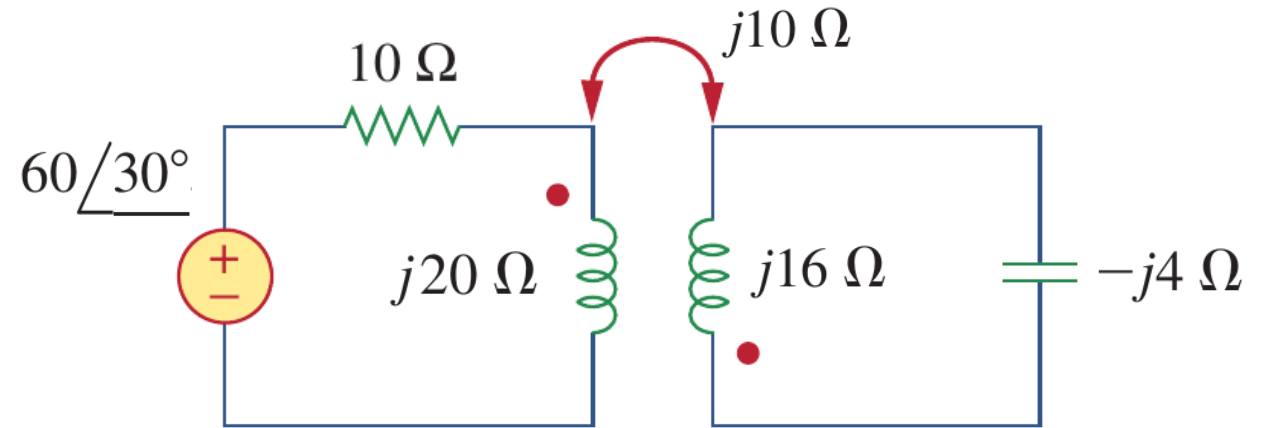
12.6 Coupling coefficient (c)

- Example 12.05:



Example 13.05: consider the following circuit. (a) determine the coupling coefficient. (b) calculate the energy stored in the coupled inductors at $t = 1\text{ s}$ if $v = 60\cos(4t + 30^\circ)\text{ V}$.

Solution: (1) transform the circuit into frequency domain and give the mesh currents:



$$60 \cos(4t + 30^\circ) \Rightarrow 60\angle 30^\circ, \quad \omega = 4 \text{ rad/s}$$

$$5 \text{ H} \Rightarrow j\omega L_1 = j20 \Omega$$

$$2.5 \text{ H} \Rightarrow j\omega M = j10 \Omega$$

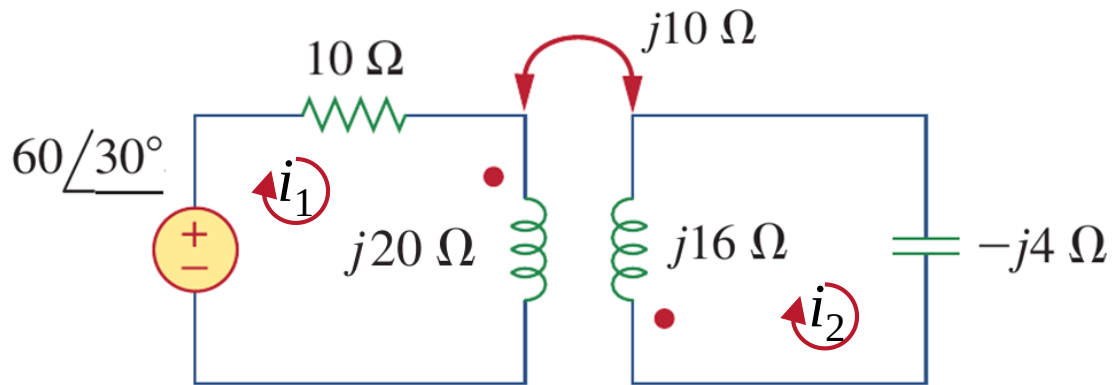
$$4 \text{ H} \Rightarrow j\omega L_2 = j16 \Omega$$

$$\frac{1}{16} \text{ F} \Rightarrow \frac{1}{j\omega C} = -j4 \Omega$$

12.6 Coupling coefficient (d)

- Example 12.05:

Solution:



(2) Determine the k :

$$k = \frac{M}{\sqrt{L_1 L_2}} = \frac{2.5}{\sqrt{20}} = 0.56$$

(3) Solve the currents based on mesh analysis:

$$\begin{cases} (10 + j20)\mathbf{I}_1 + j10\mathbf{I}_2 = 60\angle 30^\circ \\ j10\mathbf{I}_1 + (j16 - j4)\mathbf{I}_2 = 0 \Rightarrow \mathbf{I}_1 = -1.2\mathbf{I}_2 \end{cases}$$

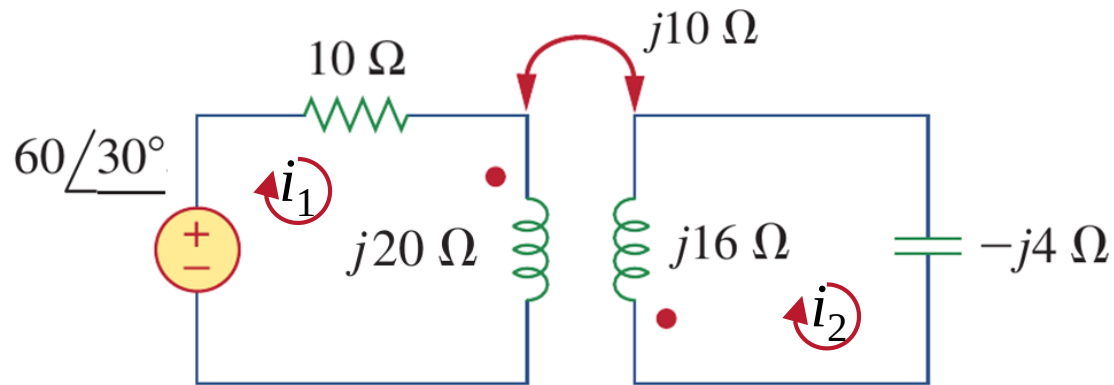
$$\Rightarrow \begin{cases} \mathbf{I}_2 = 3.254\angle 160.6^\circ \text{ A} \\ \mathbf{I}_1 = -1.2\mathbf{I}_2 = 3.905\angle -19.4^\circ \text{ A} \end{cases}$$

$$\Rightarrow \begin{cases} i_1 = 3.905 \cos(4t - 19.4^\circ) \\ i_2 = 3.254 \cos(4t + 160.6^\circ) \end{cases}$$

12.6 Coupling coefficient (e)

- Example 12.05:

Solution:



Example 12.05: consider the following circuit. (a) determine the coupling coefficient. (b) calculate the energy stored in the coupled inductors at $t = 1\text{ s}$ if $v = 60\cos(4t + 30^\circ)\text{ V}$.

(4) Calculate the energy stored at $t = 1\text{ s}$:

$$\begin{cases} i_1 = 3.905 \cos(229.2^\circ - 19.4^\circ) = -3.389\text{ A} \\ i_2 = 3.254 \cos(229.2^\circ + 160.6^\circ) = 2.824\text{ A} \end{cases}$$

$$\begin{aligned} w &= \frac{1}{2}L_1 i_1^2 + \frac{1}{2}L_2 i_2^2 + M i_1 i_2 \\ &= \frac{1}{2}(5)(-3.389)^2 + \frac{1}{2}(4)(2.824)^2 \\ &\quad + 2.5(-3.389)(2.824) \\ &= 20.73\text{ J} \end{aligned}$$

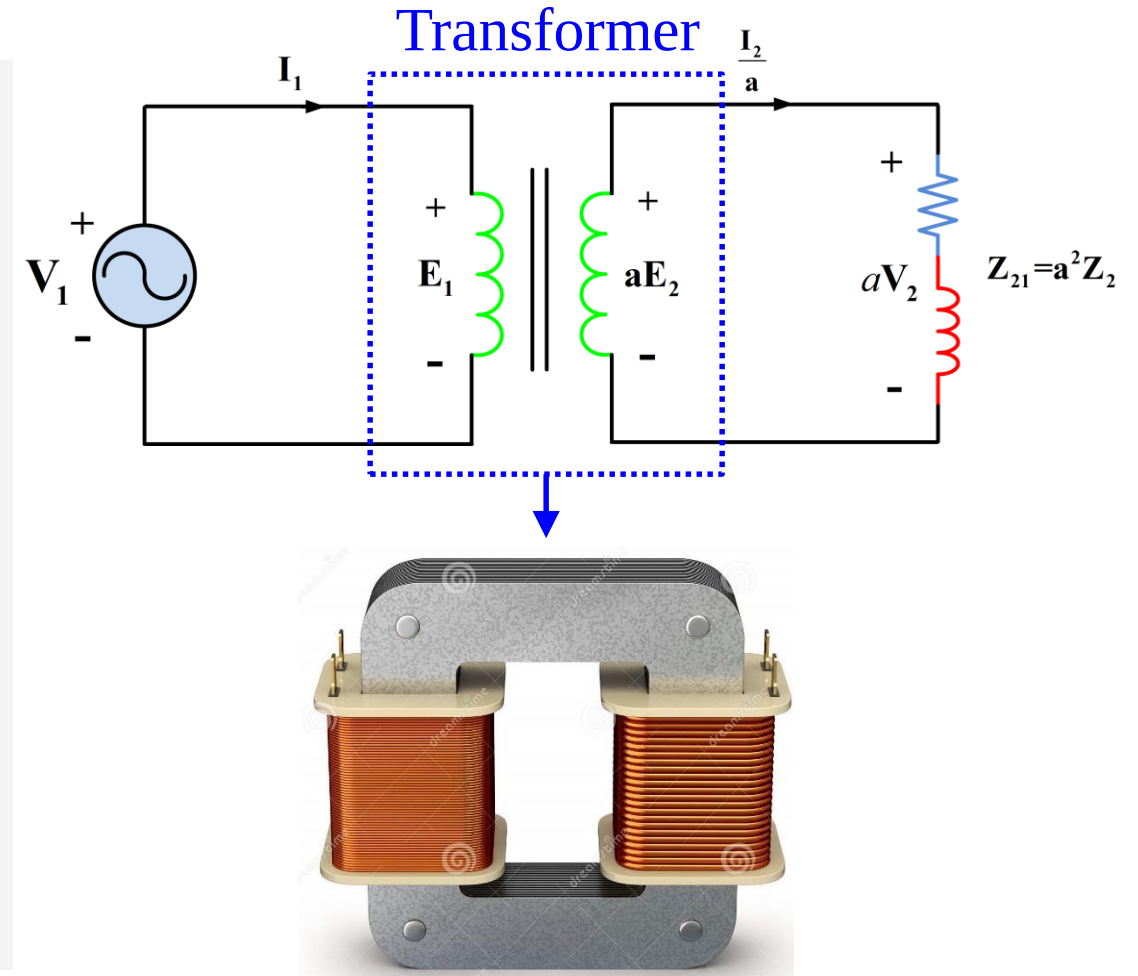
Lesson 12 Magnetically coupled circuits

12.1	Coupled circuits	Slides 15-21
12.2-12.7	Mutual inductance	Slides 23-60
12.8	Linear transformers	Slides 62-69
12.9	Ideal transformers	Slides 71-83
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12.7 Linear transformers (a)

- Transformers save the day!

- a) A transformer is a key circuit element.
- b) It is an electrical device based on magnetic coupling, using magnetically coupled coils to transfer energy from one circuit to another.
- c) A transformer is a static device and all currents and voltages are AC.
- d) It has many applications, including step-up or step-down voltage (power transfer), isolation coupling, and impedance matching (signal transfer).

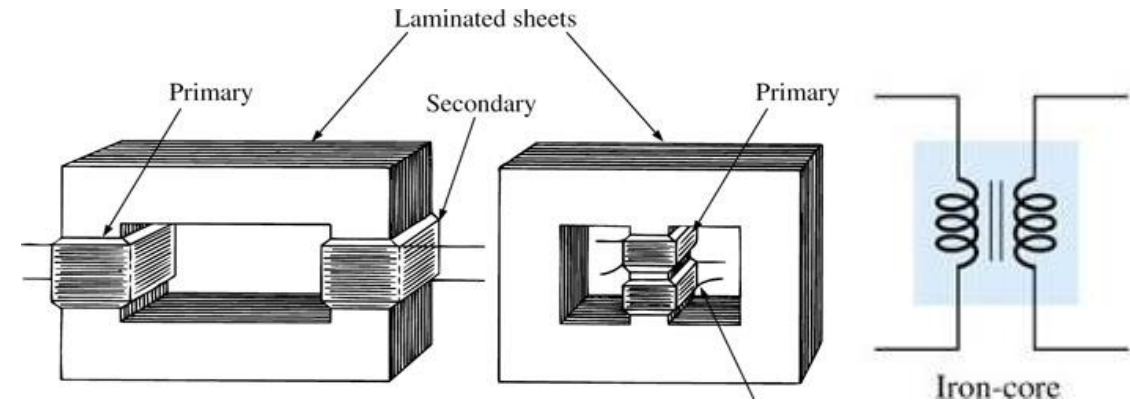


12.7 Linear transformers (b)

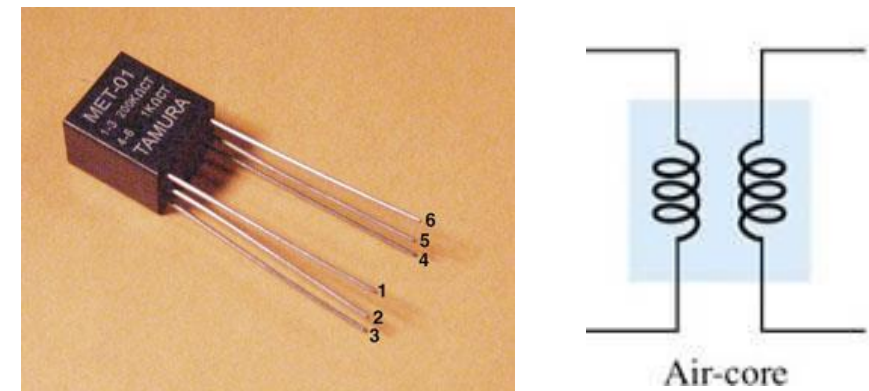
- Types of transformers

- a) Transformers are available in many different shapes and sizes.
- b) Some of the more common types include the power transformer, audio transformer, IF (intermediate frequency) transformer, and RF (radio frequency) transformer.
- c) Each is designed to fulfill a particular requirement in a specific area of application.
- d) Depending on core materials, there are **air-core** and **iron-core transformers**.

Power transformer (功率变压器)

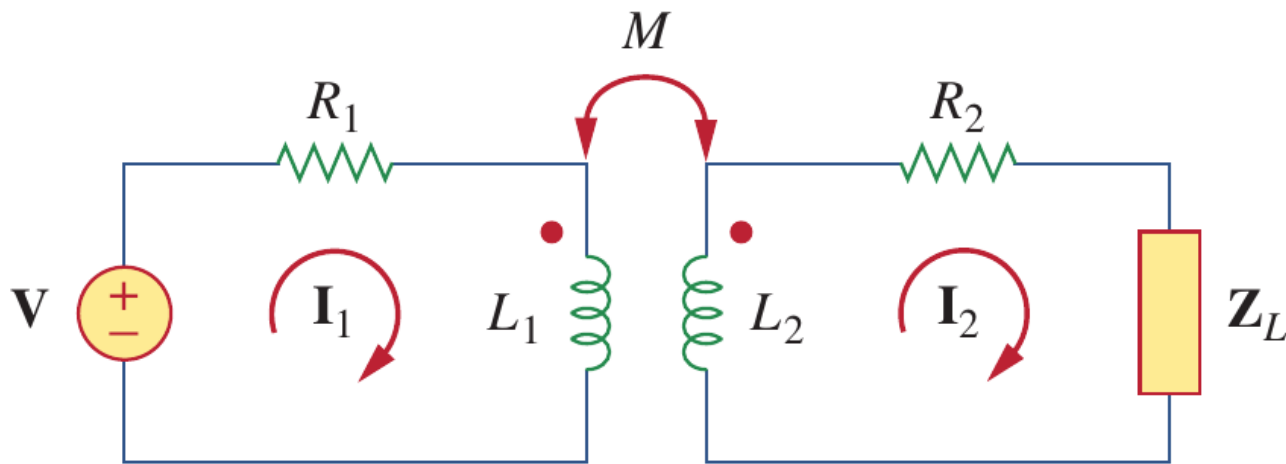


Audio transformer (音频变压器)



12.7 Linear transformers (c)

- Linear transformers:



Assume the frequency is ω , apply KVL:

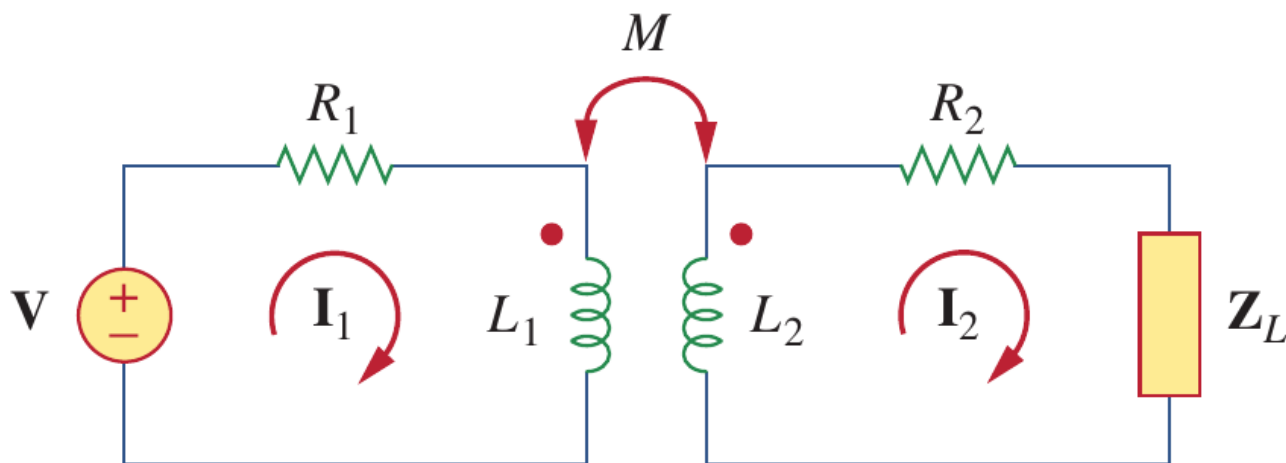
$$\mathbf{V} = (R_1 + j\omega L_1)\mathbf{I}_1 - j\omega M\mathbf{I}_2$$

$$0 = -j\omega M\mathbf{I}_1 + (R_2 + j\omega L_2 + \mathbf{Z}_L)\mathbf{I}_2$$

- The transformer is said to be linear if the coils are wound on a magnetically linear material—a material for which the magnetic permeability is constant.
- Such materials include air, plastic, Bakelite, and wood. In fact, most materials are magnetically linear.
- Linear transformers are sometimes called air-core transformers, although not all of them are necessarily air-core.

12.7 Linear transformers (d)

- Linear transformers:



Assume the frequency is ω , apply KVL:

$$\mathbf{V} = (R_1 + j\omega L_1)\mathbf{I}_1 - j\omega M\mathbf{I}_2$$

$$0 = -j\omega M\mathbf{I}_1 + (R_2 + j\omega L_2 + \mathbf{Z}_L)\mathbf{I}_2$$

The input impedance
输入阻抗

$$\mathbf{Z}_{\text{in}} = \frac{\mathbf{V}}{\mathbf{I}_1} = \boxed{R_1 + j\omega L_1} + \boxed{\frac{\omega^2 M^2}{R_2 + j\omega L_2 + \mathbf{Z}_L}}$$

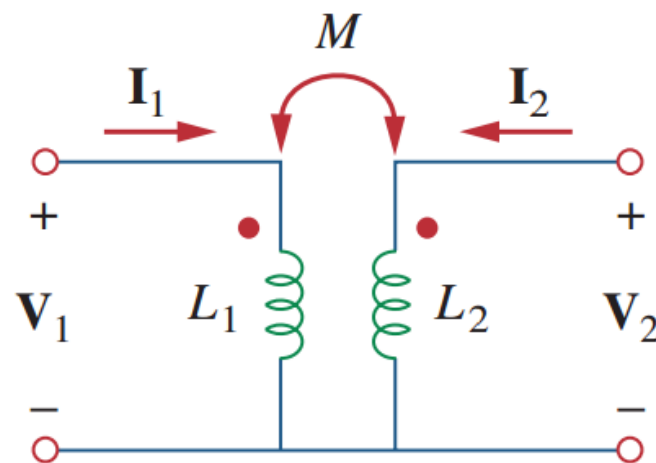
This is not affected by the location of the dots on the transformer.

Primary impedance
原边阻抗

Reflected or coupled impedance
反射阻抗/耦合阻抗

12.7 Linear transformers (e)

- Linear transformers:

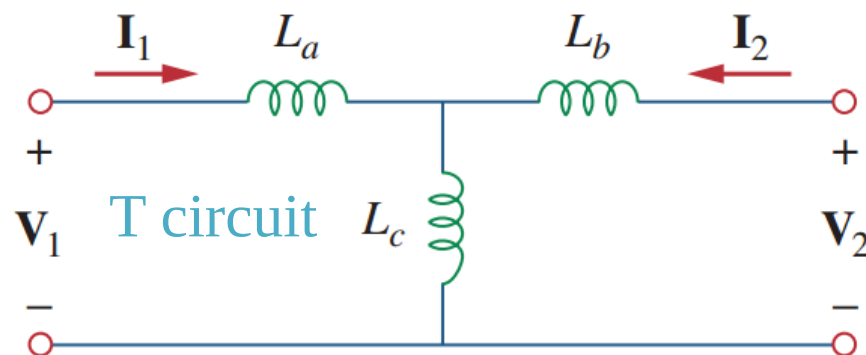


$$V_1 = j\omega L_1 I_1 + j\omega M I_2$$

$$V_2 = j\omega M I_1 + j\omega L_2 I_2$$

What if one current is into the dotted side while the other is not?

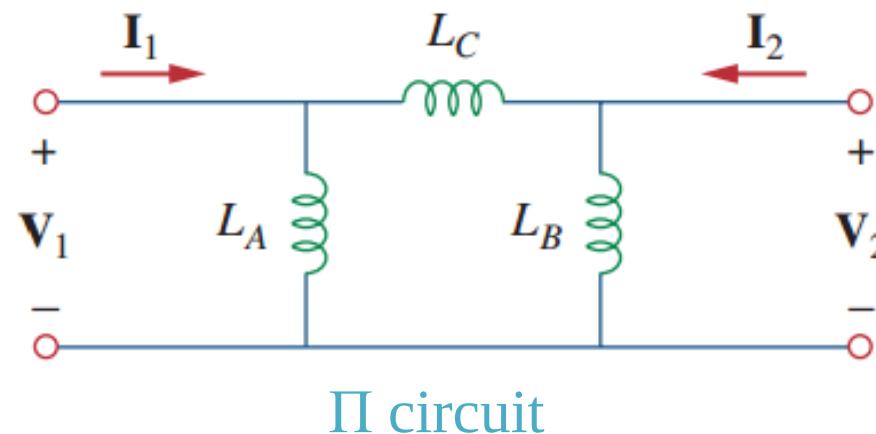
Decoupled equivalent circuits



$$L_a = L_1 - M,$$

$$L_b = L_2 - M,$$

$$L_c = M$$



$$L_A = \frac{L_1 L_2 - M^2}{L_2 - M},$$

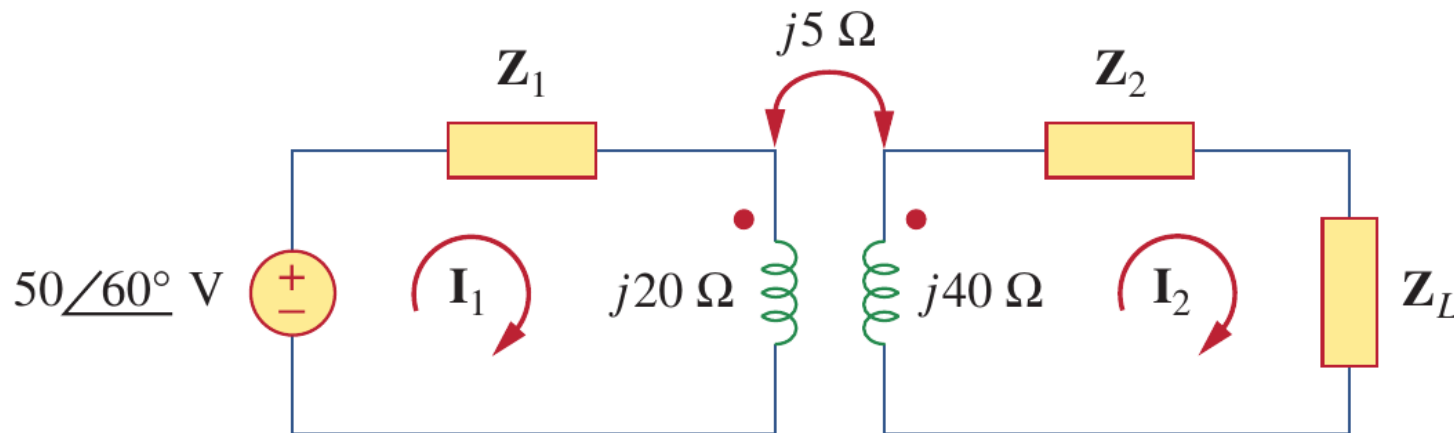
$$L_B = \frac{L_1 L_2 - M^2}{L_1 - M}$$

$$L_C = \frac{L_1 L_2 - M^2}{M}$$

12.7 Linear transformers (f)

- Example 12.06:

Example 12.06: In the following circuit, calculate the input impedance and current $\mathbf{I}_1$. Take $\mathbf{Z}_1 = 60 - j100 \, \Omega$, $\mathbf{Z}_2 = 30 + j40 \, \Omega$, $\mathbf{Z}_L = 80 + j60 \, \Omega$.



Solution:

$$\begin{aligned}\mathbf{Z}_{\text{in}} &= \mathbf{Z}_1 + j20 + \frac{(5)^2}{j40 + \mathbf{Z}_2 + \mathbf{Z}_L} \\ &= 60 - j100 + j20 + \frac{25}{110 + j140} \\ &= 60 - j80 + 0.14\angle -51.84^\circ \\ &= 60.09 - j80.11 = 100.14\angle -53.1^\circ \, \Omega\end{aligned}$$

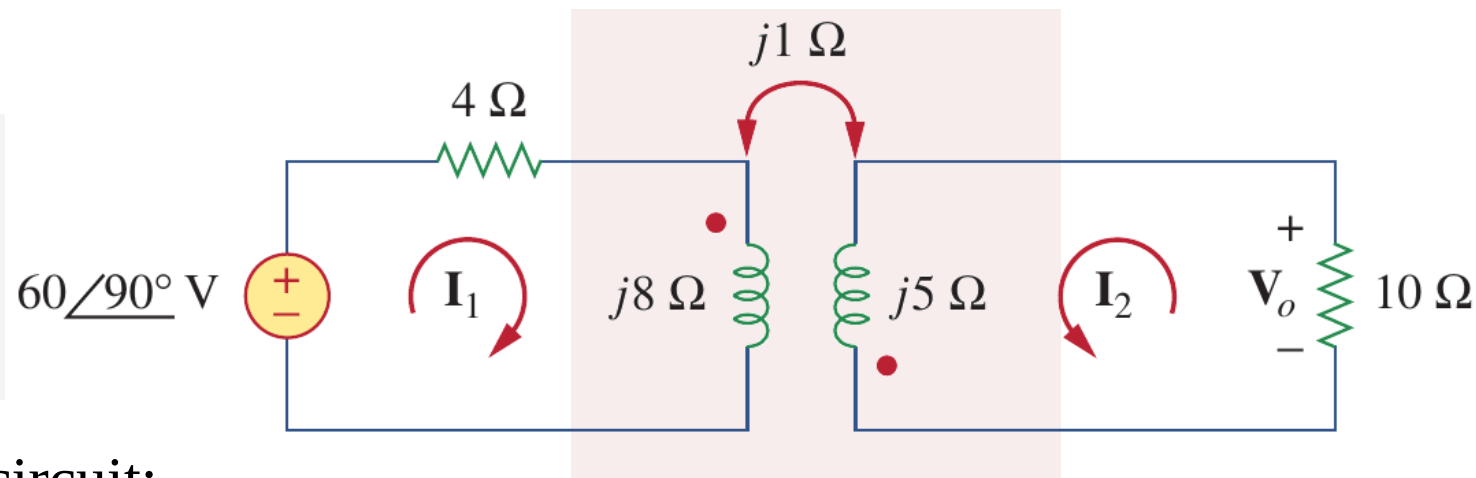
$$\mathbf{I}_1 = \frac{\mathbf{V}}{\mathbf{Z}_{\text{in}}} = \frac{50\angle 60^\circ}{100.14\angle -53.1^\circ} = 0.5\angle 113.1^\circ \text{ A}$$

$$\mathbf{Z}_{\text{in}} = \frac{\mathbf{V}}{\mathbf{I}_1} = \boxed{R_1 + j\omega L_1} + \boxed{\frac{\omega^2 M^2}{R_2 + j\omega L_2 + \mathbf{Z}_L}}$$

12.7 Linear transformers (g)

- Example 12.07:

Example 12.07: Solve for $\mathbf{I}_1$, $\mathbf{I}_2$, and $\mathbf{V}_o$ using T-equivalent circuit for the linear transformer.

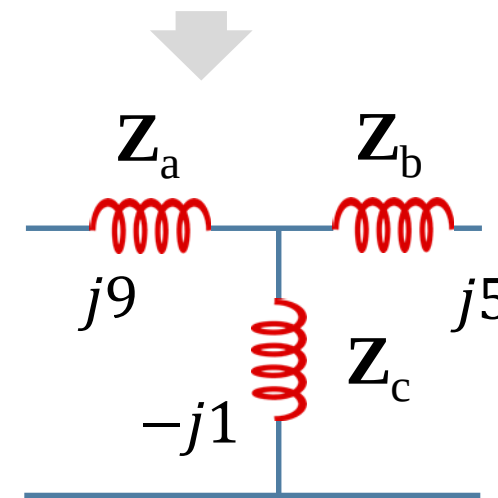


Solution: (1) Find the equivalent circuit:

$$\begin{aligned} L_a &= L_1 - (-M) \Rightarrow j\omega L_a = j\omega(L_1 - (-M)) \\ &\Rightarrow \mathbf{Z}_a = j\omega L_1 + j\omega M = j8 + j1 = j9 \end{aligned}$$

$$\begin{aligned} L_b &= L_2 - (-M) \Rightarrow j\omega L_b = j\omega(L_2 - (-M)) \\ &\Rightarrow \mathbf{Z}_b = j\omega L_2 + j\omega M = j5 + j1 = j6 \end{aligned}$$

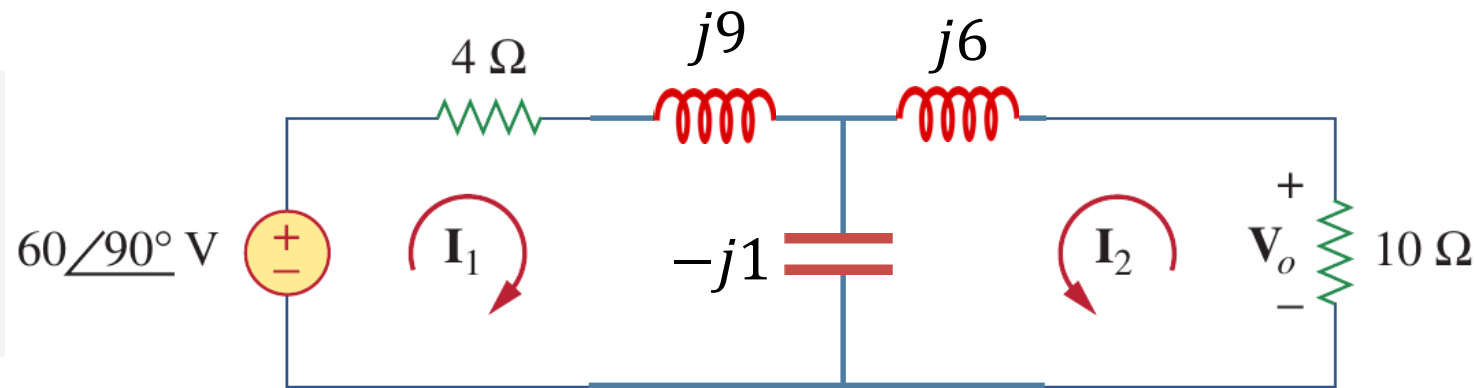
$$L_c = (-M) \Rightarrow j\omega L_c = -j\omega M \Rightarrow \mathbf{Z}_c = -j\omega M = -j1$$



12.7 Linear transformers (h)

- Example 12.07:

Example 12.07: Solve for $\mathbf{I}_1$, $\mathbf{I}_2$, and $\mathbf{V}_o$ using T-equivalent circuit for the linear transformer.



Solution: (2) Use mesh analysis:

$$\begin{cases} j6 = \mathbf{I}_1(4 + j9 - j1) + \mathbf{I}_2(-j1) \\ 0 = \mathbf{I}_1(-j1) + \mathbf{I}_2(10 + j6 - j1) \end{cases} \Rightarrow \mathbf{I}_1 = (5 - j10)\mathbf{I}_2 \Rightarrow j6 = (100 - j)\mathbf{I}_2 \approx 100\mathbf{I}_2$$

$$\mathbf{I}_2 = \frac{j6}{100} = j0.06 = 0.06\angle 90^\circ \text{ A} \quad \mathbf{I}_1 = (5 - j10)j0.06 = 0.6 + j0.3 \text{ A}$$

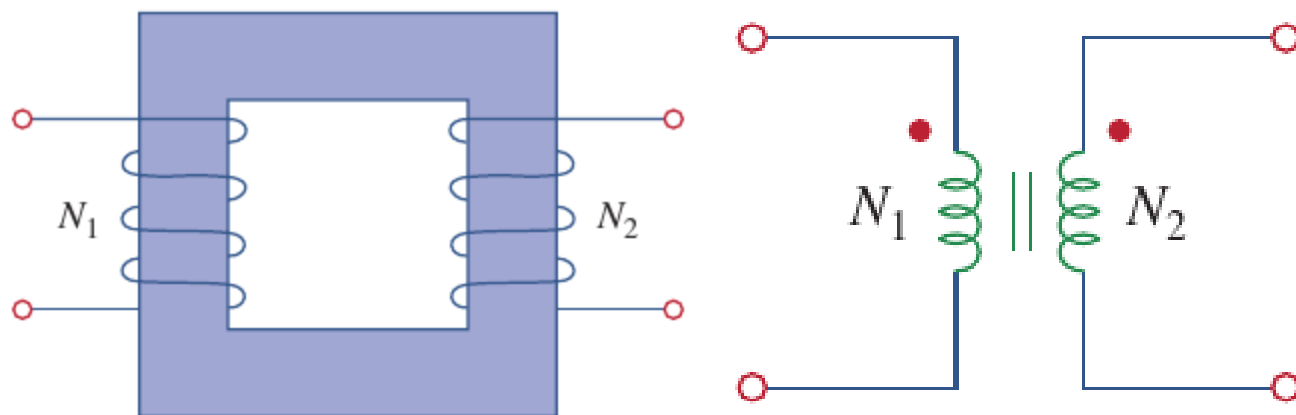
$$\mathbf{V}_o = -10\mathbf{I}_2 = -j0.6 = 0.6\angle -90^\circ \text{ V}$$

Lesson 13 Magnetically coupled circuits

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12.8 Ideal transformers (a)

- Ideal transformers (理想变压器):

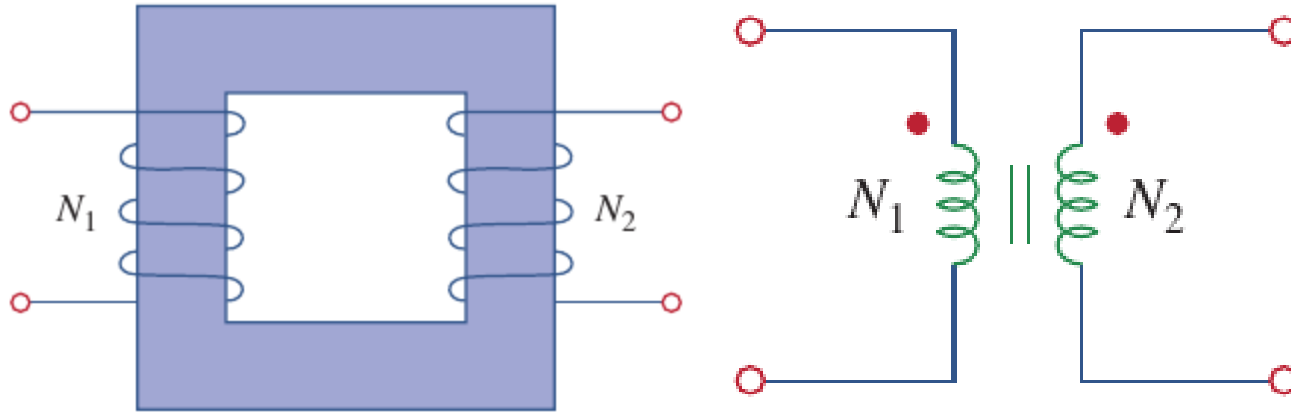


- a) Coils have very large but finite reactance: $L_1, L_2, M \rightarrow \infty$ (铁损为0)
- b) Perfect coupling: $k = 1$ (磁损为0)
- c) Coils are lossless: winding resistances are zero (铜损为0)

- a) An ideal transformer is a unity-coupled, lossless transformer in which the primary and secondary coils have infinite self-inductances.
- b) An ideal transformer is one with perfect coupling ($k = 1$). It consists of two (or more) coils with a large number of turns wound on a common core of high permeability. Because of this high permeability of the core, the flux links all the turns of both coils, thereby resulting in a perfect coupling.

12.8 Ideal transformers (b)

- Turns ratio (匝数比):



- a) Coils have very large reactance: $L_1, L_2, M \rightarrow \infty$
- b) Perfect coupling: $k = 1$
- c) Coils are lossless: winding resistances are zero

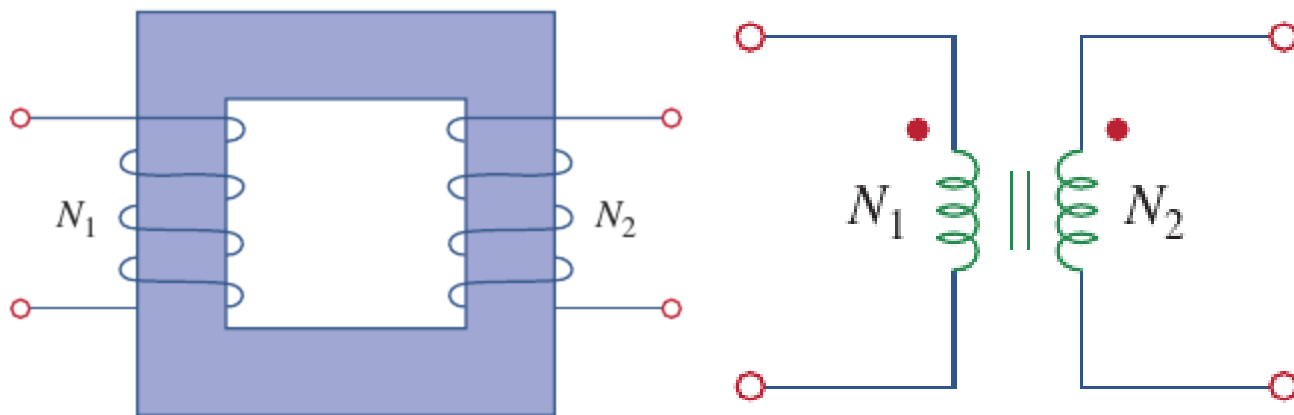
- a) When a sinusoidal voltage is applied to the primary winding, the same magnetic flux goes through both windings, and the voltages are:

$$\begin{cases} v_1 = N_1 d\phi/dt \\ v_2 = N_2 d\phi/dt \end{cases} \Rightarrow \frac{v_2}{v_1} = \frac{N_2}{N_1} = n \Rightarrow \frac{V_2}{V_1} = \frac{N_2}{N_1} = n \Rightarrow \frac{I_2}{I_1} = \frac{N_1}{N_2} = \frac{1}{n}$$

in which n is the **turns ratio** (匝数比) or the transformation ratio.

12.8 Ideal transformers (c)

- Turns ratio (匝数比):



$$\mathbf{V}_1 = j\omega L_1 \mathbf{I}_1 + j\omega M \mathbf{I}_2$$

$$\mathbf{V}_2 = j\omega M \mathbf{I}_1 + j\omega L_2 \mathbf{I}_2$$

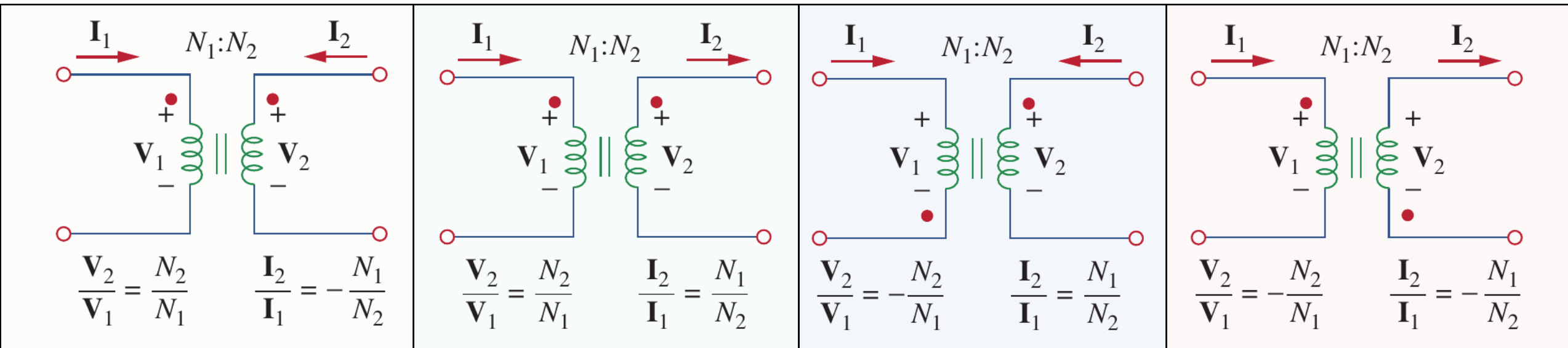
$$\mathbf{V}_2 = j\omega L_2 \mathbf{I}_2 + \frac{M \mathbf{V}_1}{L_1} - \frac{j\omega M^2 \mathbf{I}_2}{L_1} \quad \text{and } M = \sqrt{L_1 L_2} \text{ for perfect coupling}$$

$$\Rightarrow \mathbf{V}_2 = j\omega L_2 \mathbf{I}_2 + \frac{\sqrt{L_1 L_2} \mathbf{V}_1}{L_1} - \frac{j\omega L_1 L_2 \mathbf{I}_2}{L_1} \Rightarrow \mathbf{V}_2 / \mathbf{V}_1 = \sqrt{L_2 / L_1}$$

$$\text{As coils have very large reactance: } L_1, L_2, M \rightarrow \infty \Rightarrow \frac{\mathbf{V}_2}{\mathbf{V}_1} = \frac{N_2}{N_1} = n$$

12.8 Ideal transformers (d)

- Voltage and current polarities:



If V_1 and V_2 are both positive or both negative at the dotted terminals, use **+ n**.

Otherwise, use **- n**.

If I_1 and I_2 both enter into or both leave the dotted terminals, use **- n**.

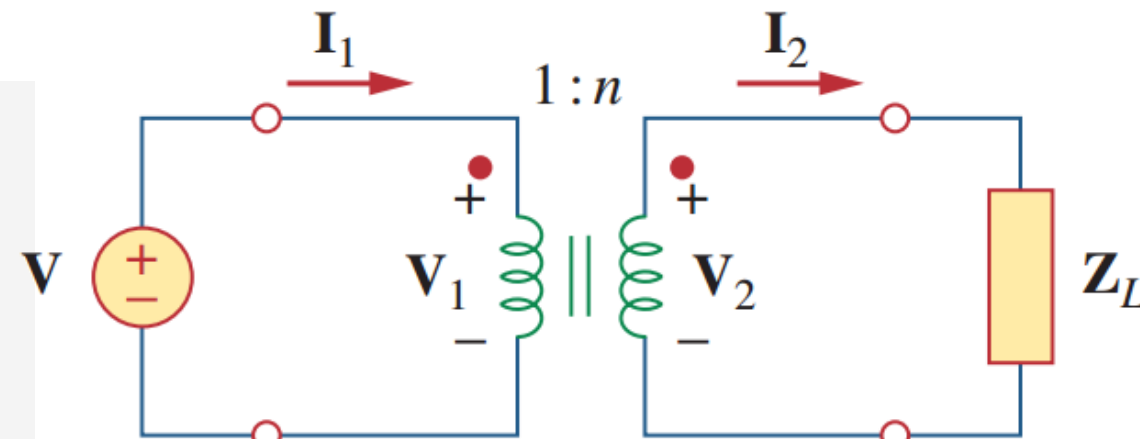
Otherwise, use **+ n**.

12.8 Ideal transformers (e)

- Different types of ideal transformers:

a) The voltage relation: $\frac{V_2}{V_1} = \frac{N_2}{N_1} = n$

- ➔ {
- $n > 1$, step-up transformer (升压变压器)
 - $n = 1$, isolation transformer (隔离变压器)
 - $n < 1$, step-down transformer (降压变压器)

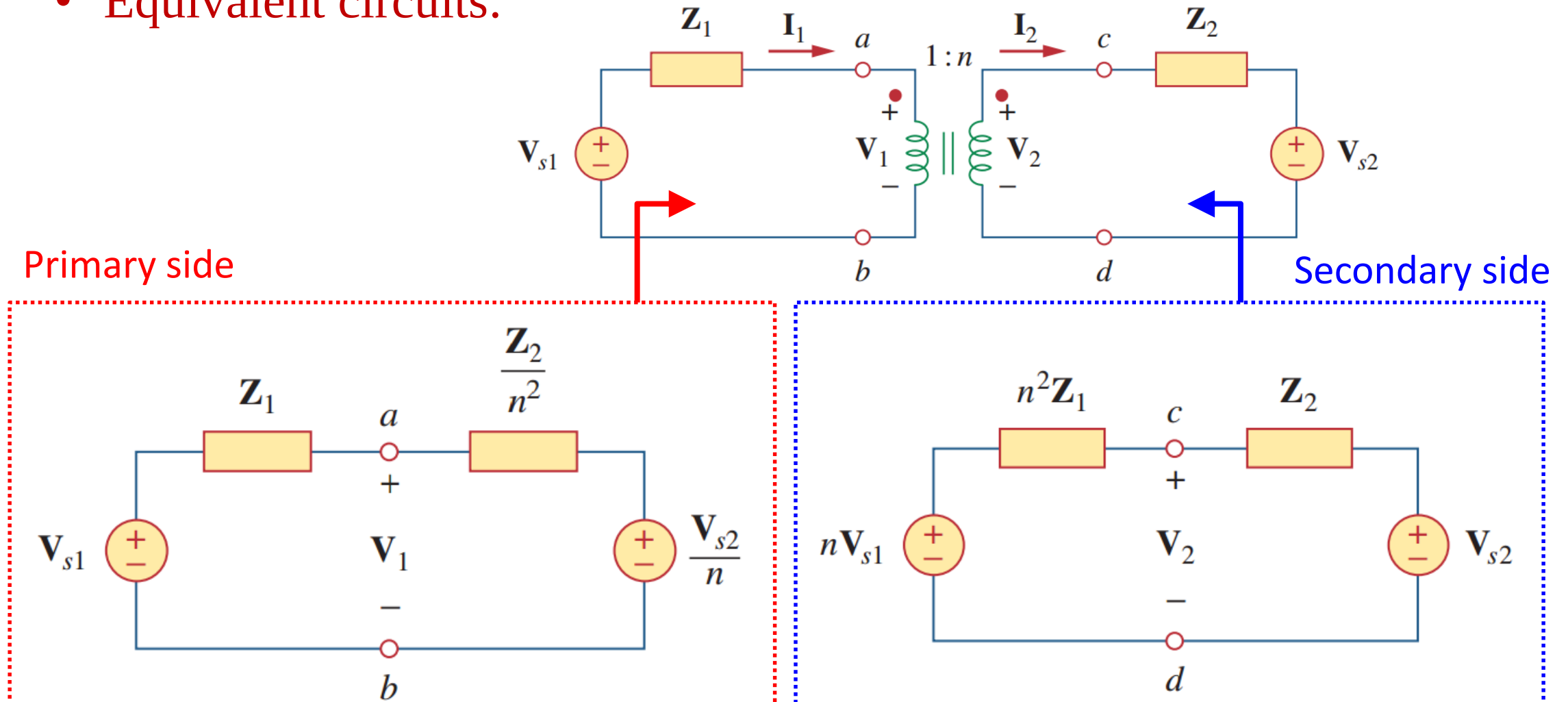


b) Reflected impedance (the equivalent impedance of load seen from the primary side):

$$Z_{\text{in}} = \frac{V_1}{I_1} = \frac{1}{n^2} \frac{V_2}{I_2} \quad \Rightarrow \quad \boxed{Z_{\text{in}} = \frac{Z_L}{n^2}}$$

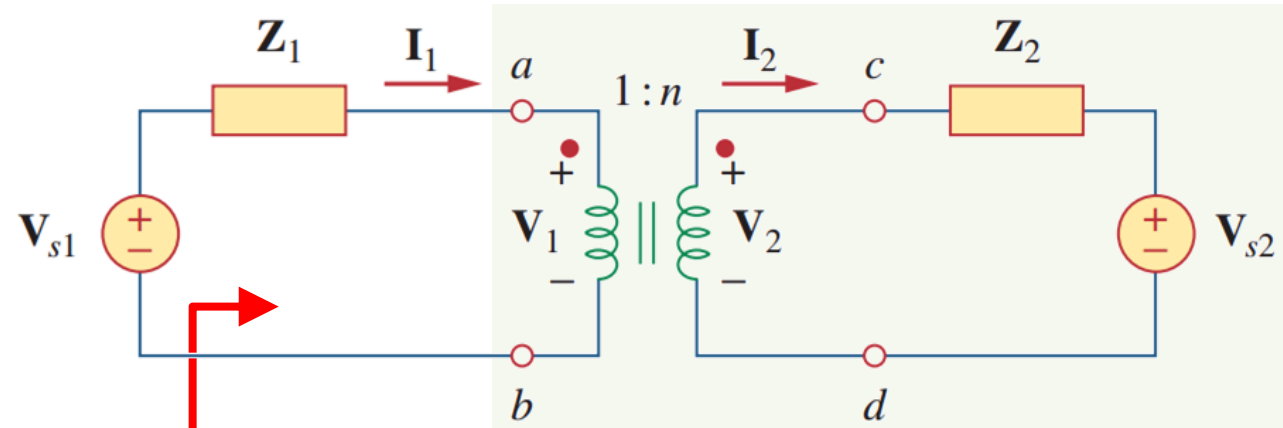
12.8 Ideal transformers (f)

- Equivalent circuits:

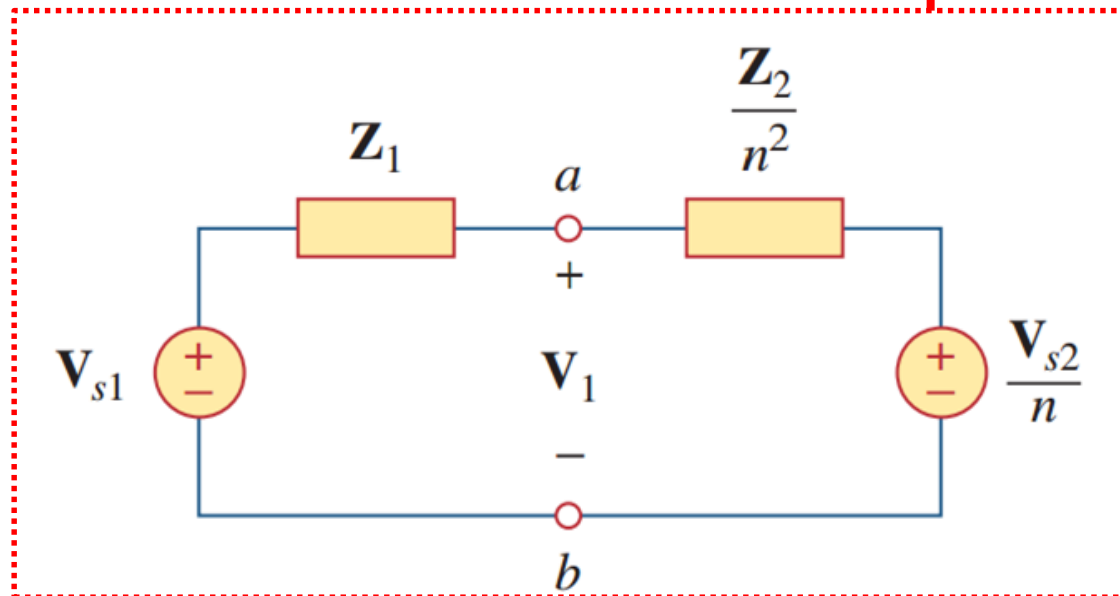


12.8 Ideal transformers (g)

- Equivalent circuits:



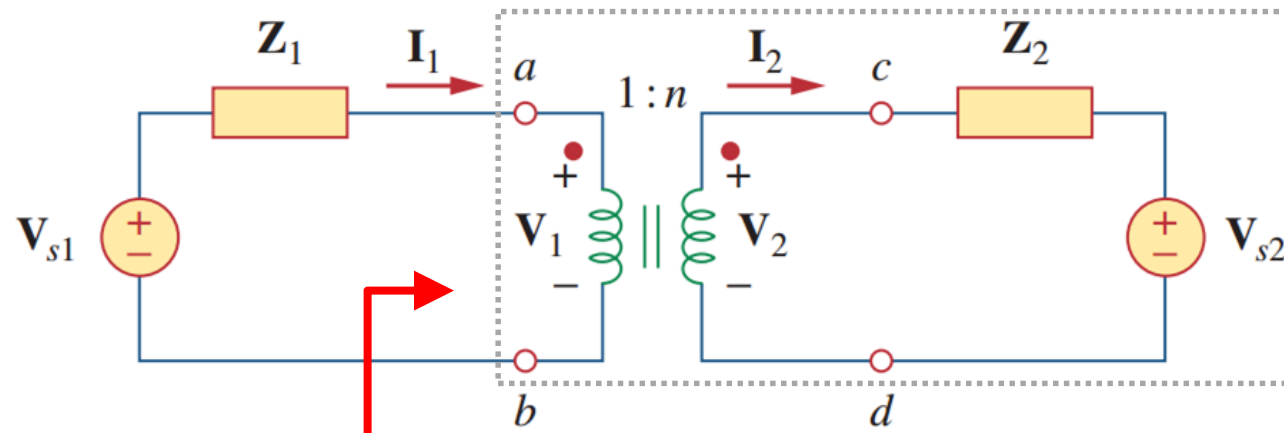
Primary side



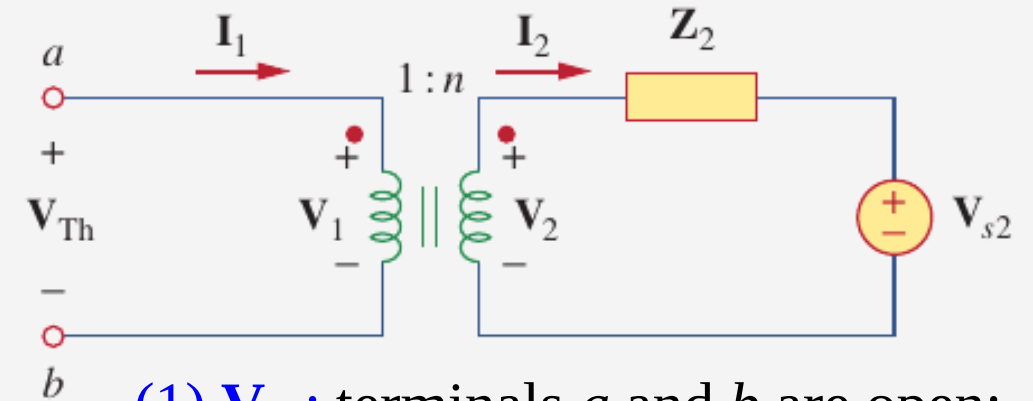
The general rule for eliminating the transformer and **reflecting the secondary circuit to the primary side** is: Divide the secondary impedance by n^2 , divide the secondary voltage by n , and multiply the secondary current by n .

12.8 Ideal transformers (h)

- Equivalent circuits:



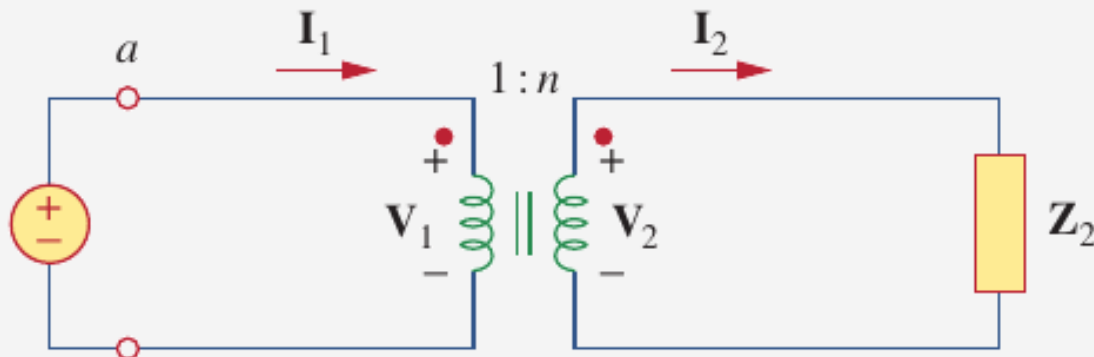
Primary side:



(1) V_{oc} : terminals a and b are open:

$$I_1 = I_2 = 0 \text{ A} \quad \& \quad V_2 = V_{s2}$$

$$V_{oc} = V_1 = V_1/n = V_{s2}/n$$

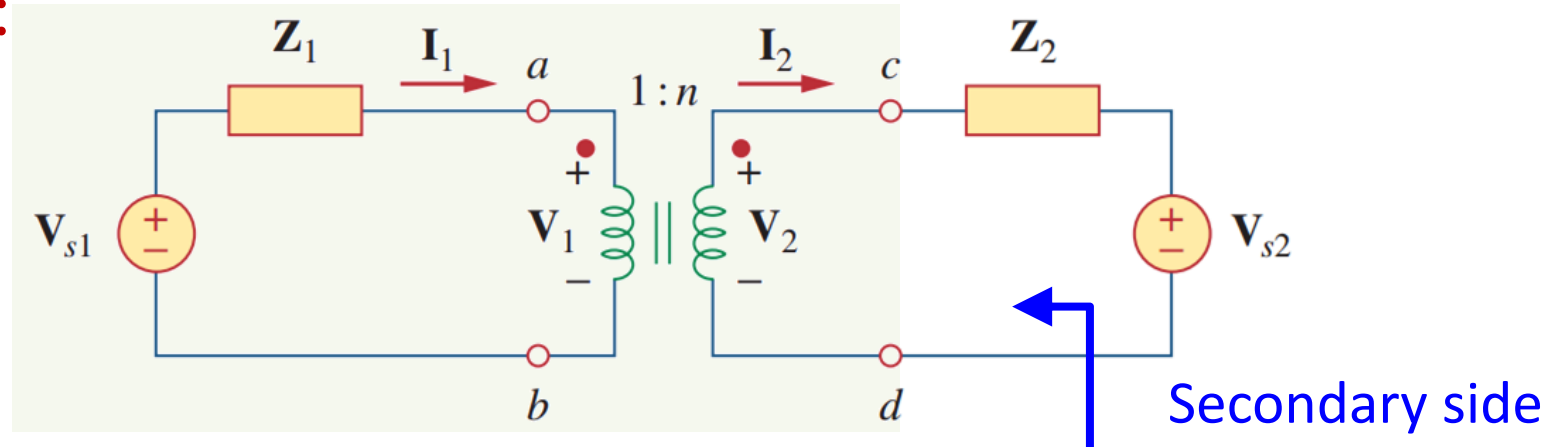


b) Z_{Th} : connect a voltage V_1 source to terminals a - b , and turn off V_{s2} ,

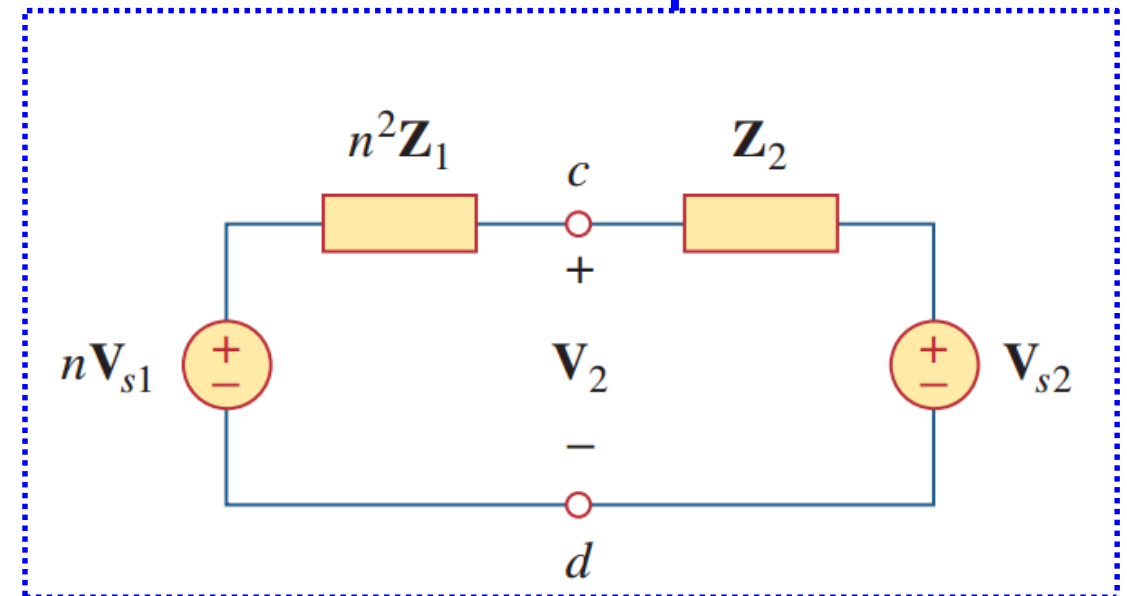
$$Z_{Th} = \frac{V_1}{I_1} = \frac{V_2/n}{nI_2} = \frac{1}{n^2} \frac{V_2}{I_2} = \frac{Z_2}{n^2}$$

12.8 Ideal transformers (i)

- Equivalent circuits:



The rule for eliminating the transformer and reflecting the primary circuit to the secondary side is: Multiply the primary impedance by n^2 , multiply the primary voltage by n , and divide the primary current by n .



12.8 Ideal transformers (j)

- Example 12.08:

Example 13.08: An ideal transformer is rated at 2400/120 V, 9.6 kVA, and has 50 turns on the secondary side. Calculate:

- (a) the turns ratio,
- (b) the number of turns on the primary side,
- (c) the current ratings for the primary and secondary windings.

Solution:

(a) This is a step-down transformer, and

$$n = \frac{V_2}{V_1} = \frac{120}{2400} = 0.05$$

(b) $N_1 = N_2/n = 50/0.05 = 1000$ turns.

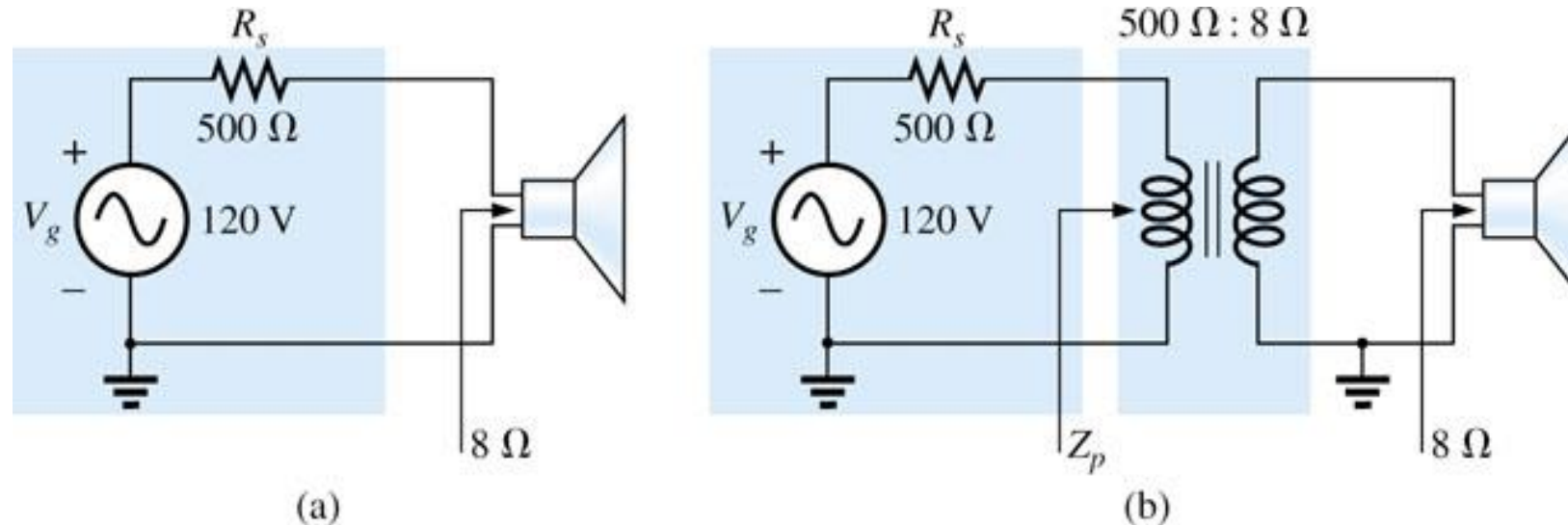
(c) $S = V_1 I_1 = V_2 I_2 = 9.6$ kVA. Hence,

$$I_1 = \frac{9,600}{V_1} = \frac{9,600}{2,400} = 4 \text{ A}$$

$$I_2 = \frac{9,600}{V_2} = \frac{9,600}{120} = 80 \text{ A}$$

13.8 Ideal transformers (k)

- Example 12.09:



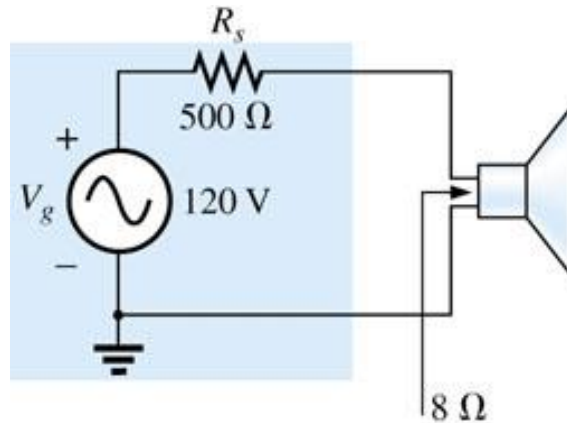
Example 12.09: Transformers can be used for impedance matching between the primary and the secondary sides in many circuits. The following figures show circuits driving the a speaker with (Fig. a) and without (Fig. b) a impedance-matching transformer.

- (a) Compare the power delivered to the speaker in the two cases and
- (b) find the approximate turns ratio for the transformer.

13.8 Ideal transformers (h)

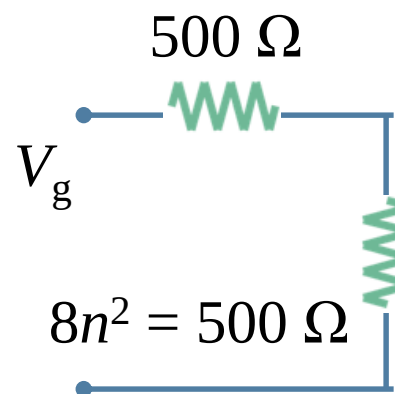
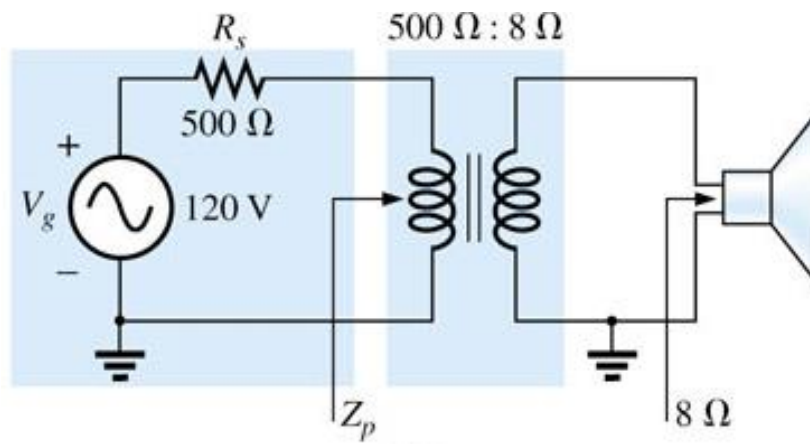
- Example 12.09:

Solution:



$$I = \frac{V}{R_{eq}} = \frac{120}{500 + 8} = 236.2 \text{ mA}$$

$$P = I^2 R_L \approx 0.45 \text{ W}$$



$$I = \frac{120}{500 + 500} = 120 \text{ mA}$$

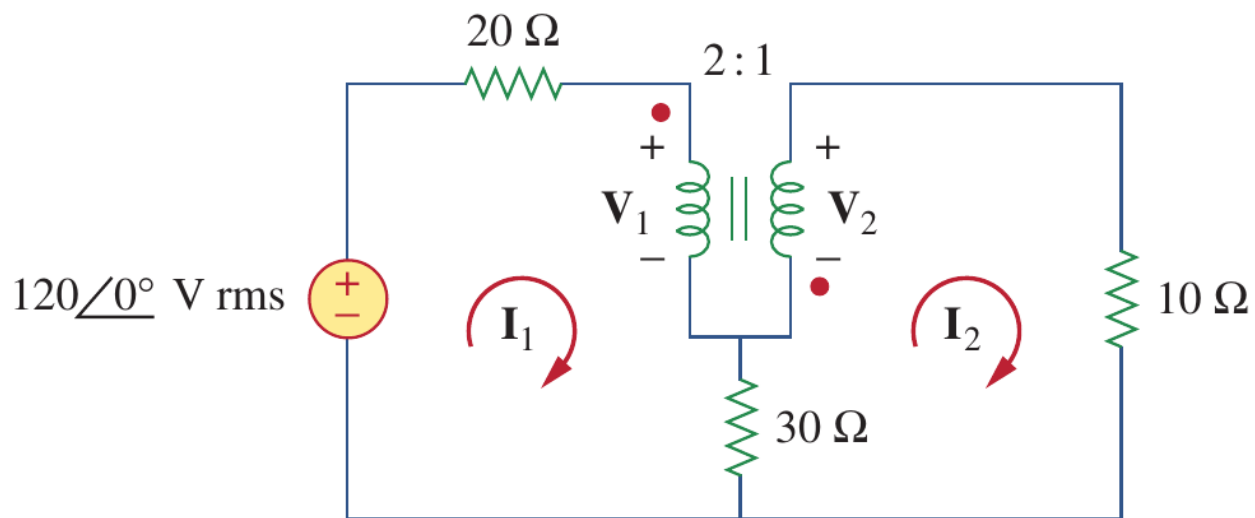
$$P = I^2 R_L = 7.2 \text{ W}$$

$$n = \sqrt{\frac{500}{8}} = 7.91 \approx 8$$

13.8 Ideal transformers (i)

- Example 12.10:

Example 12.10: Calculate the power supplied to the $10\ \Omega$ resistor in the following ideal transformer circuit.



Solution:

$$\begin{cases} -120 + (20 + 30)\mathbf{I}_1 - 30\mathbf{I}_2 + \mathbf{V}_1 = 0 \\ -\mathbf{V}_2 + (10 + 30)\mathbf{I}_2 - 30\mathbf{I}_1 = 0 \\ \mathbf{V}_2 = -\frac{1}{2}\mathbf{V}_1 \\ \mathbf{I}_2 = -2\mathbf{I}_1 \end{cases}$$

$$\mathbf{I}_2 = -\frac{120}{165} = -0.7272\text{ A}$$

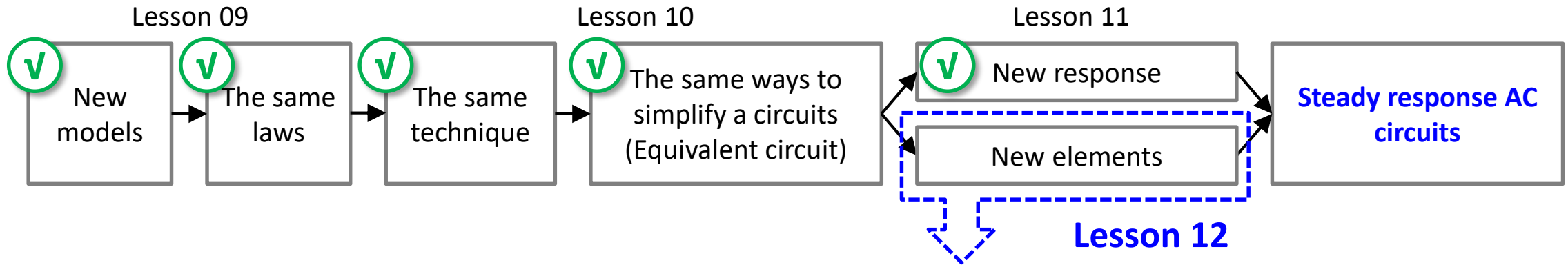
$$P = (-0.7272)^2(10) = 5.3\text{ W}$$

Lesson 12 Magnetically coupled circuits

12.1	Coupled circuits	Slides 15-21
12.2-12.7	Mutual inductance	Slides 23-60
12.8	Linear transformers	Slides 62-69
12.9	Ideal transformers	Slides 71-83
12.10	Summary and assignment	Slides 85-86

12.10 summary and assignment (a)

- EE104 road map



Transformers

- Coupling
- Mutual inductance
- Linear transformers
- Ideal transformers

12.10 summary and assignment (b)

Problems

13.25, 13.35, 13.46, 13.63, 13.65

- (a) A certain amount of practicing is necessary to sharpen your mind and to deepen your understanding.
- (b) There are plenty of problems in the textbook, try to complete as many as you can.

Alto cumulus Castelanus/Jellyfish clouds

